# MS4000

**Model C**

**Submersible motors**

**50/60 Hz**

GRUNDFOS

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# 1. Product description

The Grundfos MS4000 product range is a complete range of submersible motors available in 50 Hz and 60 Hz versions from 0.37 kW up to 7.5 kW.

The entire motor surface is made of stainless steel, so that all external components have a uniform corrosion resistance. Two material versions are available:

- a standard version made of stainless steel EN 1.4301 (AISI 304).
- an R-version made of stainless steel EN 1.4539 (AISI 904L) for aggressive liquids such as mine- and seawater.

The Grundfos MS4000 submersible motors are designed according to market standards. All Grundfos MS4000 motors are designed to fit pump ends manufactured according to NEMA standards, therefore, they can be fitted on all Grundfos SP pumps without adapters.

General characteristics of the MS4000 motor:

- 4" diameter (OD = 95 mm)
- 2-pole asynchronous motor
- hermetically sealed, canned motor with dry stator
- stator completely encapsulated in stainless steel
- cooled by pumped liquid
- hydrodynamic bearings lubricated by the motor liquid
- high efficiency
- operating media temperature up to 60 °C
- enclosure class IP68
- insulation class 150 °C (F)
- NEMA flange and shaft spline
- CE or cCSAus-approved options
- reliable cable plug system with robust and safe earth connection
- factory filled with Grundfos motor liquid SML-3, alternatively filled with demineralized water or high glycol content motor liquid (60 % HTF)
- built-in temperature transmitter (Tempcon) available on request for all motor variants
- motor temperature monitoring via a Pt1000 sensor is available as an option (used for frequency controlled installations).

TM085878

*MS4000 motors without cable*

TM087041

*MS4000 motors with cable*

## Features and benefits

The Grundfos MS4000 submersible motor offers the following features and benefits:

- high motor efficiency
- sealing system
- high reliability
- super stainless steel versions
- worldwide usage
- highly reliable thrust bearing
- monitoring of motor temperature
- additional monitoring of motor temperature.

**High motor efficiency**

The complete motor range is characterized by high efficiency which contributes to improved economy of the entire pump system.

**Sealing system**

All motors are equipped with mechanical shaft seals.

**High reliability**

The shaft seal design and material choices offer wear resistance and long operating life.

**Super stainless steel version**

Super stainless steel version EN 1.4539 (AISI 904 L) with silicon carbide based (SiC/SiC) mechanical shaft sealing system and FKM elastomer parts are used for applications in seawater and slightly contaminated environment that might contain hydrocarbons.

**Worldwide usage**

With different voltage and frequency combinations, the product range covers markets worldwide.

**Highly reliable thrust bearing**

Fitted with a sturdy Michell/Kingsbury thrust bearing, the motors offer high and reliable operation.

**Monitoring of motor temperature**

To achieve maximum protection of the motor against burnout, the motors are also available with a built-in Tempcon temperature sensor with power line communication. In combination with motor protection (MP204), the Tempcon sensor offers optimal protection for the motor. Tempcon is not suitable for frequency converter operation.

**Additional monitoring of motor temperature**

The MS4000 motors also allow for installing a Pt1000 sensor to monitor the temperature.

This solution is suitable for frequency converter operation.

## Applications

Grundfos submersible motors are designed for a wide range of applications, for example, the following:

- deep well water supply
- irrigation
- groundwater regulation
- pressure boosting
- industrial water transfer and similar applications
- fountains
- dewatering.

## Application variants

The MS4000 asynchronous motor is available in a wide range of variants for different applications.

MS4000**AZBCW____**

It is for operation in water without any appreciative content of chloride: EN 1.4301 material, elastomers in NBR and HM/Ceramic shaft seal.

The motor contains NBR elastomers, therefore it is approved for drinking water.

MS4000**AZBSW____**

It is for operation in water with abrasive particles but without appreciable chloride content: EN 1.4301 material, elastomers in NBR and Sic/Sic shaft seal.

The motor contains NBR elastomers, therefore it is approved for drinking water.

MS4000**AZECW____**

It is for operation in neutral liquids with hydrocarbon content and dissolved gasses: EN 1.4301 material, elastomers in FKM and HM/Ceramic shaft seal.

MS4000**AZESW____**

It is for operation in neutral liquids with hydrocarbon content, dissolved gasses and abrasive particles: EN 1.4301 material, elastomers in FKM and Sic/Sic shaft seal.

MS4000**ARBSW____**

It is for operation in water with chloride content (salty water) and abrasive particles: EN 1.4539 material (R version), elastomers in NBR and Sic/Sic shaft seal.

MS4000**ARESW____**

It is for operation in aggressive liquids: EN 1.4539 material (R version), elastomers in FKM, and Sic/Sic shaft seal.

MS4000**____D___**

It is for applications that do not allow the risk of contamination with SML-3 motor liquid based on glycol (e.g. chip manufacturing or RO plants): factory-filled with demineralized water.

MS4000**_____J__,** MS4000**_____X__**

The variants can be ordered with (J) or without (X) the built-in temperature sensor (Tempcon).

MS4000**_____T4**

It is for operation in liquids up to 40 °C.

MS4000**_____T6**

It is for operation in liquids up to 60 °C.

# 2. Identification

## Type key

The type key cannot be used for ordering as not all combinations are possible.

| Example | MS4000 A | R | E | S | W | D | M | X | T4 | 3 × 380-415/50 | 7.5 kW |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Pos. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |

| Pos. | Description | | |
|---|---|---|---|
| 1 | Motor type | | |
| | MS4000A | = | MS4000 asynchronous motor |
| 2 | Material version | | |
| | Z | = | EN 1.4301 |
| | R | = | EN 1.4539 |
| 3 | Elastomers | | |
| | B | = | NBR |
| | E | = | FKM |
| 4 | Shaft seal | | |
| | C | = | Hard metal/Ceramic |
| | S | = | SiC/SiC |
| 5 | Bearing system | | |
| | W | = | Ceramic |
| 6 | Motor liquid | | |
| | L | = | SML-3 |
| | D | = | Demineralised water |
| | H | = | High glycol (60 % HTF) |
| 7 | Thread | | |
| | U | = | UNF 5/16" |
| | M | = | M8 |
| 8 | Tempcon | | |
| | J | = | With |
| | X | = | Without |
| 9 | Temperature version (Max. pumped liquid temperature) | | |
| | T4 | = | 40 °C at min. 0.15 m/s |
| | T6 | = | 60 °C at min. 1.0 m/s |

| | Key | Config. code | Key | Config. code | Key | Config. code | Key | Config. code | Key | Config. code |
|---|---|---|---|---|---|---|---|---|---|---|
| Available configurations | ZBCWLMXT4 | 25 | ZBSWLMXT4 | 23 | ZESWLMXT4 | 21 | RBSWLMXT4 | 27 | RESWLMXT4 | 29 |
| | ZBCWLUXT4 | 25 | ZBSWLUXT4 | 23 | ZESWLUXT4 | 21 | RBSWLUXT4 | 27 | RESWLUXT4 | 29 |
| | ZBCWLMJT4 | 35 | ZBSWLMJT4 | 33 | ZESWLMJT4 | 31 | RBSWLMJT4 | 37 | RESWLMJT4 | 39 |
| | ZBCWLUJT4 | 35 | ZBSWLUJT4 | 33 | ZESWLUJT4 | 31 | RBSWLUJT4 | 37 | RESWLUJT4 | 39 |
| | ZBCWLMXT6 | 45 | ZBSWLMXT6 | 43 | ZESWLMXT6 | 41 | RBSWLMXT6 | 47 | RESWLMXT6 | 49 |
| | ZBCWLUXT6 | 45 | ZBSWLUXT6 | 43 | ZESWLUXT6 | 41 | RBSWLUXT6 | 47 | RESWLUXT6 | 49 |
| | ZBCWLMJT6 | 55 | ZBSWLMJT6 | 53 | ZESWLMJT6 | 51 | RBSWLMJT6 | 57 | RESWLMJT6 | 59 |
| | ZBCWLUJT6 | 55 | ZBSWLUJT6 | 53 | ZESWLUJT6 | 51 | RBSWLUJT6 | 57 | RESWLUJT6 | 59 |
| | ZBCWDMXT4 | 65 | ZBSWDMXT4 | 63 | ZESWDMXT4 | 61 | RBSWDMXT4 | 67 | RESWDMXT4 | 69 |
| | ZBCWDUXT4 | 65 | ZBSWDUXT4 | 63 | ZESWDUXT4 | 61 | RBSWDUXT4 | 67 | RESWDUXT4 | 69 |
| | ZBCWDMJT4 | 75 | ZBSWDMJT4 | 73 | ZESWDMJT4 | 71 | RBSWDMJT4 | 77 | RESWDMJT4 | 79 |
| | ZBCWDUJT4 | 75 | ZBSWDUJT4 | 73 | ZESWDUJT4 | 71 | RBSWDUJT4 | 77 | RESWDUJT4 | 79 |
| | ZBCWDMJT6 | 85 | ZBSWDMJT6 | 83 | ZESWDMJT6 | 81 | RBSWDMJT6 | 87 | RESWDMJT6 | 89 |
| | ZBCWDUJT6 | 85 | ZBSWDUJT6 | 83 | ZESWDUJT6 | 81 | RBSWDUJT6 | 87 | RESWDUJT6 | 89 |

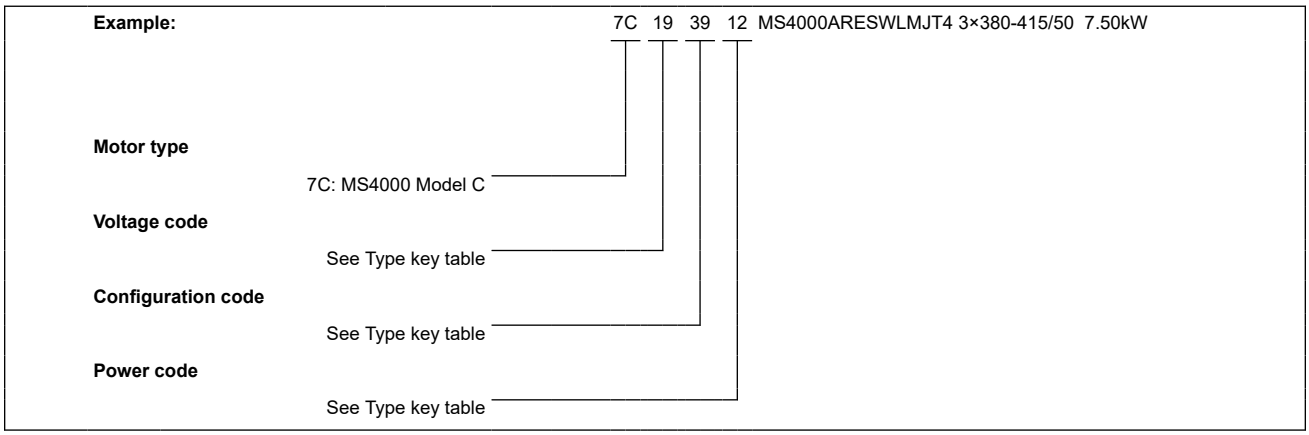
| Pos. | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 10 | Voltage | | | | | | | | |
| | | | 50 Hz | 60 Hz | Method of starting [1] | Thread | Region | Approval | Voltage code |
| | 1×220/50 PSC | = | 220-240 | - | 3-wire PSC | M8 | Global | CE | 75 |
| | 1×240/50 PSC | = | 240 | - | 3-wire PSC | M8 | Global | CE | 25 |
| | 1×220-230/50 CSCR | = | 220-230 | - | 3-wire CSCR | M8 | Global | CE | 21 |
| | 1×240/50 CSCR | = | 240 | - | 3-wire CSCR | M8 | Global | CE | 24 |
| | 1×230/60 CSCR | = | - | 230 | 3-wire CSCR | UNF 5/16 | US | cCSAus | 45 |
| | 3×220-230/50 | = | 220-230 | - | DOL | M8 | Global | CE | 10 |
| | 3×380/50 440/60 | = | 340-360-380 | 440 | DOL | M8 | Global | CE | 08 |
| | 3×380-415/50 | = | 380-400-415 | 440-460-480 | DOL | M8 | Global | CE | 19 |
| | 3×440-460/50 | = | 440-460 | - | DOL | M8 | Global | CE | 17 |
| | 3×500-525/50 | = | 500-525 | - | DOL | M8 | Global | CE | 18 |
| | 3×208-230/60 | = | - | 208-220-230 | DOL | M8 | Global | CE | 28 |
| | 3×380-415/60 | = | - | 380-400-415 | DOL | M8 | Global | CE | 33 |
| | 3×208/60 | = | - | 208 | DOL | UNF 5/16 | US | cCSAus | 32 |
| | 3×208-230/60 | = | - | 208-220-230 | DOL | UNF 5/16 | US | cCSAus | 30 |
| | 3×230/60 | = | - | 230 | DOL | UNF 5/16 | US | cCSAus | 31 |
| | 3×440-460/60 | = | - | 440-460 | DOL | UNF 5/16 | US | cCSAus | 35 |
| | 3×575/60 | = | - | 575 | DOL | UNF 5/16 | US | cCSAus | 39 |
| | 3×200/50 220/60 | = | 200 | 200-220 | DOL | M8 | Japan | CE | 09 |
| | 3×400/50 440/60 | = | 400 | 400-440 | DOL | M8 | Japan | CE | 65 |

| 11 Motor power | | | | |
|---|---|---|---|---|
| kW | hp | Service factor (60 Hz) | | Power code |
| | | Global/US | Japan | |
| 0.37 | 0.50 | 1.60 | - | 02 |
| 0.55 | 0.75 | 1.50 | - | 03 |
| 0.75 | 1.00 | 1.40 | 1.00 | 04 |
| 1.10 | 1.50 | 1.30 | 1.00 | 05 |
| 1.50 | 2.00 | 1.25 | 1.00 | 06 |
| 1.90 | 2.50 | - | 1.00 | 95 |
| 2.20 | 3.00 | 1.15 | 1.00 | 07 |
| 2.70 | 3.60 | - | 1.00 | 96 |
| 3.00 | 4.00 | 1.15 | 1.00 | 08 |
| 3.70 | 5.00 | 1.15 | 1.00 | 09 |
| 4.00 | 5.50 | 1.15 | 1.00 | 10 |
| 5.50 | 7.50 | 1.15 | 1.00 | 11 |
| 7.50 | 10.00 | 1.15 | 1.00 | 12 |

[1] Method of starting:

- DOL: Direct on line
- CSCR: Capacitor-Start Capacitor-Run induction motor, requires a Grundfos CSCR motor starter box
- PSC: Permanent Split Capacitor motor, requires a run capacitor.

## Talkative part number



# 3. Installation and operating conditions

## Pumped liquids

The MS4000 motors are produced with different materials to enable their use in various liquids.

- The standard material version (EN 1.4301/ss 316) is intended for use in freshwater without any appreciative chloride content. The R material version (EN 1.4539/ss 904L) is recommended for use in water with chloride content (for example, seawater) or in aggressive liquids (for example, mine applications).
- Elastomers made of NBR for drinking water applications, while FKM elastomers are intended for operation in neutral liquids with hydrocarbon content and dissolved gasses.
- Shaft seal with HM/ceramic faces is for use in water without abrasive particles. In water with abrasive particles, such as sand, or aggressive liquids, the Sic/Sic shaft seal is recommended.

The motors are designed for use in applications with a maximum permissible abrasive content of 200 ppm in the pumped liquid.

For special applications, make an analysis of the liquid and contact Grundfos.

### Chloride containing water

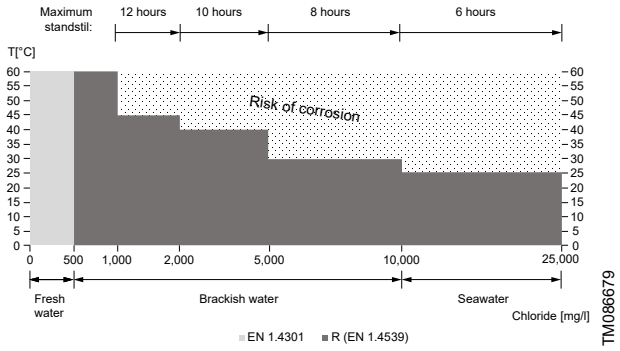
When immersed in chloride-containing water, stainless steel can be damaged by crevice or pitting corrosion.

The likelihood of corrosion depends on the following factors:

- material version (standard, R)
- chloride concentration in the water
- temperature
- oxygen concentration
- presence of biological activity
- velocity of the water around the motor
- pH value
- presence of disinfectant, such as chlorine, sulphuric acid (for example, in mining applications) and chemicals.

The increase in chloride concentration and/or temperature is unfavorable to stainless steel. Additionally, standstill time and biological activity increase the risk of corrosion. At chloride concentrations above 1000 mg/l, longer standstill periods must be avoided. All these factors must be considered when selecting the right material version and assessing the cathodic protection requirement.

The following diagram shows the allowable ranges of each material version for water with pH above 6.5.




*Corrosion resistance diagram at pH above 6.5*

| Pos. | Description |
|---|---|
| T [°C] | Pumped water temperature in °C |
| Chloride [mg/l] | Chloride concentration in mg/l |

Cathodic protection must be used when there is a risk of corrosion, that is, in standstill periods or temperatures higher than those allowed in the table above. Grundfos offers sacrificial zinc anodes to bring cathodic protection, see section Accessories. Cathodic protection is not recommended in liquids with a pH value lower than 6 or above 9.

If aggressive liquids are pumped, a flow sleeve may be required to ensure sufficient cooling (flow and temperature) around the motor since corrosion is doubled for every 10 °C rise in temperature.

To splice motor cables and drop cables, use the type KM cable termination kits in seawater applications.

## Ambient pressure

The motor alone or assembled with EPR jacketed or XLPE single leads motor cables, maximum 60 bar ~ 6.0 MPa.

Motors assembled with PTFE jacketed motor cables, maximum 20 bar ~ 2.0 MPa.

It is not recommended to use the motor for operation in a vacuum.

## Cooling

The cooling of the motor depends on the temperature and the flow velocity of the pumped liquid past the motor. It is important that the values for maximum temperature of the pumped liquid and its minimum velocity past the motor are fulfilled to ensure sufficient cooling of the motor. See the table below.

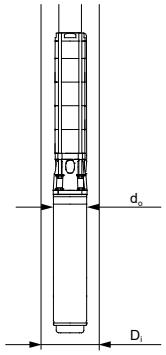
| Motor | Pumped liquid past the motor | |
|---|---|---|
| | Minimum flow velocity | Maximum temperature |
| MS4000 (T4 versions) | 0.15 m/s (0.5 ft/s) | 40 °C (~ 104 °F) |
| MS4000 (T6 versions) | 1.00 m/s (3.3 ft/s) | 60 °C (~ 140 °F) |

### Calculation of the flow velocity

$$V = \frac{Q \times 353}{D_i^2 - d_o^2} \left[ m/s \right]$$

Legends

| | | |
|---|---|---|
| v | = | flow velocity past the motor [m/s] |
| Q | = | pump rated flow [$m^3/h$] |
| $D_i$ | = | internal borehole diameter [mm] |
| $d_o$ | = | outer motor diameter [mm]. |




TM022269

*Diameter of motor and borehole*

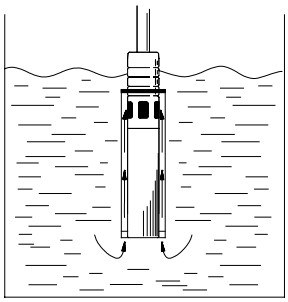
### Flow sleeve

Flow sleeves are recommended for all applications in which motor cooling is insufficient. The flow sleeve is fitted to the submersible motor so that the liquid passes close by the motor on its way toward the pump suction interconnector, ensuring the optimum cooling of the motor. The result is a general extension of motor life.

Flow sleeves are necessary to ensure sufficient cooling in the following cases:

- The velocity of the flow passing the motor is not achieved due to the difference between the borehole and motor diameters.
- The pumped liquid does not flow through the entire motor length, for instance, when the pump is located under the well screens (top feeding).
- Sedimentation, deposits, or biological growth could occur around or on the motor.
- The submersible pump is exposed to high thermal load due to the high temperature of the pumped liquid or bad cooling conditions.
- The submersible pump is installed horizontally.

Grundfos offers a complete range of stainless steel flow sleeves for both vertical and horizontal operation. The selection depends on the combined SP pump and motor assembly. Review the *SP Accessories Data Booklet* available at the *Grundfos Product Center* for more information.



TM010509

*Flow sleeve function*

### Related information

*Motor cooling*

## Temperature monitoring

Monitoring the motor temperature to detect and correct changes over time, for example, due to improper cooling, prevents motor life reduction. Deviations from the temperature under normal operation conditions could indicate changes in operation conditions or motor abnormalities that must be addressed to avoid failures.

### Temperature sensors

The MS4000 can be ordered with a built-in temperature transmitter (Tempcon) to enable monitoring of the motor temperature. The Tempcon transmits the motor temperature as a high-frequency signal through the motor power cables, and requires connection to an MP204 device for signal reading and monitoring. The Tempcon is not able to monitor the temperature with a frequency converter. The frequency converter signal will cause a fuse in the Tempcon to melt, and the fuse cannot be replaced.

Additionally, all MS4000 motors can be fitted with a Pt1000 temperature sensor. The Pt1000 can be connected to a PR 5714 relay, MP204 motor protector, frequency converter, or another device prepared for the Pt1000 reading and temperature monitoring.

### Sensor cables

Grundfos offers the Pt1000 with sensor cable up to 100 meters. For longer cables, it is recommended to splice a 20 meter Pt1000 to the required drop sensor cable length. The maximum recommended total length is 350 m.

### Temperature limits

To set the warning and alarm (trip) limits, add 5 °C and 10 °C to the stable temperature of the motor during normal operation at maximum load (highest current demand).

### Related information

*Protection*

*Frequency converter operation*

*Pt1000*

## Thrust

The thrust load capacity is related to the motor configuration and size.

| Motor type | | Motor power | | |
|---|---|---|---|---|
| Phase | Temperature version | [kW (hp)] | | Thrust |
| | | Min. | Max. | |
| Single phase | T4 | 0.37 (0.50) | 3.70 (5.00) | 4400 |
| Three phase | T4 | 0.75 (1.00) | 1.50 (2.00) | 2200 |
| | | 1.90 (2.50) | 7.50 (10.00) | 4400 |
| | T6 | 0.55 (0.75) | 1.50 (2.00) | 2200 |
| | | 1.90 (2.50) | 5.50 (7.50) | 4400 |

## Output power

The rated power (P2), service factor (SF), and SF current (Max. SF C.) are stamped on the nameplate for a respective supply (rated voltage and frequency). The SF power is the maximum power output, and equals the rated power times the SF.

The Grundfos submersible motors are designed to run continuously (S1 Duty) at SF current with SF power as long as the voltage quality and cooling requirements are met.

## Frequency of starts and stops

The MS4000 is designed for continuous as well as intermittent operation.

The maximum number of starts in the following table applies only to the motor and may be limited by the pump design and the drive system.

If the motor is stored, the shaft must be rotated manually at least once a year.

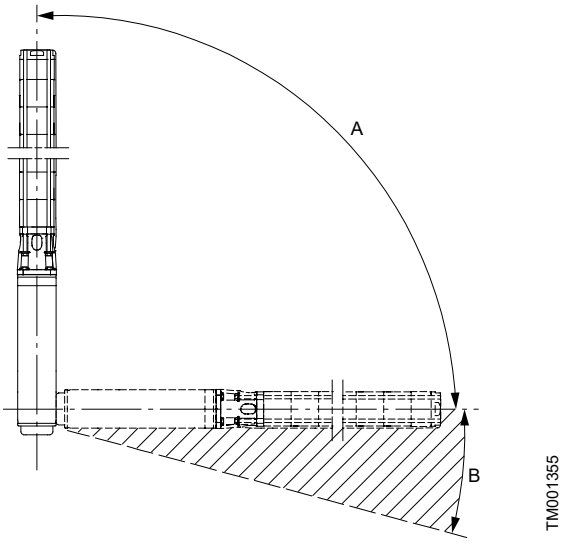
| | |
|---|---|
| Minimum number of starts | 1 per year |
| Maximum number of starts | 100 per hour |
| | 300 per day |

## Mechanical installation

The motor must be fully submerged in the pumped liquid during operation, ensuring sufficient cooling of the motor and lubrication of the shaft seal. It can be installed either horizontally or vertically.

The maximum installation depth allowed below water level is 600 m.

If the motor is installed horizontally, the shaft end must not fall below the horizontal level. The use of a flow sleeve is always recommended when the motor is installed horizontally.




*Positional requirements*

| Pos. | Description |
|---|---|
| A | Allowed |
| B | Not allowed |

## Electrical supply requirements

The electrical installation must comply with local legislation.

### Supply requirements

Measured at the motor terminals, the motor requires the following voltage quality in relation to the rated voltage stamped on the motor nameplate:

- Voltage range motors: +6 % / -10 %
- Fixed voltage motors: +10 % / -10 %.

The tolerance covers variations in the mains supply and losses in the cables.

There must be voltage symmetry in the supply network, that is, there must be the same voltage difference between the individual phases. When the motor is operating, there must be current symmetry, that is, the three phases must be evenly loaded.

The voltage and current unbalance between the phases must be within the limits below:

- maximum voltage unbalance: 2 %
- maximum current unbalance: 5 %.

Even a small voltage unbalance results in a large current unbalance, causing an uneven distribution of heat in the stator windings, leading to hot spots and local overheating. Possible current unbalance can be limited by rotating the phase sequence.

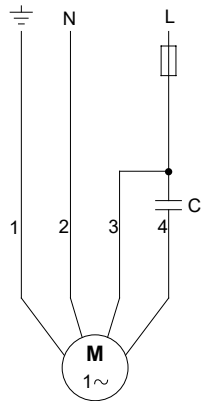
## Connection of single-phase motors

The MS4000 3-wire single-phase motors are split-phase motors with built-in separate main and auxiliary winding. The main winding is connected directly to the power supply, whereas the connection of the auxiliary winding depends on the motor type.

**Single-phase PSC motors:** the auxiliary winding must be connected via a run capacitor permanently installed in the electrical panel.

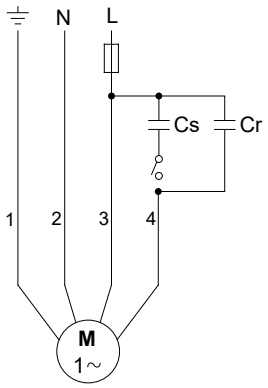
**Single-phase CSCR motors:** the auxiliary winding must be connected via two parallel branches, one with a permanently installed run capacitor and the other with a start capacitor, and a voltage relay in series. The relay must disconnect the start capacitor after the motor starts, leaving the auxiliary winding connected through the run capacitor for normal operation.

Grundfos offers run capacitors and CSCR motor starter boxes to connect the single-phase motors. The motor starter boxes include motor protection. For more details, see the accessories section.




TM001359

*PSC motors*




TM074265

*CSCR motors*

**Related information**

*CSCR motor starters*

*PSC motor capacitors*

## Protection

The MS4000 motors do not have built-in motor protection.

**Overload and short-circuit**

The MS4000 motors must be protected by a thermal relay placed on the electrical panel, starting device, or drive. A thermal relay protects against overload and short-circuit of the motor, as well as against mains failure. The thermal relay must cut out in maximum 10 seconds in case it exceeds 5 times the maximum SF current (rated current on the nameplate), IEC trip class 10.

**Surge protection**

Surge (overvoltage) protection must be installed as close to the motor or the motor starter as possible. It protects the motors against surges introduced by electronics, switch contactors, and similar factors.

The overvoltage protector does not protect against direct lightning strikes into the installation, but it protects against transients caused by lightning strikes in the vicinity of the installation.

**Overtemperature monitoring**

The thermal relay does not protect against insufficient cooling caused by, for instance, low flow past the motor, high pumped liquid temperature, top feeding, deposits on the motor, or similar factors. Direct measurement and monitoring of the motor temperature are required to avoid operating under insufficient cooling conditions.

The MS4000 motors are available with Tempcon as an option to monitor temperature via the MP 204. Additionally, all MS4000 can be fit with a Pt1000 sensor. See the section on temperature monitoring.

**Dry-running**

If the pump stops pumping water, the motor must be cut out immediately. Dry-running protection can be implemented by combining level switches, pressure/flow measurements, and CosPhi/power readings.

In addition, protection devices may be required to ensure the following:

- the supply requirements and voltage limits are not exceeded during operation.
- the phase sequence for three-phase motors is kept.

Using an MP 204 with temperature monitoring via Tempcon or Pt1000 offers complete protection of the motor, including monitoring the insulation resistance. See the section on accessories for further information.

In frequency converter applications, a sine-wave filter may be required to ensure the motor voltage limits are not exceeded, see section Frequency converter operation.

**Related information**

*Temperature monitoring*

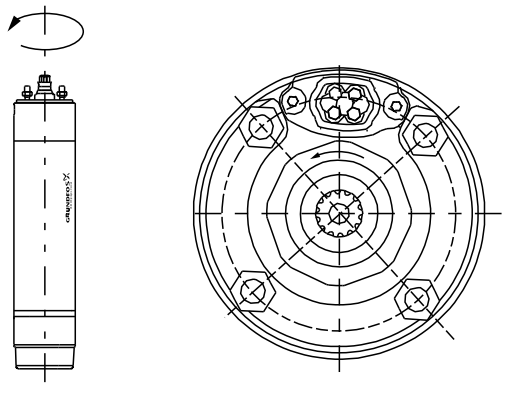
*Frequency converter operation*

## Direction of rotation

Single-phase MS4000 motors rotate counterclockwise. Three-phase MS4000 motors are bidirectional, and rotate clockwise or counterclockwise, depending on the phase rotation of the electrical supply.

If the motor is fitted to a Grundfos SP or SPA submersible pump, the correct direction of rotation is counterclockwise as seen from the shaft end.

If the motor is started without being connected to a pump, the rotating shaft end must be shielded.



TM087656

*Counterclockwise direction of rotation*

## Cables

In submersible pump applications, motor cables and drop cables can be used in combination.

*Motor cables*, if not otherwise indicated, are dimensioned for submerged operation. When a submersible motor cable is extended with a larger drop cable, the splice must always be as close to the pump outlet as possible, a maximum of 0.5 meters above the pump outlet, to ensure the motor cable is submerged.

Long motor cables that will not work completely submerged in water must be sized to work in the air.

*Drop cables* must be sized to work in the air without exceeding the allowed voltage drop. Review the cable sizing section. Grundfos offers submersible drop cables for all types of applications. The choice of submersible drop cable depends on the application and type of installation.

A cable termination kit (splice) joins the motor and the drop cable. Optionally, for an easy connection, motor cables are available with an additional plug (2-plugs motor cable) to be used in combination with a submersible drop cable with plug.

See the section on accessories for more information about motor cables, drop cables, and cable termination kits.

**Related information**

*Model C motor cables*

*Submersible drop cable*

*Submersible drop cables with plug*

*Female plug kit for drop cable*

*Cable termination kit, type KM*

*Cable termination kit, types M0 to M4*

*Cable clips*

# Frequency converter operation

Cables and other system components must be properly rated for frequency converter applications.

### Output filter

The frequency converter introduces electromagnetic noise and voltage peaks to the voltage signal. The voltage peaks are amplified in the cable between the frequency converter and the motor, and can damage the insulation system of the motor. The voltage peak levels at motor terminals depend on many factors, such as:

- voltage supply to the frequency converter
- cable length
- switching frequency
- frequency converter type.

At motor terminals, the voltage peaks (Upeak) and voltage rise time (dU/dt) must not exceed the values shown in the table *Submersible motor limits*. To ensure this, an output filter between the frequency converter and the motor may be required. If the frequency converter has a voltage boost function, the use of an output filter is always needed.

Grundfos always recommends the use of sine-wave filters, which also reduce electromagnetic noise.

The technology of Grundfos CUE or RSI frequency converter allows their use without output filters for low voltage and short cable lengths. It also allows their use for cable length up to 500 m with a sine-wave filter. For more information, see the table *CUE/RSI Supply limits*.

Local and national requirements for safety, electromagnetic interference (EMI), etc., must always be followed and can demand filtering due to, for example, electromagnetic compatibility or noise suppression.

### Submersible motor limits

| | Voltage limits at motor terminals [2] | |
|---|---|---|
| **Motor type** | Max dU/dt | Max Upeak (Line-Line) |
| | [V/us] | [V] |
| MS4000 | 6000 | 850 |

2) To be ensured regardless of the frequency converter brand/model

### CUE/RSI supply and cable length limits

| | Operating with CUE/RSI | | | | | |
|---|---|---|---|---|---|---|
| | Without filter | | | With Sine-wave filter | | |
| Motor [3] type | Max. supply to CUE/RSI | | Max. unscreened cable length | Max. supply to CUE/RSI | | Max. unscreened cable length |
| | [Vac [4]] | [Vdc [5]] | [m] | [Vac [4]] | [Vdc [5]] | [m] |
| MS4000 | 240 | 400 | 100 | 575/500 [6] | 800 | 500 |

3) Motors operating within their temperature rating

4) Single-phase or three-phase AC supply (rms Voltage)

5) Renewable energy DC supply

6) 575 V for CUE and 500 V for RSI

### Temperature monitoring

To enable monitoring of the motor temperature in applications with frequency converters, all MS4000 can be optionally fitted with a Pt1000 temperature sensor.

### Frequency range

The rated frequency must not be exceeded.

The minimum operation frequency for the MS4000 asynchronous motor is 30 Hz.

In submersible pump installations, a reduction of the rated frequency often results in an increased generation of heat in the motor. Although the motor load is reduced with its frequency, the load reduction is very small as the major part of the power input is often consumed to overcome the static head. Moreover, the flow past the motor, that is, the motor cooling, is reduced. Therefore, it is important never to adjust the frequency to a very low level as there must still be a sufficient flow of pumped liquid past the motor.

The motor must still cut out immediately if the pump stops pumping water.

The permissible frequency ranges are:

- 30-50 Hz
- 30-60 Hz.

### Ramp times

The start and stop ramp times to and from the minimum frequency must be fast enough to allow the formation and maintenance of the lubricant layer in the thrust bearing:

- start ramp from 0 to 30 Hz: maximum 3 seconds
- start ramp from 30 to 0 Hz: maximum 3 seconds.

For further details, contact your frequency converter supplier or Grundfos.

### Overload protection

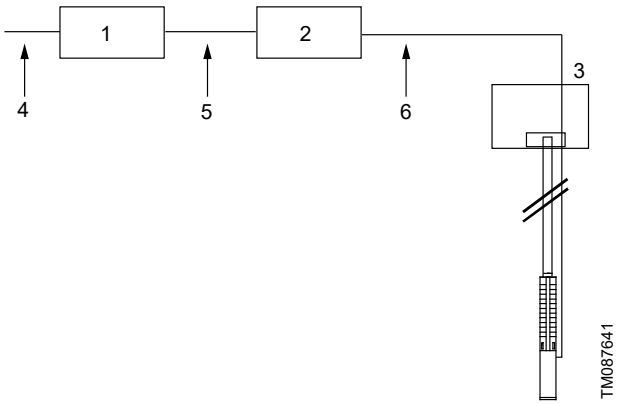
The motor must be protected against overload through setting of the output current limit of the frequency converter to the same value as the rated maximum current or the maximum actual current of the submersible motor. See section Protection in section Installation.

### Cables configuration

When a frequency converter is installed in connection with SP pumps, there are two types of installation:

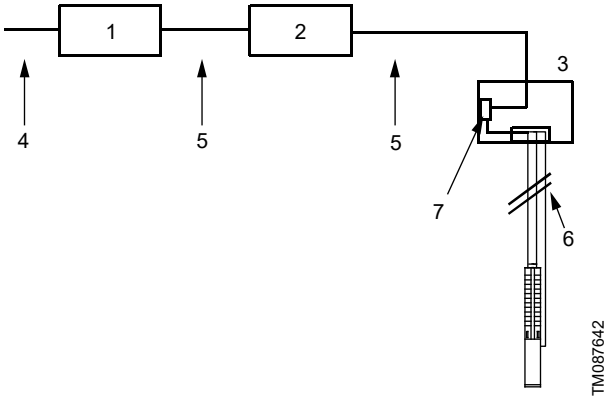
- installation in EMC-insensitive sites.
- installation in EMC-sensitive sites.

The two installation types differ in the use of screened cable. Screened cables are required in those parts of the installation where the surroundings must be protected against EMC. Drop cables inside the borehole are always unscreened. The screened cable between the frequency converter and the filter must be short.




*Example of installation in EMC-insensitive sites*

| Pos. | Description |
|---|---|
| 1 | Frequency converter |
| 2 | Sine-wave filter |
| 3 | Borehole/well head |
| 4 | Mains cable, unscreened |
| 5 | Screened cable |
| 6 | Drop cable, unscreened |




*Example of installation in EMC-sensitive sites*

| Pos. | Description |
|---|---|
| 1 | Frequency converter |
| 2 | Sine-wave filter |
| 3 | Borehole/well head |
| 4 | Mains cable, unscreened |
| 5 | Screened cable |
| 6 | Drop cable, unscreened |
| 7 | Connection box |

**Related information**

*Temperature monitoring*
*Protection*
*Accessories*

## Soft starter operation

Grundfos only recommends soft starters that control the voltage on all three phases and are provided with a bypass contactor.

Soft starters with bypass contactor will only operate during ramp-up and ramp-down. This reduces both the load on the soft starter and the energy consumption in comparison with operation without bypass contactor.

The soft starter must not be used in connection with operation via a generator.

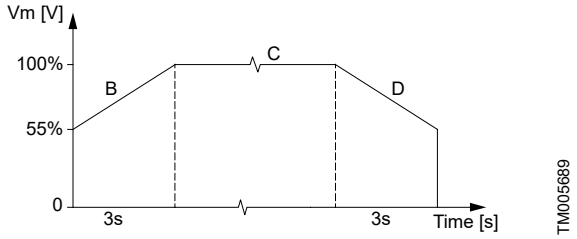
The starting voltage is minimum 55 % of the value stated on the motor nameplate.

If a high locked-rotor torque is required or if the electricity supply is not optimal, the starting voltage should be higher.

Ramp times:

- ramp-up time (until the voltage stated on the nameplate is reached): 3 seconds.
- ramp-down time: 3 seconds.

If the ramp-up and ramp-down times are followed, you will avoid unnecessary heating of the motor.




*Soft starter*

| Pos. | Description |
|---|---|
| Vm [V] | Motor rated voltage in volts |
| B | Start ramp |
| C | Operation |
| D | Stop ramp |
| Time [s] | Time in seconds |

For further details, contact your soft starter supplier or Grundfos.

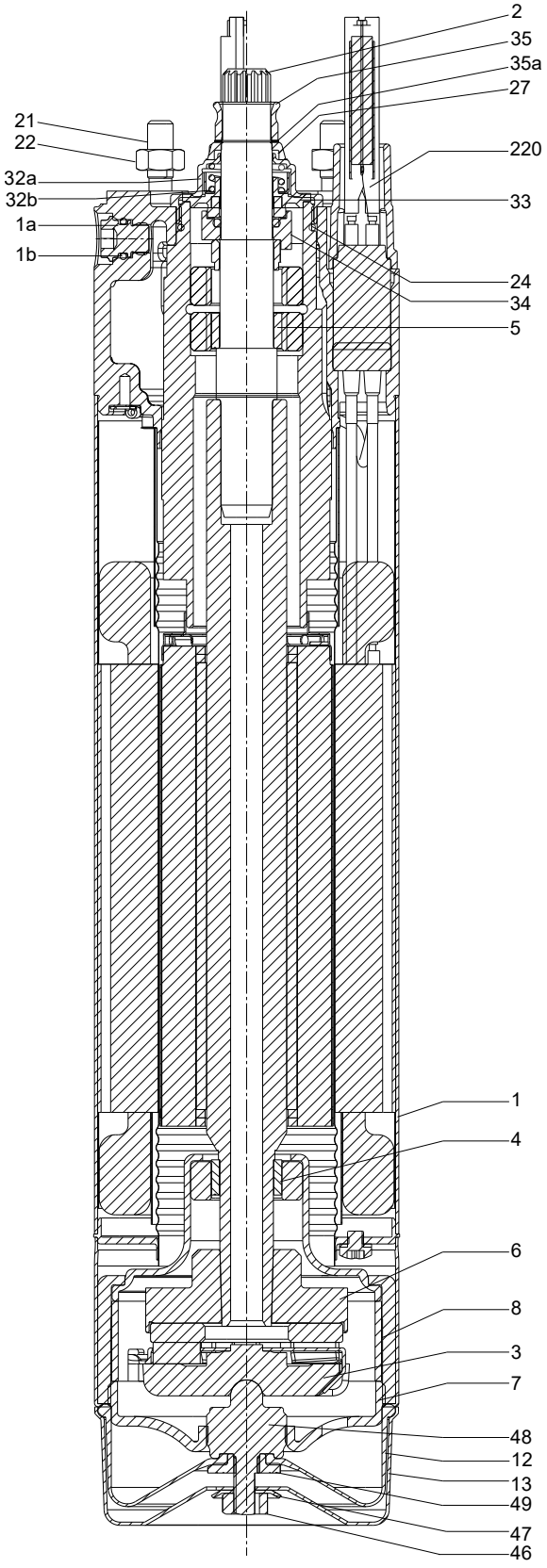
# 4. Construction

## Material specification

| Pos. | Description | | Type key MS4000_ | | | | |
|---|---|---|---|---|---|---|---|
| | | | AZBCW_ _ _ __ | AZBSW_ _ _ __ | AZESW_ _ _ __ | ARBSW_ _ _ __ | ARESW_ _ _ __ |
| 35 | Spline protector | | NBR | NBR | FKM | NBR | FKM |
| 35a | Supporting ring | | EN 1.4401 | EN 1.4401 | EN 1.4401 | EN 1.4539 | EN 1.4539 |
| 27 | Sand shield | | NBR | NBR | FKM | NBR | FKM |
| 32 | Shaft seal housing | | EN 1.4308 | EN 1.4308 | EN 1.4308 | EN 1.4517 | EN 1.4517 |
| 32b | Spring | | EN 1.4462/1.4563 | EN 1.4462/1.4563 | EN 1.4462/1.4563 | EN 1.4462/1.4563 | EN 1.4462/1.4563 |
| 32a | Supporting ring | | EN 1.4401 | EN 1.4401 | EN 1.4401 | EN 1.4539 | EN 1.4539 |
| 33 | Shaft seal, stationary | | Ceramic | Silicon Carbide | Silicon Carbide | Silicon Carbide | Silicon Carbide |
| | | | NBR | NBR | FKM | NBR | FKM |
| 34 | Shaft seal, rotating | | Tungsten Carbide | Silicon Carbide | Silicon Carbide | Silicon Carbide | Silicon Carbide |
| | | | EN 1.4308 | EN 1.4308 | EN 1.4308 | EN 1.4517 | EN 1.4517 |
| 22 | Nut | | EN 1.4401 | EN 1.4401 | EN 1.4401 | EN 1.4539 | EN 1.4539 |
| 21 | Staybolt | | EN 1.4401 | EN 1.4401 | EN 1.4401 | EN 1.4539 | EN 1.4539 |
| 220a | Screw [7)] | | EN 1.4401 | EN 1.4401 | EN 1.4401 | EN 1.4462 | EN 1.4462 |
| 220b | Retaining ring [7)] | | Polyester | Polyester | Polyester | Polyester | Polyester |
| 220 | Motor cable [7)] | | | | | | |
| 1a | Filling plug | | EN 1.4401 | EN 1.4401 | EN 1.4401 | EN 1.4462 | EN 1.4462 |
| 1b | O-ring | | EPDM | EPDM | EPDM | EPDM | EPDM |
| 1 | Stator housing | | EN 1.4301/1.4308 | EN 1.4301/1.4308 | EN 1.4301/1.4308 | EN 1.4539 | EN 1.4539 |
| 24 | O-ring | | NBR | NBR | FKM | NBR | FKM |
| 5 | Radial bearing DE | Holder | EN-GJL-150 | EN-GJL-150 | EN-GJL-150 | EN-GJL-150 | EN-GJL-150 |
| | | Stationary bush | Ceramic | Ceramic | Ceramic | Ceramic | Ceramic |
| 2 | Shaft with rotor | Shaft extension | EN 1.4057 | EN 1.4057 | EN 1.4057 | EN 1.4462 | EN 1.4462 |
| | | Bearing bush | Tungsten Carbide | Tungsten Carbide | Tungsten Carbide | Tungsten Carbide | Tungsten Carbide |
| 4 | Radial bearing NDE | Holder | EN 1.0338 | EN 1.0338 | EN 1.0338 | EN 1.0338 | EN 1.0338 |
| | | Stationary bush | Ceramic | Ceramic | Ceramic | Ceramic | Ceramic |
| 6 | Thrust bearing, rotating | Holder | EN-GJS-400-15 | EN-GJS-400-15 | EN-GJS-400-15 | EN-GJS-400-15 | EN-GJS-400-15 |
| | | Bearing disc | Ceramic | Ceramic | Ceramic | Ceramic | Ceramic |
| 8 | Clamping ring | | EN 1.0330.3 | EN 1.0330.3 | EN 1.0330.3 | EN 1.0330.3 | EN 1.0330.3 |
| 3 | Thrust bearing, stationary | Holder | EN-GJS-450-10 | EN-GJS-450-10 | EN-GJS-450-10 | EN-GJS-450-10 | EN-GJS-450-10 |
| | | Tilting pads | Carbon graphite | Carbon graphite | Carbon graphite | Carbon graphite | Carbon graphite |
| 7 | Bearing retainer | | EN 1.0330.3 | EN 1.0330.3 | EN 1.0330.3 | EN 1.0330.3 | EN 1.0330.3 |
| 48 | Adjusting screw | | EN 1.4460 | EN 1.4460 | EN 1.4460 | EN 1.4462 | EN 1.4462 |
| 12 | Diaphragm | | NBR | NBR | FKM | NBR | FKM |
| 49 | Nut for rubber diaphragm | | EN 1.4305 | EN 1.4305 | EN 1.4305 | EN 1.4539 | EN 1.4539 |
| 13 | Bottom cover | | EN 1.4301 | EN 1.4301 | EN 1.4301 | EN 1.4539 | EN 1.4539 |
| 47 | Lock washer | | EN 1.4401 | EN 1.4401 | EN 1.4401 | EN 1.4539 | EN 1.4539 |
| 46 | Nut | | EN 1.4401 | EN 1.4401 | EN 1.4401 | EN 1.4539 | EN 1.4539 |

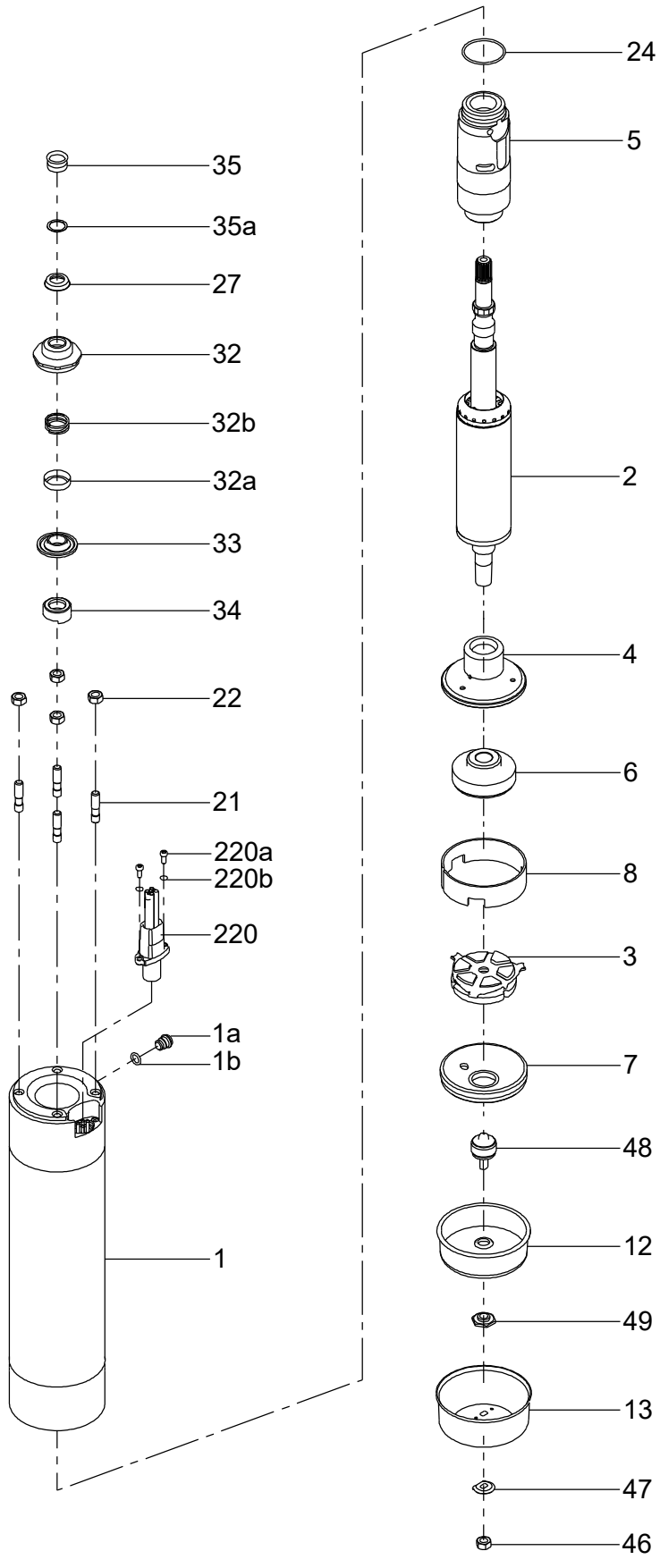
7) Motor cable components

## Sectional drawing of MS4000




*Sectional drawing of the MS4000 Model C motor*

## Exploded drawing of MS4000




*Exploded drawing of the MS4000 Model C motor*

# Description of construction

The motor is a 2-pole, asynchronous squirrel-cage, dry stator (canned) submersible motor.

- Enclosure class: IP68 according to IEC 60034-5.
- Insulation class: 155 °C (F) according to IEC 60034-1.

The entire surface is stainless steel which means that all external components have uniform corrosion resistance.

## Cable connection

The motor is connected to the drop cable via a special motor cable.

The motor cable cannot be fitted or removed when the motor and pump are assembled.

- CE-approved motors are equipped with a flat blue EPR jacketed motor cable with four cores of 1.5 $mm^2$. Motor cables with an additional plug connector to assembly the drop cable are available as option.
- cCSAus-approved motors are equipped with four single leads XLPE AWG14 motor cable.
- Round black PTFE motor cable with four cores of 2.5 mm2 is available.
- Motors for BM (booster) pumps are equipped with three single leads XLPE AWG14 motor cable.

The motor cable must work fully submerged in the pumped liquid unless specific.

## Pump connection

The motor has standardized drive-end (DE) connection, according to the NEMA standard MG1-2021, with 4 studs and a spline shaft end for mounting of the pump.

- CE-approved motors are provided with M8 thread connection studs and nuts.
- cCSAus-approved motors are provided with 5/16"-24 UNF thread connection studs and nuts.

The stainless steel splined shaft end fulfils the ANSI B92.1, 1970, class 5.

## Shaft seal

The motor is fitted with a mechanical shaft seal. A spring-loaded elastomer diaphragm is vulcanized on the stationary upper seat. The elastomer diaphragm functions as a safety valve for the motor opening at 0.5 to 1.0 bar excess pressure. The lapped stationary face seals against the rotating lower seal face.

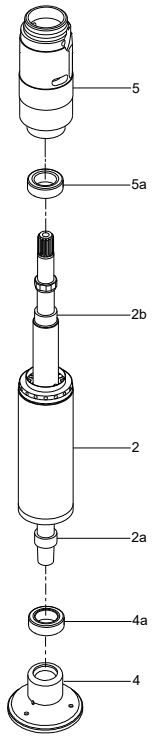
The shaft seal is available in three variants for different applications:

- hard metal (Tungsten carbide) against ceramic (Aluminium Oxide) faces with NBR elastomer, approved for drinking water
- hard metal (Tungsten carbide) against ceramic (Aluminium Oxide) faces with FKM elastomer, for operation in liquids without abrasive particles but with hydrocarbon content and dissolved gases
- silicon carbide against silicon carbide faces with NBR elastomer for operation in liquids with abrasive particles
- silicon carbide against silicon carbide faces with FKM elastomer for operation in liquids with abrasive particles, hydrocarbon content and dissolved gases.

## Radial bearing

The motor have two hydrodynamic radial bearings lubricated by the motor liquid, they are placed at the DE (top) and NDE (bottom) of the motor. The rotating bushings, fixed on the shaft, are made of tungsten carbide. The replaceable stationary bushings are made of ceramic (Aluminium Oxide) and are mounted on a bearing holder. Depending on the motor size, each radial bearing can have one or two bushings, see the table below.

| Motor type | | Motor power [kW (hp)] | | Radial bearing bushings |
|---|---|---|---|---|
| Phase | Temp. version | Min. | Max. | |
| Single-phase | T4 | 0.37 (0.50) | 0.55 (0.75) | 1 × D22 |
| | | 0.75 (1.0) | 3.70 (5.0) | 2 × D22 |
| Three-phase | T4 | 0.75 (1.0) | 2.20 (3.0) | 1 × D22 |
| | | 2.70 (3.6) | 5.50 (7.5) | 2 × D22 |
| | | 7.50 (10) | | 2 × D25 |
| | T6 | 0.55 (0.75) | 1.50 (2.0) | 1 × D22 |
| | | 1.90 (2.5) | 4.00 (5.5) | 2 × D22 |
| | | 5.50 (7.5) | | 2 × D25 |



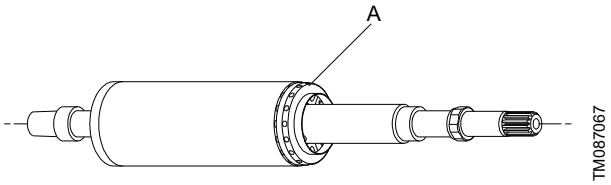

TM082540

*DE and NDE radial bearings*

| Pos. | Description |
|---|---|
| 5 | Radial bearing holder - DE |
| 5a | Stationary bush |
| 2b | Rotating bush |
| 2 | Shaft with rotor |
| 2a | Rotating bush |
| 4a | Stationary bush |
| 4 | Radial bearing holder - NDE |

## Rotor

The squirrel cage rotor is mounted on the shaft. It has brass bars on single-phase motors and copper bars on three-phase motors. To provide optimum cooling, the rotor is equipped with an impeller and has an internal channel that ensures internal circulation of the motor liquid and cooling of the motor, see section Motor cooling.




*Rotor with impeller*

| Pos. | Description |
|---|---|
| A | Rotor impeller |

**Related information**

*Motor cooling*

## Stator

The stator is hermetically encapsulated in stainless steel. Its windings are embedded in epoxy compound. This results in high mechanical stability, optimum cooling, and eliminates the risk of short circuits in the windings caused by condensing water.

## Thrust bearing

The hydrodynamic thrust bearing is of Michell/Kingsbury type, a very simple but highly efficient bearing. It fulfils the requirements specified in the NEMA MG1-2021 standard.

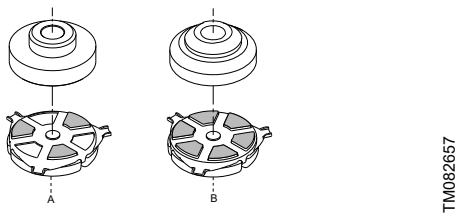
The thrust bearing consists of the following:

- a ceramic (Aluminium Oxide) rotating part with precision-ground and polished sliding surface for optimum surface finish
- a stationary part with tilting carbon pads (shoes) that have a special ground surface, ensuring optimum thrust capacity and lubrication.

The stationary part is fitted with a bearing retainer with a special adjusting screw for shaft height adjustment.

As the thrust bearing is made for bidirectional rotation, the motor can operate both clockwise and counterclockwise.

The thrust bearings are available in two sizes, rated for 2.2 and 4.4 kN thrust capacity. See the next figure.




*Thrust bearings*

| Pos. | Description |
|---|---|
| A | 2.2 kN |
| B | 4.4 kN |

The table below shows the default thrust bearing configuration. The motors can always be upgraded with a larger thrust bearing, but must not be downgraded with a smaller thrust bearing than those stated in the table.

| Motor type by phase | Motor power [kW (hp)] | | Size of thrust bearing |
|---|---|---|---|
| | Min. | Max. | [kN] |
| Single phase | 0.37 (0.5) | 3.7 (5) | 4.4 |
| Three phase | 0.55 (0.75) | 1.5 (2) | 2.2 |
| | 1.9 (2.5) | 7.5 (10) | 4.4 |

## Diaphragm

The rubber diaphragm, fitted between the stator and the bottom cover, is dimensioned to equalize volume variations caused by the temperature rise in connection with intermittent operation.

## Motor liquid

The motor is factory-filled with the Grundfos SML-3 motor liquid. The SML-3 is based on mono-propylene glycol, contains anti-corrosive and lubricating additives, and is frost-proof down to -20 °C.

Motor liquid with a high glycol content (60 % HTF) is available for motors that are stored or operated at lower temperatures (up to -55 °C).

Mixtures with more than 60 % of glycol content are not recommended.

If the motor is entirely drained of SML-3 liquid, for instance, in connection with service, it must always be refilled with SML-3. When topping up with small amounts of liquid, demineralized or clean water may be used.

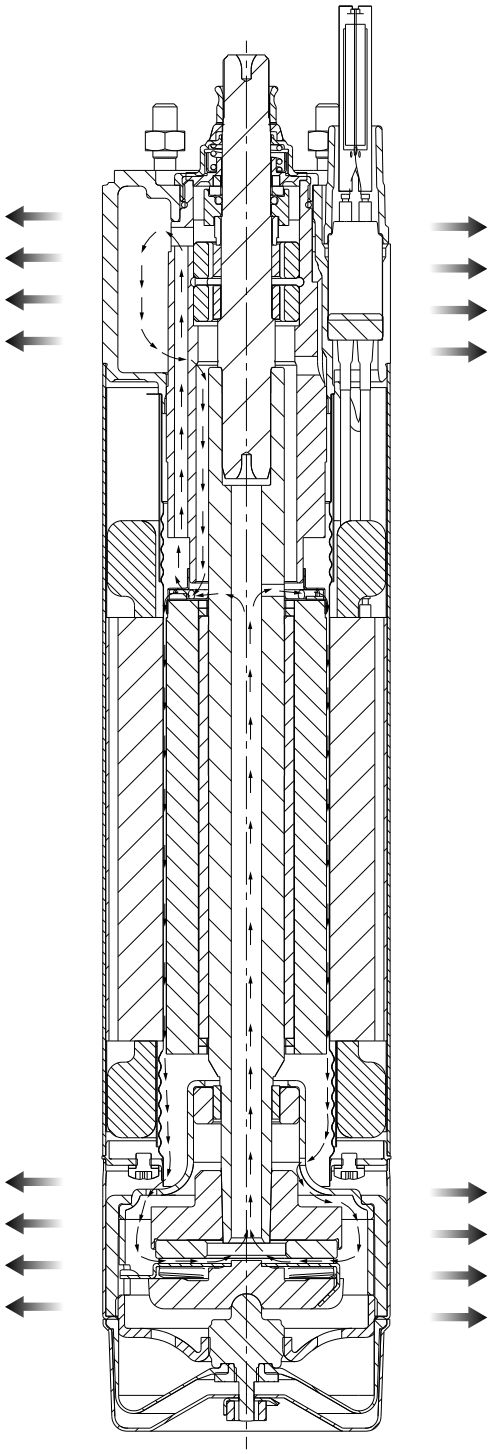
If motor liquid based on mono-propylene glycol is not allowed for special applications, the motor may be factory-filled with demineralized water upon request. A motor filled with demineralized water must be frost-protected when stored or transported in freezing temperatures.

| | Glycol content in the motor liquid | Freezing point |
|---|---|---|
| SML-3 | 36.5 % volume | -20 °C |
| High glycol (60% HTF) | 57.6 % volume | -55 °C |

## Motor cooling

The motor has cooling chambers at the top and bottom. An efficient internal circulation system transports the heat from the rotor, stator and bearings via the motor liquid to the cooling chambers. In the cooling chambers, the heat generated in the motor is carried away to the surrounding pumped liquid via the outer surface of the motor.

The temperature and flow velocity of the pumped liquid past the motor are of vital importance to the cooling process and to the life of the motor. See cooling requirements in the section Cooling.



TM082535

*Internal circulation system for cooling*

**Related information**

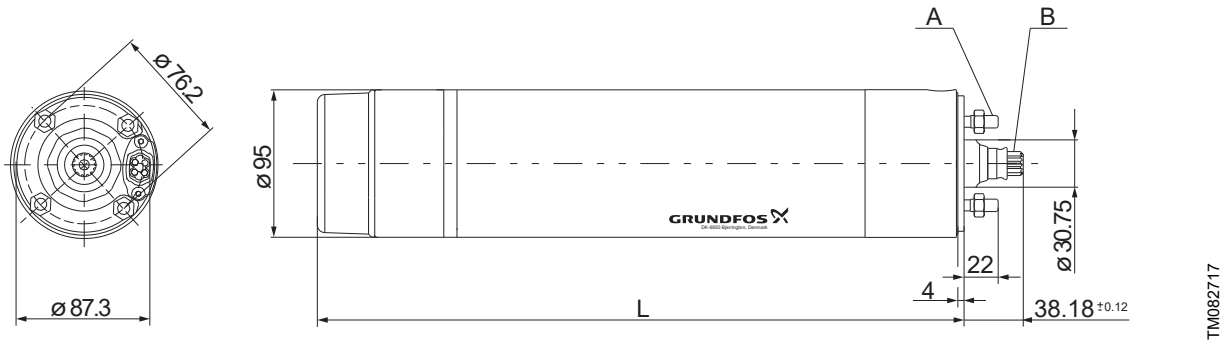
*Cooling*

*Rotor*

# 5. Technical data

## Dimensions and weights

*Dimensional drawing of MS4000*




| | | CE | cCSAus |
|---|---|---|---|
| | A = | M8 | 5/16"-24UNF |
| | B = | Spline 14x24/48-30 (ANSI B92.1 class 5) | |

| Motor power, $P_2$ | | Single-phase CSCR | | | Single-phase PSC | | | Three-phase | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T40 | | | T40 | | | T40 | | | T60 | | |
| | | L | Net weight | Shipping volume | L | Net weight | Shipping volume | L | Net weight | Shipping volume | L | Net weight | Shipping volume |
| [kW] | [hp] | [mm] | [kg] | [$m^3$] | [mm] | [kg] | [$m^3$] | [mm] | [kg] | [$m^3$] | [mm] | [kg] | [$m^3$] |
| 0.37 | 0.50 | - | - | - | 402 | 13 | 0.019 | - | - | - | - | - | - |
| 0.55 | 0.75 | - | - | - | 457 | 16 | 0.025 | - | - | - | 402 | 13 | 0.019 |
| 0.75 | 1.00 | - | - | - | 497 | 17 | 0.025 | 402 | 13 | 0.019 | 417 | 14 | 0.025 |
| 1.10 | 1.50 | - | - | - | 577 | 21 | 0.025 | 417 | 14 | 0.025 | 417 | 14 | 0.025 |
| 1.50 | 2.00 | 497 | 17 | 0.025 | 677 | 26 | 0.032 | 417 | 14 | 0.025 | 457 | 16 | 0.025 |
| 1.90 | 2.50 | - | - | - | - | - | - | 457 | 16 | 0.025 | 497 | 17 | 0.025 |
| 2.20 | 3.00 | 577 | 21 | 0.025 | 677 | 26 | 0.032 | 457 | 16 | 0.025 | 497 | 17 | 0.025 |
| 2.70 | 3.60 | - | - | - | - | - | - | 497 | 17 | 0.025 | 577 | 21 | 0.025 |
| 3.00 | 4.00 | - | - | - | - | - | - | 497 | 17 | 0.025 | 577 | 21 | 0.025 |
| 3.70 | 5.00 | 677 | 26 | 0.032 | - | - | - | 577 | 21 | 0.025 | 677 | 26 | 0.032 |
| 4.00 | 5.50 | - | - | - | - | - | - | 577 | 21 | 0.025 | 677 | 26 | 0.032 |
| 5.50 | 7.50 | - | - | - | - | - | - | 677 | 26 | 0.032 | 777 | 31 | 0.032 |
| 7.50 | 10.00 | - | - | - | - | - | - | 777 | 31 | 0.032 | - | - | - |

## Motor cables

The motors are available with four types of motor cables.
Motor cables for the model C are not interchangeable with previous models.

- Model C motor cables are available as accessories, see the section Accessories.
- Model B motor cables are available as service kits.

| Cable type | Approval |
|---|---|
| 4 G 1.5 $mm^2$ jacketed EPR flat cable | CE |
| 4 × 1 × 14 AWG single leads XLPE | cCSAus |
| 3 × 1 × 14 AWG single leads XLPE for BM (booster) | |
| 4 G 2.5 $mm^2$ jacketed PTFE round cable | - |

In submersible pump applications, motor cables and drop cables can be used in combination.
For more information about Model C motor cables, drop cables and cable termination kits, see the section Accessories.

## Moment of inertia

| Motor power, $P_2$ | | Single-phase CSCR | Single-phase PSC | Three-phase | |
|---|---|---|---|---|---|
| [kW] | [hp] | T40 | T40 | T40 | T60 |
| | | [kg·$m^2$] | [kg·$m^2$] | [kg·$m^2$] | [kg·$m^2$] |
| 0.37 | 0.50 | - | 0.0009 | - | - |
| 0.55 | 0.75 | - | 0.0014 | - | 0.0009 |
| 0.75 | 1.00 | - | 0.0017 | 0.0009 | 0.0010 |
| 1.10 | 1.50 | - | 0.0022 | 0.0010 | 0.0010 |
| 1.50 | 2.00 | 0.0017 | 0.0029 | 0.0010 | 0.0014 |
| 1.90 | 2.50 | - | - | 0.0014 | 0.0017 |
| 2.20 | 3.00 | 0.0022 | 0.0029 | 0.0014 | 0.0017 |
| 2.70 | 3.60 | - | - | 0.0017 | 0.0022 |
| 3.00 | 4.00 | - | - | 0.0017 | 0.0022 |
| 3.70 | 5.00 | 0.0029 | - | 0.0022 | 0.0029 |
| 4.00 | 5.50 | - | - | 0.0022 | 0.0029 |
| 5.50 | 7.50 | - | - | 0.0029 | 0.0036 |
| 7.50 | 10.00 | - | - | 0.0036 | - |

**Related information**

*Model C motor cables*

# 6. Electrical data

## Overview of electrical data

| Voltage code | Voltage description | | Rated voltages | | Approval | Region | Temperature version | | | | | | | | | | | | | Page |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | kW | 0.37 | 0.55 | 0.75 | 1.1 | 1.5 | 1.9 | 2.2 | 2.7 | 3.0 | 3.7 | 4.0 | 5.5 | 7.5 | |
| | | | 50 Hz | 60 Hz | | | hp | 0.50 | 0.75 | 1.00 | 1.5 | 2.0 | 2.5 | 3.0 | 3.6 | 4.0 | 5.0 | 5.5 | 7.5 | 10.0 | |
| 75 | 1×220/50 PSC | = | 220-240 | - | CE | Global | | T4 | T4 | T4 | T4 | - | - | - | - | - | - | - | - | - | **24** |
| 21 | 1×220-230/50 CSCR | = | 220-230 | - | CE | Global | | - | - | - | - | - | - | T4 | - | - | - | - | - | - | **24** |
| 24 | 1×240/50 CSCR | = | 240 | - | CE | Global | | - | - | - | - | - | - | T4 | - | - | - | - | - | - | **24** |
| 45 | 1×230/60 CSCR | = | - | 230 | cCSAus | US | | - | - | - | - | T4 | - | T4 | - | - | T4 | - | - | - | **24** |
| 10 | 3×220-230/50 | = | 220-230 | - | CE | Global | | - | - | T4<br>T6 | T4<br>T6 | T4<br>T6 | - | T4<br>T6 | - | T4<br>T6 | - | T4<br>T6 | T4<br>T6 | T4 | **25** |
| 08 | 3×380/50 440/60 | = | 340-360-380 | 440 | CE | Global | | - | - | T4 | T4 | T4 | - | T4 | - | T4 | - | T4 | T4 | T4 | **26** |
| 19 | 3×380-415/50 | = | 380-400-415 | 440-460-480 | CE | Global | | - | T6 | T4 | T4 | T4 | - | T4<br>T6 | - | T4<br>T6 | - | T4<br>T6 | T4<br>T6 | T4 | **27** |
| 18 | 3×500-525/50 | = | 500-525 | - | CE | Global | | - | - | T4 | T4 | T4 | - | T4<br>T6 | - | T4<br>T6 | - | T4<br>T6 | T4<br>T6 | T4 | **30** |
| 28 | 3×208-230/60 | = | - | 208-220-230 | CE | Global | | - | - | T4 | T4 | T4 | - | T4 | - | T4 | - | T4 | T4 | T4 | **31** |
| 33 | 3×380-415/60 | = | - | 380-400-415 | CE | Global | | - | - | T4 | T4 | T4 | - | T4 | - | T4 | - | T4 | T4<br>T6 | T4 | **32** |
| 30 | 3×208-230/60 | = | - | 208-220-230 | cCSAus | US | | - | - | - | - | - | - | T4 | - | T4 | T4 | - | T4 | T4 | **33** |
| 35 | 3×440-460/60 | = | - | 440-460 | cCSAus | US | | - | - | T4 | T4 | T4 | - | T4<br>T6 | - | T4<br>T6 | T4<br>T6 | - | T4<br>T6 | T4 | **34** |
| 39 | 3×575/60 | = | - | 575 | cCSAus | US | | - | - | T4 | T4 | T4 | - | T4<br>T6 | - | T4<br>T6 | T4<br>T6 | - | T4<br>T6 | T4 | **35** |
| 09 | 3×200/50 220/60 | = | 200 | 200-220 | CE | Japan | | - | - | T4 | T4 | T4<br>T6 | T4<br>T6 | T4<br>T6 | T4<br>T6 | T4<br>T6 | T4<br>T6 | T4<br>T6 | T4<br>T6 | - | **35** |
| 65 | 3×400/50 440/60 | = | 400 | 400-440 | CE | Japan | | - | - | T4 | T4 | T4<br>T6 | T4<br>T6 | T4<br>T6 | T4<br>T6 | T4<br>T6 | T4<br>T6 | T4<br>T6 | T4<br>T6 | T4 | **37** |

## Electrical data tables

The abbreviations used in the following tables are explained below.

| Category | Abbr. | Explanation |
|---|---|---|
| Power | FL | Power at full load (rated load) |
| | SF | Power at service factor load |
| Current | FL | Current at full load (rated current at 50 Hz) |
| | SF | Current at service factor load (rated current at 60 Hz) |
| | FL Main | Current at full load in the main winding |
| | FL Aux | Current at full load in the auxiliary winding |
| | LR | Locked rotor current |
| Motor efficiency η | SF | Motor efficiency at service factor load |
| | FL | Motor efficiency at full load |
| | ¾FL | Motor efficiency at 75 % of the full load |
| | ½FL | Motor efficiency at 50 % of the full load |
| Cos φ | SF | Power factor at service factor load |
| | FL | Power factor at full load |
| | ¾FL | Power factor at 75 % of the full load |
| | ½FL | Power factor at 50 % of the full load |
| Torque | FL | Torque at full load |
| | LR | Locked rotor torque provided in % of FL torque |
| | BD | Break down torque provided in % of FL torque |
| Speed | SF | Rotation speed at service factor load |
| | FL | Rotation speed at full load |

## Voltage code 75

### 1 × 220 V PSC, 50 Hz, T40

| Power | Current | | | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | FL Main | FL Aux | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.37 | 3.3 | 2.1 | 1.8 | 12 | 57.7 | 51.6 | 41.4 | 0.99 | 0.98 | 0.95 | 1.3 | 70 | 220 | 2800 |
| 0.55 | 4.5 | 3.1 | 2.2 | 17 | 63.4 | 58.4 | 48.7 | 0.98 | 0.97 | 0.94 | 1.9 | 60 | 240 | 2790 |
| 0.75 | 5.8 | 3.7 | 3.3 | 22 | 66.1 | 60.9 | 51.1 | 0.99 | 0.98 | 0.97 | 2.6 | 70 | 230 | 2790 |
| 1.10 | 8.7 | 6.1 | 4.4 | 34 | 68.2 | 63.6 | 54.2 | 0.97 | 0.95 | 0.91 | 3.8 | 60 | 250 | 2790 |

### 1 × 240 V PSC, 50 Hz, T40

| Power | Current | | | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | FL Main | FL Aux | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.37 | 3.3 | 2.6 | 2.0 | 13 | 54.4 | 47.1 | 36.5 | 0.94 | 0.90 | 0.83 | 1.3 | 80 | 280 | 2830 |
| 0.55 | 4.4 | 3.5 | 2.5 | 19 | 61.1 | 54.7 | 44.0 | 0.93 | 0.89 | 0.82 | 1.9 | 70 | 280 | 2830 |
| 0.75 | 5.9 | 4.5 | 3.7 | 24 | 63.2 | 56.7 | 46.0 | 0.95 | 0.91 | 0.84 | 2.5 | 80 | 300 | 2830 |
| 1.10 | 9.1 | 7.4 | 4.9 | 37 | 65.2 | 58.8 | 48.2 | 0.90 | 0.84 | 0.75 | 3.8 | 70 | 300 | 2830 |

## Voltage code 21

### 1 × 220 V CSCR, 50 Hz, T40

| Power | Current | | | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | FL Main | FL Aux | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 2.2 | 14.6 | 10.3 | 6.0 | 48 | 75 | 74.1 | 67.9 | 0.99 | 0.99 | 0.99 | 7.4 | 140 | 170 | 2850 |

### 1 × 230 V CSCR, 50 Hz, T40

| Power | Current | | | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | FL Main | FL Aux | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 2.2 | 14.0 | 9.6 | 6.3 | 50 | 75.5 | 73.4 | 66.1 | 0.99 | 0.99 | 0.99 | 7.3 | 150 | 190 | 2870 |

## Voltage code 24

### 1 × 240 V  CSCR, 50 Hz, T40

| Power | Current | | | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | FL Main | FL Aux | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 2.2 | 13.2 | 9.0 | 6.5 | 47 | 75.5 | 73.4 | 66.2 | 0.99 | 0.99 | 0.99 | 7.3 | 150 | 190 | 2860 |

## Voltage code 45

### 1 × 230 V CSCR, 60 Hz, T40

| Power | | | Current | | | | | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed of rotation | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | SF | SF | SF Main | SF Aux | FL | FL Main | FL Aux | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [A] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 1.50 | 2.00 | 1.88 | 13.2 | 11.0 | 2.5 | 10.6 | 9.0 | 2.6 | 40 | 74.5 | 73.6 | 70.1 | 62.4 | 0.92 | 0.90 | 0.86 | 0.78 | 4.1 | 80 | 250 | 3470 | 3500 |
| 2.20 | 3.00 | 2.53 | 17.0 | 13.3 | 4.5 | 14.4 | 11.4 | 4.6 | 55 | 75.1 | 74.9 | 72.3 | 65.4 | 0.96 | 0.96 | 0.96 | 0.94 | 6.0 | 160 | 220 | 3460 | 3480 |
| 3.70 | 5.00 | 4.26 | 27.5 | 16.0 | 12.2 | 24.1 | 13.2 | 12.4 | 86 | 73.8 | 72.4 | 67.6 | 58.3 | 0.99 | 0.99 | 0.98 | 0.97 | 10.1 | 160 | 230 | 3460 | 3480 |

## Voltage code 10

### 3 × 220 V, 50 Hz, T40

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 3.4 | 16 | 73.7 | 72.7 | 69.1 | 0.84 | 0.79 | 0.70 | 2.5 | 160 | 240 | 2870 |
| 1.10 | 5.0 | 24 | 74.9 | 74.0 | 70.3 | 0.81 | 0.74 | 0.62 | 3.7 | 160 | 250 | 2870 |
| 1.50 | 7.2 | 30 | 73.6 | 72.9 | 68.3 | 0.78 | 0.68 | 0.54 | 5.0 | 150 | 230 | 2860 |
| 2.20 | 11.2 | 49 | 74.3 | 73.1 | 67.6 | 0.74 | 0.63 | 0.49 | 7.3 | 180 | 260 | 2860 |
| 3.00 | 14.2 | 62 | 75.1 | 74.8 | 70.8 | 0.77 | 0.67 | 0.52 | 10.0 | 170 | 240 | 2860 |
| 4.00 | 17.4 | 88 | 77.9 | 79.0 | 76.6 | 0.80 | 0.71 | 0.57 | 13.4 | 190 | 250 | 2860 |
| 5.50 | 23.6 | 117 | 79.5 | 80.5 | 78.3 | 0.81 | 0.72 | 0.57 | 18.4 | 180 | 240 | 2860 |
| 7.50 | 32.8 | 160 | 76.7 | 77.4 | 74.8 | 0.84 | 0.77 | 0.63 | 25.3 | 200 | 250 | 2830 |

### 3 × 230 V, 50 Hz, T40

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 3.4 | 17 | 72.9 | 71.1 | 66.8 | 0.82 | 0.76 | 0.66 | 2.5 | 170 | 260 | 2880 |
| 1.10 | 5.0 | 25 | 75.0 | 73.2 | 69.1 | 0.78 | 0.70 | 0.57 | 3.7 | 180 | 270 | 2880 |
| 1.50 | 7.4 | 32 | 70.1 | 67.5 | 60.6 | 0.76 | 0.66 | 0.53 | 5.0 | 170 | 250 | 2860 |
| 2.20 | 11.6 | 52 | 69.1 | 65.2 | 56.9 | 0.69 | 0.58 | 0.45 | 7.3 | 200 | 290 | 2870 |
| 3.00 | 14.6 | 65 | 74.4 | 72.8 | 67.3 | 0.77 | 0.67 | 0.53 | 10.0 | 190 | 260 | 2870 |
| 4.00 | 17.6 | 99 | 76.4 | 74.8 | 69.6 | 0.77 | 0.67 | 0.53 | 13.3 | 210 | 280 | 2870 |
| 5.50 | 24.2 | 123 | 77.7 | 76.8 | 72.6 | 0.81 | 0.72 | 0.58 | 18.3 | 200 | 270 | 2880 |
| 7.50 | 33.1 | 168 | 76.8 | 76.5 | 72.5 | 0.80 | 0.71 | 0.56 | 25.2 | 220 | 280 | 2850 |

### 3 × 220 V, 50 Hz, T60

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 3.6 | 19 | 69.0 | 66.4 | 59.9 | 0.82 | 0.77 | 0.67 | 2.5 | 190 | 280 | 2880 |
| 1.10 | 5.1 | 24 | 71.0 | 69.6 | 64.1 | 0.83 | 0.76 | 0.65 | 3.7 | 160 | 250 | 2860 |
| 1.50 | 7.4 | 39 | 72.3 | 69.7 | 63.2 | 0.77 | 0.69 | 0.56 | 5.0 | 200 | 300 | 2880 |
| 2.20 | 9.5 | 45 | 74.5 | 74.6 | 71.0 | 0.85 | 0.80 | 0.70 | 7.3 | 160 | 240 | 2850 |
| 3.00 | 12.7 | 66 | 76.2 | 75.9 | 72.3 | 0.85 | 0.80 | 0.70 | 10.0 | 180 | 250 | 2860 |
| 4.00 | 16.6 | 89 | 77.5 | 77.4 | 74.2 | 0.86 | 0.81 | 0.71 | 13.1 | 190 | 260 | 2870 |
| 5.50 | 22.0 | 130 | 78.0 | 78.1 | 75.0 | 0.84 | 0.78 | 0.67 | 18.4 | 210 | 280 | 2850 |

### 3 × 230 V, 50 Hz, T60

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 3.6 | 20 | 69.1 | 65.8 | 58.7 | 0.79 | 0.73 | 0.63 | 2.5 | 210 | 320 | 2890 |
| 1.10 | 5.1 | 25 | 71.0 | 68.7 | 62.4 | 0.79 | 0.71 | 0.59 | 3.7 | 180 | 270 | 2870 |
| 1.50 | 7.7 | 41 | 71.3 | 67.8 | 60.3 | 0.72 | 0.62 | 0.50 | 5.0 | 220 | 330 | 2890 |
| 2.20 | 9.3 | 48 | 75.1 | 74.3 | 69.9 | 0.83 | 0.76 | 0.65 | 7.3 | 180 | 270 | 2870 |
| 3.00 | 12.5 | 69 | 76.6 | 75.6 | 71.2 | 0.83 | 0.76 | 0.64 | 10.0 | 200 | 280 | 2880 |
| 4.00 | 15.1 | 94 | 78.0 | 77.2 | 73.2 | 0.83 | 0.77 | 0.66 | 13.3 | 210 | 280 | 2890 |
| 5.50 | 21.8 | 137 | 78.2 | 77.5 | 73.5 | 0.81 | 0.73 | 0.60 | 18.3 | 240 | 310 | 2870 |

## Voltage code 08

### 3 × 340 V, 50 Hz, T40

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 2.4 | 9 | 67.5 | 68.0 | 64.2 | 0.86 | 0.83 | 0.76 | 2.5 | 140 | 190 | 2810 |
| 1.10 | 3.4 | 13 | 69.7 | 70.1 | 66.2 | 0.86 | 0.82 | 0.74 | 3.7 | 140 | 210 | 2820 |
| 1.50 | 4.6 | 17 | 70.6 | 71.4 | 67.8 | 0.86 | 0.80 | 0.69 | 5.0 | 130 | 200 | 2810 |
| 2.20 | 6.7 | 26 | 71.9 | 72.4 | 68.6 | 0.85 | 0.78 | 0.66 | 7.5 | 140 | 200 | 2820 |
| 3.00 | 8.5 | 35 | 74.1 | 75.5 | 73.1 | 0.87 | 0.81 | 0.70 | 10.2 | 140 | 200 | 2820 |
| 4.00 | 11.3 | 44 | 74.3 | 76.7 | 75.4 | 0.88 | 0.85 | 0.76 | 13.6 | 130 | 190 | 2800 |
| 5.50 | 15.0 | 65 | 76.4 | 78.1 | 76.5 | 0.88 | 0.83 | 0.73 | 18.5 | 150 | 210 | 2830 |
| 7.50 | 20.9 | 97 | 74.9 | 77.7 | 77.1 | 0.88 | 0.84 | 0.74 | 25.8 | 160 | 210 | 2780 |

### 3 × 360 V, 50 Hz, T40

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 2.2 | 9.5 | 69.3 | 68.5 | 63.7 | 0.85 | 0.81 | 0.72 | 2.5 | 150 | 230 | 2840 |
| 1.10 | 3.1 | 14 | 71.0 | 70.1 | 65.0 | 0.84 | 0.78 | 0.68 | 3.7 | 150 | 230 | 2850 |
| 1.50 | 4.5 | 18 | 71.5 | 70.6 | 65.4 | 0.82 | 0.74 | 0.61 | 5.0 | 150 | 230 | 2840 |
| 2.20 | 6.6 | 27 | 72.2 | 70.9 | 65.4 | 0.80 | 0.71 | 0.57 | 7.4 | 160 | 230 | 2850 |
| 3.00 | 8.3 | 37 | 75.0 | 74.8 | 70.8 | 0.83 | 0.75 | 0.61 | 10.1 | 160 | 230 | 2850 |
| 4.00 | 10.6 | 47 | 76.1 | 76.9 | 74.4 | 0.86 | 0.81 | 0.70 | 13.5 | 160 | 220 | 2840 |
| 5.50 | 14.5 | 70 | 77.6 | 77.9 | 75.0 | 0.85 | 0.78 | 0.66 | 18.4 | 180 | 240 | 2860 |
| 7.50 | 19.8 | 94 | 76.6 | 77.8 | 75.6 | 0.86 | 0.79 | 0.66 | 25.4 | 190 | 240 | 2820 |

### 3 × 380 V, 50 Hz, T40

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 2.0 | 10 | 72.8 | 71.9 | 68.1 | 0.84 | 0.79 | 0.69 | 2.5 | 170 | 250 | 2870 |
| 1.10 | 3.0 | 15 | 74.9 | 74.0 | 70.3 | 0.82 | 0.74 | 0.62 | 3.7 | 170 | 260 | 2870 |
| 1.50 | 4.3 | 19 | 73.6 | 72.7 | 68.3 | 0.78 | 0.68 | 0.54 | 5.0 | 170 | 250 | 2860 |
| 2.20 | 7.0 | 29 | 74.3 | 73.1 | 67.6 | 0.74 | 0.63 | 0.49 | 7.3 | 180 | 260 | 2870 |
| 3.00 | 8.8 | 40 | 75.1 | 74.8 | 70.8 | 0.77 | 0.67 | 0.52 | 10.0 | 190 | 260 | 2860 |
| 4.00 | 10.8 | 50 | 77.9 | 79.0 | 76.6 | 0.80 | 0.71 | 0.57 | 13.4 | 180 | 250 | 2860 |
| 5.50 | 14.6 | 75 | 79.5 | 80.5 | 78.3 | 0.81 | 0.72 | 0.57 | 18.3 | 200 | 270 | 2870 |
| 7.50 | 20.4 | 100 | 77.5 | 78.7 | 76.8 | 0.84 | 0.73 | 0.56 | 25.5 | 210 | 270 | 2850 |

### 3 × 440 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.75 | 1.00 | 1.05 | 1.8 | 2.3 | 11 | 71.1 | 71.1 | 68.3 | 61.6 | 0.87 | 0.83 | 0.78 | 0.69 | 2.1 | 190 | 300 | 3420 | 3480 |
| 1.10 | 1.50 | 1.43 | 2.6 | 3.3 | 15 | 74.8 | 74.8 | 72.5 | 66.7 | 0.84 | 0.80 | 0.73 | 0.62 | 3.0 | 200 | 320 | 3440 | 3480 |
| 1.50 | 2.00 | 1.88 | 3.8 | 4.4 | 20 | 75.1 | 75.3 | 73.0 | 66.8 | 0.82 | 0.75 | 0.66 | 0.54 | 4.1 | 190 | 310 | 3440 | 3480 |
| 2.20 | 3.00 | 2.53 | 5.8 | 6.1 | 31 | 76.5 | 76.1 | 73.9 | 68.2 | 0.75 | 0.71 | 0.61 | 0.48 | 6.0 | 210 | 320 | 3460 | 3480 |
| 3.00 | 4.00 | 3.45 | 7.3 | 7.8 | 42 | 75.2 | 74.5 | 71.6 | 65.0 | 0.78 | 0.74 | 0.65 | 0.52 | 8.2 | 220 | 320 | 3460 | 3480 |
| 4.00 | 5.50 | 4.60 | 9.1 | 9.9 | 53 | 79.7 | 80.1 | 79.5 | 76.0 | 0.82 | 0.78 | 0.70 | 0.57 | 11.0 | 210 | 300 | 3460 | 3480 |
| 5.50 | 7.50 | 6.33 | 12.2 | 13.2 | 79 | 81.0 | 81.4 | 81.0 | 77.8 | 0.82 | 0.79 | 0.70 | 0.57 | 15.0 | 240 | 330 | 3460 | 3480 |
| 7.50 | 10.00 | 8.63 | 16.6 | 19.2 | 106 | 79.9 | 80.2 | 79.7 | 76.3 | 0.84 | 0.78 | 0.70 | 0.57 | 20.7 | 250 | 330 | 3440 | 3470 |

## Voltage code 19

### 3 × 380 V, 50 Hz, T40

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 2.1 | 8.5 | 71.7 | 71.6 | 68.2 | 0.86 | 0.82 | 0.74 | 2.5 | 140 | 200 | 2850 |
| 1.10 | 2.8 | 13 | 74.5 | 74.6 | 71.8 | 0.85 | 0.79 | 0.69 | 3.7 | 140 | 220 | 2850 |
| 1.50 | 4.0 | 16 | 73.0 | 73.8 | 70.6 | 0.83 | 0.75 | 0.62 | 5.0 | 140 | 210 | 2840 |
| 2.20 | 6.0 | 25 | 75.5 | 75.8 | 72.2 | 0.80 | 0.70 | 0.56 | 7.4 | 150 | 220 | 2850 |
| 3.00 | 7.7 | 33 | 75.0 | 75.9 | 73.8 | 0.86 | 0.79 | 0.66 | 10.1 | 150 | 220 | 2830 |
| 4.00 | 9.8 | 43 | 77.6 | 80.3 | 78.6 | 0.85 | 0.78 | 0.65 | 13.5 | 150 | 210 | 2830 |
| 5.50 | 13.0 | 61 | 79.4 | 81.6 | 80.3 | 0.85 | 0.78 | 0.65 | 18.5 | 160 | 220 | 2840 |
| 7.50 | 18.4 | 86 | 78.0 | 80.1 | 78.5 | 0.83 | 0.74 | 0.59 | 25.5 | 180 | 230 | 2830 |

### 3 × 400 V, 50 Hz, T40

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 2.0 | 9 | 72.8 | 71.9 | 68.1 | 0.84 | 0.79 | 0.69 | 2.5 | 180 | 280 | 2870 |
| 1.10 | 2.8 | 13 | 74.9 | 74.0 | 70.3 | 0.81 | 0.74 | 0.62 | 3.7 | 160 | 240 | 2870 |
| 1.50 | 4.0 | 17 | 73.6 | 72.9 | 68.3 | 0.78 | 0.68 | 0.54 | 5.0 | 150 | 230 | 2860 |
| 2.20 | 6.0 | 26 | 74.3 | 73.1 | 67.6 | 0.74 | 0.63 | 0.49 | 7.4 | 170 | 250 | 2860 |
| 3.00 | 7.9 | 35 | 75.5 | 74.9 | 70.9 | 0.81 | 0.73 | 0.58 | 10.0 | 170 | 250 | 2850 |
| 4.00 | 9.6 | 46 | 77.9 | 79.0 | 76.6 | 0.80 | 0.71 | 0.57 | 13.4 | 170 | 240 | 2860 |
| 5.50 | 13.0 | 65 | 79.5 | 80.5 | 78.3 | 0.81 | 0.72 | 0.57 | 18.4 | 180 | 250 | 2860 |
| 7.50 | 19.3 | 92 | 77.9 | 78.4 | 75.2 | 0.77 | 0.67 | 0.52 | 25.3 | 200 | 260 | 2850 |

### 3 × 415 V, 50 Hz, T40

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 1.9 | 9.5 | 72.9 | 71.4 | 66.8 | 0.82 | 0.76 | 0.66 | 2.5 | 170 | 260 | 2880 |
| 1.10 | 2.8 | 14 | 75.0 | 73.5 | 69.2 | 0.78 | 0.70 | 0.57 | 3.7 | 170 | 260 | 2880 |
| 1.50 | 4.1 | 18 | 72.4 | 70.2 | 63.9 | 0.74 | 0.64 | 0.50 | 5.0 | 170 | 260 | 2870 |
| 2.20 | 6.3 | 28 | 73.0 | 70.8 | 64.2 | 0.69 | 0.57 | 0.44 | 7.3 | 190 | 280 | 2880 |
| 3.00 | 8.1 | 37 | 75.1 | 73.5 | 68.3 | 0.77 | 0.67 | 0.52 | 10.0 | 190 | 270 | 2870 |
| 4.00 | 9.8 | 48 | 77.7 | 77.6 | 73.8 | 0.77 | 0.66 | 0.52 | 13.3 | 190 | 260 | 2870 |
| 5.50 | 13.4 | 68 | 79.7 | 79.2 | 76.0 | 0.76 | 0.66 | 0.51 | 18.3 | 200 | 270 | 2880 |
| 7.50 | 19.6 | 96 | 77.7 | 76.9 | 72.3 | 0.72 | 0.61 | 0.46 | 25.1 | 230 | 290 | 2860 |

### 3 × 440 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.75 | 1.00 | 1.05 | 1.7 | 2.3 | 9 | 65.4 | 72.4 | 70.1 | 56.6 | 0.87 | 0.85 | 0.80 | 0.72 | 2.1 | 160 | 230 | 3460 | 3450 |
| 1.10 | 1.50 | 1.43 | 2.5 | 3.0 | 13 | 74.5 | 74.9 | 73.2 | 67.8 | 0.87 | 0.83 | 0.77 | 0.67 | 3.0 | 160 | 260 | 3420 | 3470 |
| 1.50 | 2.00 | 1.88 | 3.5 | 4.1 | 17 | 74.4 | 75.6 | 74.3 | 68.9 | 0.85 | 0.80 | 0.73 | 0.61 | 4.2 | 150 | 250 | 3420 | 3460 |
| 2.20 | 3.00 | 2.53 | 5.3 | 5.7 | 26 | 77.2 | 77.3 | 75.9 | 71.0 | 0.80 | 0.76 | 0.67 | 0.54 | 6.1 | 170 | 270 | 3440 | 3470 |
| 3.00 | 4.00 | 3.45 | 6.7 | 7.4 | 34 | 75.8 | 76.0 | 74.6 | 69.8 | 0.86 | 0.83 | 0.77 | 0.66 | 8.3 | 170 | 260 | 3430 | 3460 |
| 4.00 | 5.50 | 4.60 | 8.7 | 9.4 | 45 | 79.0 | 79.9 | 79.7 | 76.3 | 0.85 | 0.82 | 0.76 | 0.64 | 11.1 | 170 | 250 | 3440 | 3460 |
| 5.50 | 7.50 | 6.33 | 11.4 | 12.8 | 63 | 78.2 | 79.0 | 79.9 | 78.9 | 0.87 | 0.84 | 0.77 | 0.63 | 15.2 | 180 | 260 | 3440 | 3460 |
| 7.50 | 10.00 | 8.63 | 16.0 | 18.0 | 91 | 80.4 | 81.3 | 81.5 | 78.8 | 0.83 | 0.79 | 0.71 | 0.57 | 20.8 | 210 | 280 | 3430 | 3460 |

## 3 × 460 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.75 | 1.00 | 1.05 | 1.8 | 2.2 | 9.5 | 71.1 | 71.3 | 68.3 | 61.6 | 0.86 | 0.83 | 0.78 | 0.69 | 2.1 | 170 | 270 | 3420 | 3470 |
| 1.10 | 1.50 | 1.43 | 2.5 | 3.0 | 14 | 74.8 | 74.8 | 72.5 | 66.7 | 0.80 | 0.80 | 0.73 | 0.62 | 3.0 | 180 | 290 | 3440 | 3480 |
| 1.50 | 2.00 | 1.88 | 3.6 | 4.1 | 18 | 75.1 | 75.3 | 73.0 | 66.8 | 0.82 | 0.75 | 0.66 | 0.54 | 4.1 | 180 | 280 | 3440 | 3480 |
| 2.20 | 3.00 | 2.53 | 5.5 | 5.8 | 28 | 73.5 | 72.7 | 69.6 | 62.6 | 0.78 | 0.74 | 0.64 | 0.52 | 6.0 | 190 | 300 | 3460 | 3480 |
| 3.00 | 4.00 | 3.45 | 6.6 | 7.5 | 36 | 76.2 | 75.9 | 73.8 | 68.1 | 0.83 | 0.80 | 0.72 | 0.59 | 8.2 | 200 | 290 | 3450 | 3470 |
| 4.00 | 5.50 | 4.60 | 8.7 | 9.4 | 47 | 79.7 | 80.2 | 79.6 | 75.9 | 0.82 | 0.78 | 0.70 | 0.57 | 11.0 | 190 | 280 | 3460 | 3480 |
| 5.50 | 7.50 | 6.33 | 11.7 | 12.6 | 67 | 81.0 | 81.4 | 81.0 | 77.8 | 0.82 | 0.79 | 0.70 | 0.57 | 15.1 | 210 | 290 | 3460 | 3480 |
| 7.50 | 10.00 | 8.63 | 16.9 | 18.6 | 97 | 80.3 | 81.0 | 80.6 | 76.7 | 0.78 | 0.74 | 0.64 | 0.51 | 20.7 | 230 | 310 | 3470 | 3450 |

## 3 × 480 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.75 | 1.00 | 1.05 | 1.8 | 2.1 | 10 | 73.0 | 72.3 | 69.1 | 62.3 | 0.85 | 0.80 | 0.74 | 0.65 | 2.1 | 190 | 300 | 3440 | 3480 |
| 1.10 | 1.50 | 1.43 | 2.5 | 3.0 | 14 | 75.6 | 74.8 | 72.0 | 65.6 | 0.81 | 0.75 | 0.68 | 0.56 | 3.0 | 200 | 320 | 3460 | 3490 |
| 1.50 | 2.00 | 1.88 | 3.8 | 4.2 | 19 | 74.0 | 73.6 | 71.0 | 64.6 | 0.77 | 0.70 | 0.60 | 0.47 | 4.1 | 190 | 310 | 3450 | 3490 |
| 2.20 | 3.00 | 2.53 | 5.8 | 6.0 | 29 | 75.4 | 71.7 | 71.7 | 65.0 | 0.70 | 0.65 | 0.54 | 0.42 | 6.0 | 220 | 330 | 3470 | 3490 |
| 3.00 | 4.00 | 3.45 | 7.0 | 7.8 | 38 | 75.9 | 75.2 | 72.4 | 65.8 | 0.78 | 0.74 | 0.65 | 0.52 | 8.2 | 220 | 320 | 3470 | 3490 |
| 4.00 | 5.50 | 4.60 | 9.0 | 9.4 | 50 | 79.5 | 79.5 | 78.0 | 73.0 | 0.77 | 0.73 | 0.64 | 0.51 | 11.0 | 220 | 310 | 3470 | 3490 |
| 5.50 | 7.50 | 6.33 | 12.1 | 12.8 | 71 | 80.6 | 80.7 | 79.5 | 75.2 | 0.78 | 0.73 | 0.64 | 0.50 | 15.0 | 230 | 320 | 3470 | 3490 |
| 7.50 | 10.00 | 8.63 | 17.5 | 18.6 | 102 | 80.0 | 79.9 | 78.4 | 73.5 | 0.73 | 0.68 | 0.58 | 0.44 | 20.6 | 260 | 350 | 3460 | 3490 |

## 3 × 380 V, 50 Hz, T60

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.55 | 1.5 | 6.5 | 65.0 | 64.2 | 59.3 | 0.85 | 0.82 | 0.75 | 1.8 | 150 | 210 | 2830 |
| 2.20 | 5.8 | 26 | 76.9 | 77.2 | 74.0 | 0.84 | 0.77 | 0.65 | 7.4 | 150 | 230 | 2870 |
| 3.00 | 7.8 | 37 | 79.1 | 79.5 | 76.8 | 0.83 | 0.77 | 0.65 | 10.0 | 170 | 240 | 2880 |
| 4.00 | 10.2 | 45 | 78.6 | 79.7 | 77.4 | 0.86 | 0.82 | 0.72 | 13.4 | 160 | 220 | 2860 |
| 5.50 | 14.2 | 66 | 79.7 | 80.9 | 78.7 | 0.83 | 0.75 | 0.62 | 18.6 | 180 | 240 | 2860 |

## 3 × 400 V, 50 Hz, T60

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.55 | 1.4 | 7 | 66.4 | 64.7 | 59.0 | 0.84 | 0.80 | 0.73 | 1.8 | 170 | 230 | 2850 |
| 2.20 | 5.9 | 27 | 77.1 | 76.5 | 72.4 | 0.80 | 0.72 | 0.59 | 7.3 | 180 | 260 | 2880 |
| 3.00 | 7.8 | 39 | 79.5 | 79.0 | 75.1 | 0.79 | 0.71 | 0.58 | 10.0 | 200 | 280 | 2890 |
| 4.00 | 9.8 | 48 | 79.3 | 79.6 | 76.6 | 0.84 | 0.78 | 0.67 | 13.3 | 180 | 250 | 2880 |
| 5.50 | 14.4 | 71 | 79.9 | 79.8 | 76.8 | 0.78 | 0.69 | 0.55 | 18.4 | 210 | 270 | 2880 |

## 3 × 415 V, 50 Hz, T60

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.55 | 1.4 | 7 | 67.1 | 64.9 | 58.8 | 0.83 | 0.79 | 0.70 | 1.8 | 180 | 260 | 2870 |
| 2.20 | 6.0 | 29 | 76.6 | 75.3 | 70.4 | 0.76 | 0.67 | 0.54 | 7.3 | 190 | 280 | 2890 |
| 3.00 | 7.8 | 41 | 78.9 | 77.6 | 73.4 | 0.76 | 0.67 | 0.54 | 10.0 | 210 | 300 | 2900 |
| 4.00 | 9.7 | 50 | 79.6 | 79.0 | 75.3 | 0.82 | 0.74 | 0.62 | 13.3 | 200 | 270 | 2890 |
| 5.50 | 14.6 | 74 | 79.1 | 79.1 | 74.8 | 0.74 | 0.64 | 0.50 | 18.4 | 230 | 290 | 2890 |

## 3 × 440 V, 60 Hz, T60

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.55 | 0.75 | 0.83 | 1.5 | 2.1 | 6.7 | 61.2 | 61.9 | 58.6 | 51.3 | 0.86 | 0.84 | 0.81 | 0.75 | 1.5 | 170 | 240 | 3320 | 3440 |
| 2.20 | 3.00 | 2.53 | 5.0 | 5.8 | 27 | 73.6 | 73.5 | 71.6 | 66.1 | 0.86 | 0.84 | 0.79 | 0.70 | 6.1 | 180 | 270 | 3450 | 3470 |
| 3.00 | 4.00 | 3.45 | 6.7 | 7.6 | 38 | 75.7 | 75.4 | 73.4 | 67.9 | 0.86 | 0.84 | 0.79 | 0.71 | 8.2 | 200 | 290 | 3450 | 3480 |
| 4.00 | 5.50 | 4.60 | 8.4 | 9.9 | 47 | 75.9 | 76.2 | 75.1 | 70.7 | 0.88 | 0.86 | 0.83 | 0.76 | 11.0 | 180 | 260 | 3440 | 3460 |
| 5.50 | 7.50 | 6.33 | 11.6 | 13.8 | 70 | 76.8 | 77.1 | 76.2 | 71.9 | 0.87 | 0.86 | 0.81 | 0.72 | 15.2 | 210 | 290 | 3460 | 3480 |

## 3 × 460 V, 60 Hz, T60

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.55 | 0.75 | 0.83 | 1.4 | 1.9 | 7 | 63.6 | 62.5 | 58.7 | 51.1 | 0.86 | 0.83 | 0.79 | 0.73 | 1.5 | 190 | 270 | 3369 | 3460 |
| 2.20 | 3.00 | 2.53 | 5.0 | 5.7 | 28 | 74.3 | 73.8 | 71.3 | 65.2 | 0.84 | 0.82 | 0.76 | 0.66 | 6.0 | 200 | 310 | 3460 | 3480 |
| 3.00 | 4.00 | 3.45 | 7.6 | 7.6 | 41 | 76.2 | 75.6 | 78.2 | 67.1 | 0.84 | 0.82 | 0.76 | 0.66 | 8.2 | 220 | 320 | 3470 | 3490 |
| 4.00 | 5.50 | 4.60 | 8.4 | 9.6 | 49 | 76.8 | 76.7 | 75.1 | 70.1 | 0.87 | 0.85 | 0.80 | 0.71 | 11.0 | 200 | 290 | 3460 | 3480 |
| 5.50 | 7.50 | 6.33 | 12.2 | 13.8 | 74 | 81.2 | 81.3 | 80.0 | 75.6 | 0.80 | 0.76 | 0.68 | 0.55 | 15.2 | 230 | 320 | 3470 | 3490 |

## 3 × 480 V, 60 Hz, T60

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.55 | 0.75 | 0.83 | 1.4 | 1.8 | 7.3 | 65.3 | 62.9 | 58.7 | 50.7 | 0.86 | 0.82 | 0.78 | 0.71 | 1.5 | 200 | 290 | 3400 | 3480 |
| 2.20 | 3.00 | 2.53 | 5.0 | 5.7 | 30 | 74.4 | 73.6 | 70.6 | 63.9 | 0.82 | 0.79 | 0.72 | 0.60 | 6.0 | 220 | 340 | 3480 | 3500 |
| 3.00 | 4.00 | 3.45 | 6.6 | 7.6 | 43 | 76.4 | 75.5 | 72.5 | 66.0 | 0.82 | 0.79 | 0.72 | 0.61 | 8.2 | 250 | 360 | 3480 | 3500 |
| 4.00 | 5.50 | 4.60 | 8.2 | 9.5 | 52 | 77.4 | 77.0 | 74.9 | 69.4 | 0.85 | 0.83 | 0.77 | 0.68 | 10.9 | 220 | 320 | 3470 | 3490 |
| 5.50 | 7.50 | 6.33 | 12.9 | 14.2 | 78 | 80.9 | 80.6 | 78.7 | 73.6 | 0.75 | 0.71 | 0.62 | 0.49 | 15.1 | 260 | 360 | 3480 | 3500 |

## Voltage code 18

### 3 × 500 V, 50 Hz, T40

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 1.5 | 7 | 73.7 | 72.7 | 69.1 | 0.84 | 0.79 | 0.69 | 2.5 | 160 | 240 | 2870 |
| 1.10 | 2.2 | 11 | 74.9 | 74.0 | 75.0 | 0.81 | 0.74 | 0.62 | 3.7 | 160 | 240 | 2870 |
| 1.50 | 3.2 | 14 | 73.6 | 72.9 | 68.3 | 0.78 | 0.68 | 0.54 | 5.0 | 150 | 230 | 2860 |
| 2.20 | 4.5 | 21 | 72.3 | 70.5 | 64.7 | 0.74 | 0.63 | 0.49 | 7.4 | 160 | 240 | 2850 |
| 3.00 | 6.3 | 28 | 75.1 | 74.8 | 70.8 | 0.77 | 0.67 | 0.52 | 10.0 | 170 | 240 | 2860 |
| 4.00 | 7.7 | 37 | 78.0 | 79.0 | 76.6 | 0.80 | 0.71 | 0.57 | 13.4 | 170 | 240 | 2860 |
| 5.50 | 10.4 | 52 | 79.5 | 80.5 | 78.5 | 0.81 | 0.72 | 0.57 | 18.3 | 180 | 250 | 2860 |
| 7.50 | 15.0 | 73 | 77.9 | 78.4 | 75.2 | 0.77 | 0.67 | 0.52 | 25.3 | 200 | 260 | 2850 |

### 3 × 525 V, 50 Hz, T40

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 1.5 | 7.5 | 72.9 | 71.4 | 66.8 | 0.82 | 0.76 | 0.66 | 2.5 | 180 | 260 | 2880 |
| 1.10 | 2.2 | 11 | 75.0 | 73.5 | 69.1 | 0.78 | 0.70 | 0.57 | 3.7 | 170 | 270 | 2880 |
| 1.50 | 3.3 | 14 | 72.9 | 71.4 | 66.0 | 0.73 | 0.64 | 0.51 | 5.0 | 170 | 260 | 2870 |
| 2.20 | 4.8 | 22 | 70.6 | 67.5 | 60.0 | 0.72 | 0.61 | 0.48 | 7.4 | 190 | 270 | 2870 |
| 3.00 | 6.4 | 29 | 74.5 | 72.9 | 67.1 | 0.72 | 0.61 | 0.47 | 10.0 | 190 | 270 | 2870 |
| 4.00 | 7.8 | 39 | 77.7 | 77.6 | 73.8 | 0.77 | 0.66 | 0.52 | 13.3 | 200 | 270 | 2870 |
| 5.50 | 10.6 | 56 | 79.2 | 79.2 | 76.0 | 0.76 | 0.66 | 0.51 | 18.2 | 210 | 280 | 2880 |
| 7.50 | 15.4 | 78 | 77.7 | 77.1 | 72.3 | 0.72 | 0.62 | 0.46 | 25.1 | 230 | 290 | 2870 |

### 3 × 500 V, 50 Hz, T60

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 2.20 | 4.7 | 22 | 77.1 | 76.5 | 72.4 | 0.80 | 0.72 | 0.59 | 7.3 | 180 | 270 | 2880 |
| 3.00 | 6.2 | 30 | 79.5 | 79.0 | 75.1 | 0.79 | 0.71 | 0.58 | 10.0 | 180 | 260 | 2890 |
| 4.00 | 7.8 | 39 | 79.3 | 79.6 | 76.6 | 0.84 | 0.78 | 0.67 | 13.3 | 180 | 250 | 2880 |
| 5.50 | 11.6 | 54 | 79.9 | 79.8 | 76.8 | 0.78 | 0.69 | 0.55 | 18.5 | 200 | 260 | 2880 |

### 3 × 525 V, 50 Hz, T60

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 2.20 | 4.7 | 24 | 76.6 | 75.3 | 70.4 | 0.76 | 0.67 | 0.54 | 7.3 | 210 | 300 | 2890 |
| 3.00 | 6.2 | 32 | 78.9 | 77.6 | 73.3 | 0.76 | 0.67 | 0.54 | 10.0 | 200 | 290 | 2900 |
| 4.00 | 7.7 | 41 | 79.6 | 79.0 | 75.3 | 0.82 | 0.74 | 0.62 | 13.2 | 210 | 280 | 2890 |
| 5.50 | 11.6 | 58 | 79.7 | 79.1 | 74.8 | 0.74 | 0.64 | 0.50 | 18.4 | 230 | 290 | 2890 |

## Voltage code 28

### 3 × 208 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.75 | 1.00 | 1.05 | 3.9 | 5.1 | 19 | 67.1 | 68.1 | 65.7 | 59.5 | 0.87 | 0.84 | 0.80 | 0.73 | 2.1 | 160 | 230 | 3370 | 3450 |
| 1.10 | 1.50 | 1.43 | 5.8 | 6.9 | 24 | 65.7 | 67.5 | 65.9 | 60.1 | 0.87 | 0.86 | 0.83 | 0.77 | 3.1 | 140 | 220 | 3340 | 3420 |
| 1.50 | 2.00 | 1.88 | 7.9 | 9.4 | 31 | 67.9 | 69.8 | 68.8 | 63.7 | 0.88 | 0.86 | 0.81 | 0.73 | 4.2 | 130 | 210 | 3350 | 3420 |
| 2.20 | 3.00 | 2.53 | 10.8 | 11.9 | 48 | 72.2 | 72.6 | 71.1 | 65.9 | 0.86 | 0.84 | 0.79 | 0.68 | 6.1 | 150 | 230 | 3410 | 3440 |
| 3.00 | 4.00 | 3.45 | 13.9 | 15.1 | 63 | 73.3 | 74.3 | 74.1 | 70.1 | 0.88 | 0.86 | 0.82 | 0.73 | 8.4 | 150 | 220 | 3400 | 3430 |
| 4.00 | 5.50 | 4.60 | 18.1 | 19.3 | 84 | 78.9 | 79.9 | 79.9 | 77.1 | 0.88 | 0.87 | 0.83 | 0.76 | 11.1 | 150 | 220 | 3400 | 3430 |
| 5.50 | 7.50 | 6.33 | 24.5 | 27.0 | 119 | 75.8 | 76.9 | 77.0 | 73.8 | 0.88 | 0.87 | 0.83 | 0.76 | 15.3 | 160 | 230 | 3420 | 3440 |
| 7.50 | 10.00 | 8.63 | 33.0 | 35.0 | 164 | 77.6 | 79.5 | 80.5 | 78.3 | 0.88 | 0.87 | 0.82 | 0.73 | 20.9 | 180 | 250 | 3370 | 3410 |

### 3 × 220 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.75 | 1.00 | 1.05 | 3.6 | 4.7 | 20 | 72.0 | 72.4 | 70.1 | 64.0 | 0.87 | 0.85 | 0.80 | 0.72 | 2.1 | 180 | 290 | 3400 | 3470 |
| 1.10 | 1.50 | 1.43 | 5.0 | 6.3 | 25 | 74.5 | 74.9 | 73.2 | 67.8 | 0.86 | 0.82 | 0.77 | 0.67 | 3.1 | 150 | 250 | 3390 | 3470 |
| 1.50 | 2.00 | 1.88 | 7.5 | 7.5 | 32 | 74.4 | 75.6 | 74.3 | 68.9 | 0.85 | 0.80 | 0.73 | 0.60 | 4.2 | 150 | 240 | 3460 | 3420 |
| 2.20 | 3.00 | 2.53 | 10.6 | 11.4 | 52 | 72.9 | 72.5 | 70.0 | 63.7 | 0.83 | 0.80 | 0.72 | 0.60 | 6.1 | 170 | 270 | 3440 | 3460 |
| 3.00 | 4.00 | 3.45 | 13.0 | 14.0 | 68 | 78.2 | 79.0 | 78.7 | 74.8 | 0.82 | 0.79 | 0.70 | 0.57 | 8.3 | 170 | 260 | 3440 | 3460 |
| 4.00 | 5.50 | 4.60 | 16.8 | 18.1 | 90 | 79.0 | 79.9 | 79.3 | 71.1 | 0.85 | 0.82 | 0.76 | 0.64 | 11.1 | 170 | 250 | 3440 | 3460 |
| 5.50 | 7.50 | 6.33 | 23.4 | 24.5 | 128 | 78.2 | 79.0 | 79.9 | 78.9 | 0.87 | 0.84 | 0.77 | 0.64 | 15.2 | 180 | 260 | 3440 | 3470 |
| 7.50 | 10.00 | 8.63 | 31.4 | 33.0 | 177 | 79.4 | 80.4 | 80.3 | 77.2 | 0.86 | 0.84 | 0.78 | 0.66 | 20.8 | 200 | 280 | 3410 | 3450 |

### 3 × 230 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.75 | 1.00 | 1.05 | 3.5 | 4.4 | 21 | 71.1 | 71.1 | 68.3 | 61.6 | 0.88 | 0.83 | 0.78 | 0.69 | 2.1 | 200 | 310 | 3430 | 3480 |
| 1.10 | 1.50 | 1.43 | 5.4 | 6.0 | 27 | 74.8 | 74.8 | 72.5 | 66.7 | 0.84 | 0.80 | 0.73 | 0.62 | 3.0 | 180 | 290 | 3420 | 3440 |
| 1.50 | 2.00 | 1.88 | 7.1 | 8.1 | 34 | 75.1 | 75.3 | 73.0 | 66.8 | 0.82 | 0.79 | 0.75 | 0.54 | 4.1 | 170 | 270 | 3440 | 3460 |
| 2.20 | 3.00 | 2.53 | 11.1 | 11.6 | 54 | 72.5 | 71.6 | 68.4 | 61.1 | 0.78 | 0.74 | 0.65 | 0.53 | 6.0 | 190 | 300 | 3460 | 3480 |
| 3.00 | 4.00 | 3.45 | 13.7 | 15.0 | 72 | 75.2 | 75.0 | 72.9 | 67.1 | 0.83 | 0.80 | 0.72 | 0.60 | 8.3 | 190 | 290 | 3450 | 3470 |
| 4.00 | 5.50 | 4.60 | 17.0 | 17.5 | 95 | 77.1 | 77.0 | 75.3 | 70.3 | 0.85 | 0.83 | 0.76 | 0.66 | 11.0 | 200 | 290 | 3450 | 3470 |
| 5.50 | 7.50 | 6.33 | 23.8 | 25.0 | 135 | 81.0 | 81.4 | 81.0 | 77.8 | 0.82 | 0.79 | 0.70 | 0.57 | 15.1 | 210 | 290 | 3460 | 3480 |
| 7.50 | 10.00 | 8.63 | 30.5 | 32.3 | 187 | 80.0 | 80.5 | 79.7 | 75.7 | 0.84 | 0.81 | 0.73 | 0.60 | 20.7 | 230 | 310 | 3740 | 3470 |

## Voltage code 33

### 3 × 380V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.75 | 1.00 | 1.05 | 2.2 | 2.7 | 12 | 71.1 | 70.7 | 67.8 | 61.3 | 0.86 | 0.81 | 0.76 | 0.68 | 2.1 | 180 | 290 | 3420 | 3480 |
| 1.10 | 1.50 | 1.43 | 3.0 | 3.7 | 17 | 74.8 | 74.8 | 72.5 | 66.7 | 0.86 | 0.80 | 0.73 | 0.62 | 3.0 | 190 | 300 | 3480 | 3480 |
| 1.50 | 2.00 | 1.88 | 4.4 | 5.1 | 22 | 75.5 | 74.9 | 72.7 | 67.2 | 0.81 | 0.76 | 0.67 | 0.53 | 4.1 | 180 | 290 | 3440 | 3470 |
| 2.20 | 3.00 | 2.53 | 6.3 | 6.8 | 34 | 76.5 | 76.2 | 74.0 | 68.1 | 0.75 | 0.71 | 0.61 | 0.48 | 6.0 | 200 | 310 | 3460 | 3480 |
| 3.00 | 4.00 | 3.45 | 7.8 | 8.1 | 43 | 77.3 | 77.3 | 75.7 | 70.7 | 0.83 | 0.80 | 0.72 | 0.59 | 8.3 | 190 | 290 | 3450 | 3470 |
| 4.00 | 5.50 | 4.60 | 10.4 | 11.4 | 58 | 79.7 | 80.2 | 79.6 | 75.9 | 0.82 | 0.78 | 0.70 | 0.57 | 11.0 | 200 | 290 | 3450 | 3470 |
| 5.50 | 7.50 | 6.33 | 14.2 | 15.0 | 84 | 81.0 | 81.4 | 81.0 | 77.8 | 0.82 | 0.79 | 0.70 | 0.57 | 15.1 | 220 | 300 | 3460 | 3480 |
| 7.50 | 10.00 | 8.63 | 18.1 | 19.1 | 104 | 78.6 | 79.4 | 79.4 | 76.3 | 0.86 | 0.84 | 0.78 | 0.67 | 20.8 | 210 | 280 | 3410 | 3440 |

### 3 × 400 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.75 | 1.00 | 1.05 | 2.1 | 2.7 | 12 | 72.3 | 70.7 | 67.3 | 60.1 | 0.84 | 0.80 | 0.73 | 0.63 | 2.1 | 200 | 320 | 3440 | 3490 |
| 1.10 | 1.50 | 1.43 | 3.1 | 3.7 | 18 | 73.4 | 71.7 | 68.1 | 60.7 | 0.81 | 0.75 | 0.67 | 0.56 | 3.0 | 210 | 340 | 3460 | 3500 |
| 1.50 | 2.00 | 1.88 | 4.6 | 5.2 | 23 | 73.2 | 71.6 | 67.9 | 60.2 | 0.77 | 0.70 | 0.60 | 0.48 | 4.1 | 200 | 320 | 3450 | 3490 |
| 2.20 | 3.00 | 2.53 | 7.0 | 7.3 | 36 | 72.1 | 70.6 | 66.0 | 57.6 | 0.69 | 0.64 | 0.54 | 0.43 | 6.0 | 230 | 350 | 3480 | 3490 |
| 3.00 | 4.00 | 3.45 | 8.3 | 9.0 | 46 | 77.0 | 76.5 | 73.9 | 67.7 | 0.78 | 0.73 | 0.64 | 0.51 | 8.2 | 220 | 330 | 3470 | 3490 |
| 4.00 | 5.50 | 4.60 | 10.4 | 11.4 | 62 | 78.6 | 78.2 | 76.1 | 70.6 | 0.81 | 0.77 | 0.68 | 0.55 | 11.0 | 220 | 320 | 3470 | 3490 |
| 5.50 | 7.50 | 6.33 | 14.8 | 15.2 | 90 | 76.9 | 76.1 | 73.4 | 67.0 | 0.80 | 0.76 | 0.68 | 0.55 | 15.0 | 250 | 340 | 3480 | 3500 |
| 7.50 | 10.00 | 8.63 | 17.7 | 18.6 | 111 | 79.3 | 79.6 | 78.6 | 74.6 | 0.84 | 0.80 | 0.72 | 0.59 | 20.7 | 240 | 320 | 3430 | 3460 |

### 3 × 415 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.75 | 1.00 | 1.30 | 2.1 | 2.7 | 13 | 72.2 | 70.5 | 61.3 | 58.9 | 0.83 | 077 | 0.69 | 0.59 | 2.1 | 230 | 370 | 3460 | 3490 |
| 1.10 | 1.50 | 1.43 | 3.2 | 3.7 | 19 | 73.1 | 70.9 | 66.7 | 58.8 | 0.78 | 0.71 | 0.62 | 0.51 | 3.0 | 230 | 370 | 3470 | 3500 |
| 1.50 | 2.00 | 1.88 | 4.8 | 5.4 | 24 | 72.2 | 69.9 | 65.4 | 57.0 | 0.72 | 0.64 | 0.54 | 0.43 | 4.1 | 220 | 350 | 3460 | 3490 |
| 2.20 | 3.00 | 2.53 | 7.7 | 7.9 | 38 | 69.9 | 67.9 | 62.6 | 53.3 | 0.63 | 0.58 | 0.49 | 0.39 | 6.0 | 250 | 380 | 3490 | 3500 |
| 3.00 | 4.00 | 3.45 | 8.6 | 9.4 | 48 | 76.2 | 75.3 | 72.0 | 64.9 | 0.73 | 0.68 | 0.58 | 0.45 | 8.2 | 240 | 350 | 3480 | 3500 |
| 4.00 | 5.50 | 4.60 | 10.6 | 11.6 | 64 | 78.1 | 77.4 | 74.7 | 68.5 | 0.76 | 0.72 | 0.62 | 0.49 | 10.9 | 250 | 350 | 3480 | 3500 |
| 5.50 | 7.50 | 6.33 | 14.6 | 15.6 | 95 | 76.3 | 75.3 | 71.9 | 64.9 | 0.75 | 0.71 | 0.62 | 0.49 | 15.0 | 270 | 370 | 3490 | 3500 |
| 7.50 | 10.00 | 8.63 | 18.7 | 20.6 | 116 | 79.3 | 79.2 | 77.7 | 72.8 | 0.80 | 0.76 | 0.67 | 0.54 | 20.6 | 260 | 350 | 3450 | 3470 |

### 3 × 380 V, 60 Hz, T60

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 5.50 | 7.50 | 6.33 | 13.3 | 14.2 | 75 | 77.4 | 78.1 | 77.8 | 74.5 | 0.88 | 0.87 | 0.83 | 0.74 | 15.3 | 190 | 270 | 3410 | 3440 |

### 3 × 400V, 60 Hz, T60

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 5.50 | 7.50 | 6.33 | 12.9 | 13.7 | 80 | 78.5 | 78.8 | 77.8 | 73.8 | 0.87 | 0.85 | 0.80 | 0.70 | 15.2 | 220 | 300 | 3440 | 3460 |

## 3 × 415 V, 60 Hz, T60

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 5.50 | 7.50 | 6.33 | 12.5 | 13.4 | 83.6 | 79.1 | 79.1 | 77.6 | 73.0 | 0.85 | 0.83 | 0.77 | 0.67 | 15.1 | 240 | 330 | 3450 | 3470 |

## Voltage code 30

## 3 × 208 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 2.20 | 3.00 | 2.53 | 10.8 | 11.9 | 48 | 72.2 | 72.6 | 71.1 | 65.9 | 0.86 | 0.84 | 0.79 | 0.68 | 6.1 | 150 | 230 | 3410 | 3440 |
| 3.00 | 4.00 | 3.45 | 13.9 | 15.1 | 63 | 73.3 | 74.3 | 74.1 | 70.1 | 0.88 | 0.86 | 0.82 | 0.73 | 8.4 | 150 | 220 | 3400 | 3430 |
| 3.70 | 5.00 | 4.26 | 17.0 | 18.6 | 84 | 75.4 | 76.0 | 75.2 | 71.1 | 0.88 | 0.86 | 0.82 | 0.74 | 10.3 | 160 | 240 | 3420 | 3450 |
| 5.50 | 7.50 | 6.33 | 24.5 | 27.0 | 119 | 75.8 | 76.9 | 77.0 | 73.8 | 0.88 | 0.87 | 0.83 | 0.76 | 15.3 | 160 | 230 | 3420 | 3440 |
| 7.50 | 10.00 | 8.63 | 32.8 | 35.0 | 164 | 77.6 | 79.5 | 80.5 | 78.3 | 0.88 | 0.87 | 0.82 | 0.73 | 20.9 | 170 | 240 | 3370 | 3420 |

## 3 × 220 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 2.20 | 3.00 | 2.53 | 10.6 | 11.4 | 52 | 72.9 | 72.5 | 70.0 | 63.7 | 0.83 | 0.80 | 0.72 | 0.60 | 6.1 | 170 | 270 | 3440 | 3460 |
| 3.00 | 4.00 | 3.45 | 13.0 | 14.0 | 68 | 78.2 | 79.0 | 78.7 | 74.8 | 0.82 | 0.79 | 0.70 | 0.57 | 8.3 | 180 | 260 | 3440 | 3460 |
| 3.70 | 5.00 | 4.26 | 16.4 | 17.8 | 89 | 77.1 | 76.7 | 75.1 | 70.2 | 0.84 | 0.84 | 0.79 | 0.69 | 10.2 | 200 | 280 | 3440 | 3470 |
| 5.50 | 7.50 | 6.33 | 23.4 | 24.5 | 128 | 78.2 | 79.0 | 79.9 | 78.9 | 0.87 | 0.84 | 0.77 | 0.64 | 15.2 | 190 | 270 | 3440 | 3470 |
| 7.50 | 10.00 | 8.63 | 31.4 | 33.0 | 177 | 79.4 | 80.4 | 80.3 | 77.2 | 0.86 | 0.84 | 0.78 | 0.66 | 20.8 | 200 | 280 | 3410 | 3450 |

## 3 × 230 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 2.20 | 3.00 | 2.53 | 11.1 | 11.6 | 54 | 72.5 | 71.6 | 68.4 | 61.1 | 0.78 | 0.74 | 0.65 | 0.53 | 6.0 | 190 | 300 | 3460 | 3480 |
| 3.00 | 4.00 | 3.45 | 13.7 | 15.0 | 72 | 75.2 | 75.0 | 72.9 | 67.1 | 0.83 | 0.80 | 0.72 | 0.60 | 8.3 | 190 | 290 | 3450 | 3470 |
| 3.70 | 5.00 | 4.26 | 16.1 | 17.4 | 95 | 81.5 | 81.6 | 80.4 | 76.0 | 0.79 | 0.76 | 0.67 | 0.54 | 10.2 | 210 | 310 | 3460 | 3480 |
| 5.50 | 7.50 | 6.33 | 23.8 | 25.0 | 135 | 81.0 | 81.4 | 81.0 | 77.8 | 0.82 | 0.79 | 0.70 | 0.57 | 15.1 | 210 | 290 | 3460 | 3480 |
| 7.50 | 10.00 | 8.63 | 30.5 | 32.3 | 187 | 80.0 | 80.5 | 79.7 | 75.7 | 0.84 | 0.81 | 0.73 | 0.60 | 20.7 | 230 | 310 | 3740 | 3470 |

## Voltage code 35

### 3 × 440 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.75 | 1.00 | 1.05 | 1.7 | 2.3 | 9 | 65.4 | 72.4 | 70.1 | 56.6 | 0.87 | 0.85 | 0.80 | 0.72 | 2.1 | 160 | 230 | 3360 | 3450 |
| 1.10 | 1.50 | 1.43 | 2.5 | 3.0 | 13 | 74.5 | 74.9 | 73.2 | 67.8 | 0.87 | 0.83 | 0.77 | 0.67 | 3.0 | 160 | 260 | 3420 | 3470 |
| 1.50 | 2.00 | 1.88 | 3.5 | 4.1 | 17 | 74.4 | 75.6 | 74.3 | 68.9 | 0.85 | 0.80 | 0.73 | 0.61 | 4.2 | 150 | 250 | 3420 | 3460 |
| 2.20 | 3.00 | 2.53 | 5.3 | 5.7 | 26 | 77.2 | 77.3 | 75.9 | 71.0 | 0.80 | 0.76 | 0.67 | 0.54 | 6.1 | 170 | 270 | 3440 | 3470 |
| 3.00 | 4.00 | 3.45 | 6.7 | 7.4 | 34 | 75.8 | 76.0 | 74.6 | 69.8 | 0.86 | 0.83 | 0.77 | 0.66 | 8.3 | 170 | 260 | 3430 | 3460 |
| 3.70 | 5.00 | 4.26 | 8.1 | 8.6 | 45 | 79.6 | 80.1 | 79.3 | 75.3 | 0.83 | 0.81 | 0.74 | 0.61 | 10.2 | 190 | 270 | 3450 | 3470 |
| 5.50 | 7.50 | 6.33 | 11.4 | 12.8 | 63 | 78.2 | 79.0 | 79.9 | 78.9 | 0.87 | 0.84 | 0.77 | 0.63 | 15.2 | 180 | 260 | 3440 | 3460 |
| 7.50 | 10.00 | 8.63 | 16.0 | 18.0 | 91 | 80.4 | 81.3 | 81.5 | 78.8 | 0.83 | 0.79 | 0.71 | 0.57 | 20.8 | 210 | 280 | 3430 | 3460 |

### 3 × 460 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.75 | 1.00 | 1.05 | 1.8 | 2.2 | 10 | 71.1 | 71.3 | 68.3 | 61.6 | 0.86 | 0.83 | 0.78 | 0.69 | 2.1 | 170 | 270 | 3420 | 3470 |
| 1.10 | 1.50 | 1.43 | 2.5 | 3.0 | 14 | 74.8 | 74.8 | 72.5 | 66.7 | 0.80 | 0.80 | 0.73 | 0.62 | 3.0 | 180 | 290 | 3440 | 3480 |
| 1.50 | 2.00 | 1.88 | 3.6 | 4.1 | 18 | 75.1 | 75.3 | 73.0 | 66.8 | 0.82 | 0.75 | 0.66 | 0.54 | 4.1 | 180 | 280 | 3440 | 3480 |
| 2.20 | 3.00 | 2.53 | 5.5 | 5.8 | 28 | 73.5 | 72.7 | 69.6 | 62.6 | 0.78 | 0.74 | 0.64 | 0.52 | 6.0 | 190 | 300 | 3460 | 3480 |
| 3.00 | 4.00 | 3.45 | 6.6 | 7.5 | 36 | 76.2 | 75.9 | 73.8 | 68.1 | 0.83 | 0.80 | 0.72 | 0.59 | 8.2 | 200 | 290 | 3450 | 3470 |
| 3.70 | 5.00 | 4.26 | 8.1 | 8.6 | 47 | 80.0 | 80.2 | 79.1 | 74.9 | 0.80 | 0.76 | 0.67 | 0.54 | 10.2 | 210 | 300 | 3460 | 3480 |
| 5.50 | 7.50 | 6.33 | 11.7 | 12.6 | 67 | 81.0 | 81.4 | 81.0 | 77.8 | 0.82 | 0.79 | 0.70 | 0.57 | 15.1 | 210 | 290 | 3460 | 3480 |
| 7.50 | 10.00 | 8.63 | 16.9 | 18.6 | 97 | 80.3 | 81.0 | 80.6 | 76.7 | 0.78 | 0.74 | 0.64 | 0.51 | 20.7 | 230 | 310 | 3470 | 3450 |

### 3 × 440 V, 60 Hz, T60

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 2.20 | 3.00 | 2.53 | 5.0 | 5.8 | 27 | 73.6 | 73.5 | 71.6 | 66.1 | 0.86 | 0.84 | 0.79 | 0.70 | 6.1 | 180 | 270 | 3450 | 3470 |
| 3.00 | 4.00 | 3.45 | 6.7 | 7.6 | 38 | 75.7 | 75.4 | 73.4 | 67.9 | 0.86 | 0.84 | 0.79 | 0.71 | 8.2 | 200 | 290 | 3450 | 3480 |
| 3.70 | 5.00 | 4.26 | 8.0 | 9.1 | 47 | 76.1 | 76.1 | 74.5 | 69.5 | 0.87 | 0.86 | 0.82 | 0.74 | 10.2 | 190 | 280 | 3460 | 3480 |
| 5.50 | 7.50 | 6.33 | 11.6 | 13.8 | 70 | 76.8 | 77.1 | 76.2 | 71.9 | 0.87 | 0.86 | 0.81 | 0.72 | 15.2 | 210 | 290 | 3460 | 3480 |

### 3 × 460 V, 60 Hz, T60

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 2.20 | 3.00 | 2.53 | 5.0 | 5.7 | 28 | 74.3 | 73.8 | 71.3 | 65.2 | 0.84 | 0.82 | 0.76 | 0.66 | 6.0 | 200 | 310 | 3460 | 3480 |
| 3.00 | 4.00 | 3.45 | 7.6 | 7.6 | 41 | 76.2 | 75.6 | 78.2 | 67.1 | 0.84 | 0.82 | 0.76 | 0.66 | 8.2 | 220 | 320 | 3470 | 3490 |
| 3.70 | 5.00 | 4.26 | 9.1 | 9.1 | 49 | 76.8 | 76.5 | 74.4 | 68.8 | 0.86 | 0.84 | 0.79 | 0.70 | 10.1 | 220 | 320 | 3470 | 3490 |
| 5.50 | 7.50 | 6.33 | 12.2 | 13.8 | 74 | 81.2 | 81.3 | 80.0 | 75.6 | 0.80 | 0.76 | 0.68 | 0.55 | 15.2 | 230 | 320 | 3470 | 3490 |

## Voltage code 39

### 3 × 575 V, 60 Hz, T40

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 0.75 | 1.00 | 1.05 | 1.5 | 1.8 | 8 | 71.1 | 71.1 | 68.3 | 61.6 | 0.85 | 0.83 | 0.78 | 0.69 | 2.1 | 170 | 280 | 3470 | 3400 |
| 1.10 | 1.50 | 1.43 | 2.0 | 2.4 | 11 | 74.8 | 74.8 | 72.5 | 66.7 | 0.85 | 0.84 | 0.73 | 0.62 | 3.0 | 180 | 290 | 3440 | 3480 |
| 1.50 | 2.00 | 1.88 | 2.9 | 3.3 | 14 | 75.1 | 75.3 | 73.0 | 66.8 | 0.81 | 0.75 | 0.66 | 0.54 | 4.1 | 170 | 280 | 3480 | 3480 |
| 2.20 | 3.00 | 2.53 | 4.3 | 4.7 | 21 | 73.9 | 73.3 | 70.5 | 63.9 | 0.80 | 0.76 | 0.67 | 0.54 | 6.0 | 190 | 290 | 3450 | 3470 |
| 3.00 | 4.00 | 3.45 | 5.6 | 6.0 | 29 | 75.1 | 74.9 | 72.8 | 67.0 | 0.83 | 0.80 | 0.72 | 0.60 | 8.3 | 190 | 280 | 3450 | 3470 |
| 3.70 | 5.00 | 4.26 | 6.6 | 6.9 | 38 | 80.4 | 80.5 | 79.4 | 75.4 | 0.80 | 0.76 | 0.67 | 0.54 | 10.2 | 210 | 300 | 3470 | 3490 |
| 5.50 | 7.50 | 6.33 | 9.0 | 10.0 | 54 | 81.0 | 81.4 | 81.0 | 77.8 | 0.82 | 0.78 | 0.70 | 0.57 | 15.1 | 210 | 290 | 3460 | 3480 |
| 7.50 | 10.00 | 8.63 | 13.2 | 14.4 | 77 | 80.3 | 81.0 | 80.6 | 76.7 | 0.79 | 0.74 | 0.64 | 0.51 | 20.7 | 230 | 310 | 3450 | 3470 |

### 3 × 575 V, 60 Hz, T60

| Power | | | Current | | | Motor efficiency η | | | | Cos φ | | | | Torque | | | Speed | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | SF | FL | SF | LR | SF | FL | ¾FL | ½FL | SF | FL | ¾FL | ½FL | FL | LR | BD | SF | FL |
| [kW] | [hp] | [kW] | [A] | [A] | [A] | [%] | [%] | [%] | [%] | [-] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] | [rpm] |
| 2.20 | 3.00 | 2.53 | 4.0 | 4.6 | 23 | 74.7 | 74.2 | 71.6 | 65.4 | 0.84 | 0.82 | 0.75 | 0.65 | 6.0 | 210 | 320 | 3460 | 3480 |
| 3.00 | 4.00 | 3.45 | 5.2 | 6.1 | 31 | 75.6 | 75.2 | 72.9 | 67.2 | 0.83 | 0.83 | 0.78 | 0.68 | 8.2 | 200 | 300 | 3460 | 3480 |
| 3.70 | 5.00 | 4.26 | 6.3 | 7.2 | 40 | 77.0 | 76.6 | 74.5 | 68.9 | 0.86 | 0.84 | 0.78 | 0.69 | 10.1 | 220 | 320 | 3470 | 3490 |
| 5.50 | 7.50 | 6.33 | 9.1 | 10.3 | 57 | 77.4 | 77.6 | 76.3 | 71.7 | 0.86 | 0.85 | 0.80 | 0.70 | 15.2 | 230 | 310 | 3430 | 3460 |

## Voltage code 09

### 3 × 200 V, 50 Hz, T40

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 3.8 | 20 | 72.8 | 71.9 | 68.1 | 0.84 | 0.79 | 0.69 | 2.5 | 180 | 280 | 2870 |
| 1.10 | 5.5 | 26 | 74.9 | 74.0 | 70.3 | 0.82 | 0.74 | 0.62 | 3.7 | 160 | 240 | 2870 |
| 1.50 | 7.9 | 33 | 73.6 | 72.9 | 68.3 | 0.78 | 0.68 | 0.54 | 5.0 | 150 | 230 | 2860 |
| 1.90 | 10.2 | 44 | 72.3 | 70.7 | 65.1 | 0.80 | 0.72 | 0.58 | 6.3 | 170 | 250 | 2860 |
| 2.20 | 12.2 | 52 | 71.7 | 69.4 | 62.8 | 0.77 | 0.63 | 0.52 | 7.4 | 170 | 250 | 2860 |
| 2.70 | 13.6 | 64 | 74.9 | 74.3 | 69.8 | 0.82 | 0.74 | 0.61 | 9.0 | 170 | 250 | 2860 |
| 3.00 | 15.6 | 69 | 75.1 | 74.8 | 70.8 | 0.77 | 0.67 | 0.52 | 10.0 | 170 | 240 | 2860 |
| 3.70 | 18.1 | 90 | 76.8 | 76.3 | 72.3 | 0.83 | 0.75 | 0.62 | 12.3 | 190 | 260 | 2870 |
| 4.00 | 19.2 | 92 | 77.9 | 79.0 | 76.6 | 0.80 | 0.71 | 0.57 | 13.4 | 180 | 240 | 2860 |
| 5.50 | 24.3 | 130 | 77.4 | 77.5 | 74.1 | 0.84 | 0.77 | 0.64 | 18.3 | 180 | 250 | 2860 |

## 3 × 200 V, 60 Hz, T40

| Power | | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [hp] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 1.00 | 3.9 | 18 | 70.2 | 69.4 | 64.7 | 0.87 | 0.84 | 0.78 | 2.1 | 150 | 210 | 3430 |
| 1.10 | 1.50 | 5.3 | 23 | 72.8 | 72.3 | 68.2 | 0.86 | 0.83 | 0.76 | 3.1 | 130 | 180 | 3430 |
| 1.50 | 2.00 | 7.0 | 29 | 75.0 | 75.8 | 72.8 | 0.86 | 0.82 | 0.72 | 4.2 | 120 | 200 | 3430 |
| 1.90 | 2.50 | 9.5 | 39 | 71.6 | 71.5 | 67.5 | 0.87 | 0.84 | 0.76 | 5.3 | 130 | 210 | 3420 |
| 2.20 | 3.00 | 10.8 | 46 | 72.0 | 71.4 | 66.8 | 0.86 | 0.82 | 0.73 | 6.1 | 130 | 210 | 3440 |
| 2.70 | 3.60 | 13.1 | 55 | 73.2 | 73.5 | 69.9 | 0.88 | 0.84 | 0.77 | 7.5 | 130 | 210 | 3420 |
| 3.00 | 4.00 | 14.2 | 60 | 77.5 | 78.8 | 76.5 | 0.86 | 0.80 | 0.70 | 8.4 | 130 | 200 | 3430 |
| 3.70 | 5.00 | 17.6 | 79 | 74.9 | 74.9 | 71.3 | 0.87 | 0.84 | 0.77 | 10.3 | 140 | 220 | 3430 |
| 4.00 | 5.50 | 18.8 | 79 | 78.7 | 80.1 | 79.0 | 0.87 | 0.82 | 0.74 | 11.2 | 140 | 200 | 3430 |
| 5.50 | 7.50 | 26.1 | 113 | 75.8 | 76.7 | 74.1 | 0.88 | 0.85 | 0.78 | 15.3 | 140 | 210 | 3430 |

## 3 × 220 V, 60 Hz, T40

| Power | | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [hp] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 1.00 | 3.6 | 20 | 72.4 | 70.1 | 64.0 | 0.85 | 0.80 | 0.72 | 2.1 | 180 | 290 | 3470 |
| 1.10 | 1.50 | 5.0 | 25 | 74.9 | 73.2 | 67.8 | 0.82 | 0.77 | 0.67 | 3.1 | 150 | 250 | 3470 |
| 1.50 | 2.00 | 6.8 | 32 | 75.6 | 74.3 | 68.9 | 0.80 | 0.73 | 0.60 | 4.2 | 150 | 240 | 3460 |
| 1.90 | 2.50 | 9.0 | 44 | 73.0 | 71.0 | 65.3 | 0.82 | 0.76 | 0.65 | 5.2 | 170 | 260 | 3460 |
| 2.20 | 3.00 | 10.6 | 52 | 72.5 | 70.0 | 63.7 | 0.80 | 0.72 | 0.60 | 6.1 | 170 | 270 | 3460 |
| 2.70 | 3.60 | 12.2 | 62 | 75.0 | 73.5 | 68.4 | 0.84 | 0.78 | 0.68 | 7.4 | 170 | 260 | 3460 |
| 3.00 | 4.00 | 13.0 | 68 | 79.0 | 78.7 | 74.8 | 0.79 | 0.70 | 0.57 | 8.3 | 170 | 260 | 3460 |
| 3.70 | 5.00 | 16.2 | 89 | 76.5 | 75.0 | 70.0 | 0.84 | 0.79 | 0.69 | 10.2 | 190 | 270 | 3470 |
| 4.00 | 5.50 | 16.8 | 90 | 79.9 | 79.3 | 71.1 | 0.82 | 0.76 | 0.64 | 11.1 | 170 | 250 | 3460 |
| 5.50 | 7.50 | 23.4 | 128 | 79.0 | 79.9 | 78.9 | 0.84 | 0.77 | 0.64 | 15.2 | 180 | 260 | 3470 |

## 3 × 200 V, 50 Hz, T60

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 1.50 | 7.4 | 36 | 71.6 | 69.8 | 66.2 | 0.85 | 0.79 | 0.67 | 5.0 | 170 | 260 | 2860 |
| 1.90 | 9.4 | 47 | 74.2 | 73.7 | 70.0 | 0.85 | 0.79 | 0.69 | 6.3 | 170 | 260 | 2860 |
| 2.20 | 10.4 | 55 | 73.2 | 71.6 | 68.2 | 0.85 | 0.79 | 0.67 | 7.3 | 180 | 270 | 2870 |
| 2.70 | 12.9 | 67 | 76.3 | 76.2 | 73.2 | 0.85 | 0.80 | 0.70 | 9.0 | 180 | 260 | 2870 |
| 3.00 | 13.3 | 76 | 76.7 | 76.3 | 72.5 | 0.84 | 0.78 | 0.66 | 10.0 | 190 | 270 | 2870 |
| 3.70 | 17.3 | 87 | 76.8 | 77.2 | 74.8 | 0.87 | 0.83 | 0.73 | 12.3 | 180 | 250 | 2860 |
| 4.00 | 18.0 | 95 | 75.3 | 74.2 | 71.6 | 0.88 | 0.84 | 0.74 | 13.3 | 180 | 250 | 2860 |
| 5.50 | 26.5 | 156 | 78.2 | 77.6 | 73.7 | 0.81 | 0.74 | 0.61 | 18.3 | 230 | 300 | 2870 |

## 3 × 200 V, 60 Hz, T60

| Power | | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [hp] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 1.50 | 2.00 | 7.6 | 32 | 68.8 | 67.8 | 62.7 | 0.87 | 0.84 | 0.79 | 4.2 | 140 | 210 | 3410 |
| 1.90 | 2.50 | 9.7 | 41 | 70.4 | 69.9 | 65.4 | 0.87 | 0.84 | 0.78 | 5.3 | 140 | 210 | 3420 |
| 2.20 | 3.00 | 10.6 | 48 | 72.4 | 71.8 | 67.5 | 0.87 | 0.84 | 0.78 | 6.1 | 140 | 220 | 3430 |
| 2.70 | 3.60 | 13.2 | 58 | 73.1 | 73.1 | 69.3 | 0.87 | 0.85 | 0.79 | 7.5 | 140 | 220 | 3430 |
| 3.00 | 4.00 | 13.2 | 66 | 74.3 | 74.1 | 70.4 | 0.87 | 0.84 | 0.78 | 8.3 | 150 | 230 | 3440 |
| 3.70 | 5.00 | 18.0 | 75 | 73.1 | 73.5 | 70.3 | 0.88 | 0.86 | 0.80 | 10.3 | 130 | 200 | 3420 |
| 4.00 | 5.50 | 18.8 | 82 | 73.8 | 74.3 | 71.2 | 0.88 | 0.86 | 0.80 | 11.2 | 140 | 210 | 3420 |
| 5.50 | 7.50 | 25.5 | 136 | 77.5 | 77.6 | 74.5 | 0.87 | 0.83 | 0.75 | 15.2 | 180 | 260 | 3440 |

## 3 × 220 V, 60 Hz, T60

| Power | | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [hp] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 1.50 | 2.00 | 7.0 | 36 | 70.7 | 68.1 | 61.8 | 0.84 | 0.80 | 0.73 | 4.1 | 170 | 270 | 3460 |
| 1.90 | 2.50 | 8.8 | 46 | 72.4 | 70.4 | 64.7 | 0.85 | 0.81 | 0.73 | 5.2 | 170 | 270 | 3460 |
| 2.20 | 3.00 | 10.0 | 54 | 74.0 | 72.0 | 66.4 | 0.84 | 0.79 | 0.70 | 6.1 | 180 | 280 | 3470 |
| 2.70 | 3.60 | 11.9 | 65 | 75.2 | 73.7 | 68.7 | 0.86 | 0.82 | 0.74 | 7.4 | 180 | 270 | 3470 |
| 3.00 | 4.00 | 12.1 | 75 | 75.9 | 74.3 | 69.3 | 0.84 | 0.79 | 0.70 | 8.2 | 190 | 290 | 3470 |
| 3.70 | 5.00 | 16.0 | 85 | 75.6 | 74.5 | 70.0 | 0.86 | 0.83 | 0.76 | 10.2 | 170 | 260 | 3460 |
| 4.00 | 5.50 | 16.6 | 93 | 76.2 | 75.1 | 70.7 | 0.86 | 0.83 | 0.76 | 11.0 | 180 | 260 | 3470 |
| 5.50 | 7.50 | 23.6 | 154 | 78.8 | 77.5 | 73.0 | 0.84 | 0.78 | 0.67 | 15.1 | 230 | 330 | 3470 |

## Voltage code 65

## 3 × 400 V, 50 Hz, T40

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 2.0 | 9 | 72.8 | 71.9 | 68.1 | 0.84 | 0.79 | 0.69 | 2.5 | 180 | 280 | 2870 |
| 1.10 | 2.8 | 13 | 74.9 | 74.0 | 70.3 | 0.81 | 0.74 | 0.62 | 3.7 | 160 | 240 | 2870 |
| 1.50 | 4.0 | 17 | 73.6 | 72.9 | 68.2 | 0.78 | 0.68 | 0.54 | 5.0 | 150 | 230 | 2860 |
| 1.90 | 5.1 | 22 | 72.3 | 70.7 | 65.1 | 0.80 | 0.72 | 0.58 | 6.3 | 170 | 250 | 2860 |
| 2.20 | 6.0 | 26 | 74.3 | 73.1 | 67.6 | 0.76 | 0.63 | 0.49 | 7.4 | 170 | 250 | 2860 |
| 2.70 | 6.8 | 32 | 74.9 | 74.3 | 69.8 | 0.82 | 0.74 | 0.61 | 9.0 | 170 | 250 | 2860 |
| 3.00 | 7.9 | 35 | 75.5 | 74.9 | 70.9 | 0.81 | 0.73 | 0.58 | 10.0 | 170 | 250 | 2850 |
| 3.70 | 9.1 | 45 | 76.8 | 76.3 | 72.3 | 0.83 | 0.75 | 0.62 | 12.3 | 190 | 260 | 2870 |
| 4.00 | 9.6 | 46 | 77.9 | 79.0 | 76.6 | 0.80 | 0.71 | 0.57 | 13.4 | 170 | 240 | 2860 |
| 5.50 | 13.0 | 65 | 79.5 | 80.5 | 78.3 | 0.81 | 0.72 | 0.57 | 18.4 | 180 | 250 | 2860 |
| 7.50 | 19.3 | 92 | 77.9 | 78.4 | 75.2 | 0.77 | 0.67 | 0.52 | 25.3 | 200 | 260 | 2850 |

## 3 × 400 V, 60 Hz, T40

| Power | | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [hp] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 1.00 | 1.9 | 8 | 70.2 | 69.4 | 64.7 | 0.87 | 0.84 | 0.78 | 2.1 | 130 | 190 | 3430 |
| 1.10 | 1.50 | 2.7 | 12 | 72.8 | 72.3 | 68.2 | 0.86 | 0.83 | 0.76 | 3.1 | 130 | 200 | 3430 |
| 1.50 | 2.00 | 3.5 | 15 | 75.0 | 75.8 | 72.8 | 0.86 | 0.81 | 0.72 | 4.2 | 120 | 200 | 3430 |
| 1.90 | 2.50 | 4.8 | 20 | 71.6 | 71.5 | 67.5 | 0.87 | 0.84 | 0.76 | 5.3 | 130 | 210 | 3420 |
| 2.20 | 3.00 | 5.4 | 23 | 73.1 | 72.8 | 68.7 | 0.86 | 0.81 | 0.72 | 6.1 | 130 | 210 | 3430 |
| 2.70 | 3.60 | 6.5 | 28 | 73.2 | 73.5 | 69.9 | 0.88 | 0.84 | 0.77 | 7.5 | 130 | 210 | 3420 |
| 3.00 | 4.00 | 7.1 | 30 | 74.4 | 74.8 | 71.5 | 0.87 | 0.84 | 0.77 | 8.4 | 130 | 210 | 3410 |
| 3.70 | 5.00 | 8.7 | 40 | 74.9 | 74.9 | 71.3 | 0.87 | 0.84 | 0.77 | 10.3 | 140 | 220 | 3430 |
| 4.00 | 5.50 | 9.4 | 40 | 78.6 | 80.4 | 78.8 | 0.87 | 0.83 | 0.75 | 11.2 | 130 | 200 | 3430 |
| 5.50 | 7.50 | 11.8 | 56 | 80.0 | 81.6 | 80.3 | 0.87 | 0.85 | 0.77 | 15.4 | 140 | 210 | 3440 |
| 7.50 | 10.00 | 16.4 | 81 | 79.9 | 82.6 | 82.3 | 0.86 | 0.80 | 0.69 | 21.1 | 160 | 230 | 3420 |

## 3 × 440 V, 60 Hz, T40

| Power | | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [hp] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 0.75 | 1.00 | 1.7 | 9 | 72.4 | 70.1 | 56.6 | 0.85 | 0.80 | 0.72 | 2.1 | 160 | 230 | 3470 |
| 1.10 | 1.50 | 2.5 | 13 | 74.9 | 73.2 | 67.8 | 0.83 | 0.77 | 0.67 | 3.0 | 160 | 260 | 3470 |
| 1.50 | 2.00 | 3.5 | 17 | 75.6 | 74.3 | 68.9 | 0.80 | 0.73 | 0.61 | 4.2 | 150 | 250 | 3460 |
| 1.90 | 2.50 | 4.5 | 22 | 73.0 | 71.0 | 65.3 | 0.82 | 0.76 | 0.65 | 5.2 | 170 | 260 | 3460 |
| 2.20 | 3.00 | 5.3 | 26 | 77.3 | 75.9 | 71.0 | 0.76 | 0.67 | 0.54 | 6.1 | 170 | 270 | 3470 |
| 2.70 | 3.60 | 6.1 | 31 | 75.0 | 73.5 | 68.4 | 0.84 | 0.78 | 0.68 | 7.4 | 170 | 260 | 3460 |
| 3.00 | 4.00 | 6.7 | 34 | 76.0 | 74.6 | 69.8 | 0.83 | 0.77 | 0.66 | 8.3 | 170 | 260 | 3460 |
| 3.70 | 5.00 | 8.1 | 45 | 80.1 | 79.3 | 75.3 | 0.81 | 0.74 | 0.61 | 10.2 | 190 | 270 | 3470 |
| 4.00 | 5.50 | 8.7 | 45 | 79.9 | 79.7 | 76.3 | 0.82 | 0.76 | 0.64 | 11.1 | 170 | 250 | 3460 |
| 5.50 | 7.50 | 11.4 | 63 | 79.0 | 79.9 | 78.9 | 0.84 | 0.77 | 0.63 | 15.2 | 180 | 260 | 3460 |
| 7.50 | 10.00 | 16.0 | 91 | 81.3 | 81.5 | 78.8 | 0.79 | 0.71 | 0.57 | 20.8 | 210 | 280 | 3460 |

## 3 × 400 V, 50 Hz, T60

| Power | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 1.50 | 3.8 | 18 | 71.6 | 69.8 | 66.2 | 0.85 | 0.79 | 0.67 | 5.0 | 170 | 260 | 2860 |
| 1.90 | 4.7 | 24 | 74.3 | 73.8 | 70.0 | 0.85 | 0.79 | 0.68 | 6.3 | 170 | 260 | 2860 |
| 2.20 | 5.9 | 27 | 77.1 | 76.5 | 72.4 | 0.80 | 0.72 | 0.59 | 7.3 | 180 | 260 | 2880 |
| 2.70 | 6.5 | 33 | 76.3 | 76.2 | 73.2 | 0.85 | 0.80 | 0.70 | 9.0 | 180 | 260 | 2870 |
| 3.00 | 6.7 | 38 | 76.0 | 76.3 | 72.5 | 0.84 | 0.76 | 0.66 | 10.0 | 190 | 270 | 2870 |
| 3.70 | 8.6 | 44 | 76.8 | 77.2 | 74.8 | 0.87 | 0.83 | 0.73 | 12.3 | 180 | 250 | 2860 |
| 4.00 | 9.8 | 48 | 79.3 | 79.6 | 76.6 | 0.84 | 0.78 | 0.67 | 13.3 | 180 | 250 | 2880 |
| 5.50 | 14.4 | 71 | 79.9 | 79.8 | 76.8 | 0.78 | 0.69 | 0.55 | 18.4 | 210 | 270 | 2880 |

## 3 × 400 V, 60 Hz, T60

| Power | | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [hp] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 1.50 | 2.00 | 4.0 | 16 | 68.8 | 67.8 | 62.7 | 0.87 | 0.84 | 0.79 | 4.2 | 140 | 210 | 3410 |
| 1.90 | 2.50 | 4.8 | 21 | 70.4 | 69.9 | 65.4 | 0.87 | 0.84 | 0.78 | 5.3 | 140 | 210 | 3420 |
| 2.20 | 3.00 | 5.4 | 24 | 71.8 | 71.4 | 67.2 | 0.87 | 0.84 | 0.78 | 6.1 | 140 | 220 | 3420 |
| 2.70 | 3.60 | 6.6 | 29 | 73.1 | 73.1 | 69.3 | 0.87 | 0.85 | 0.79 | 7.5 | 140 | 220 | 3430 |
| 3.00 | 4.00 | 6.7 | 33 | 74.3 | 74.1 | 70.4 | 0.87 | 0.84 | 0.78 | 8.3 | 150 | 230 | 3440 |
| 3.70 | 5.00 | 9.0 | 38 | 73.1 | 73.5 | 70.3 | 0.88 | 0.86 | 0.8 | 10.3 | 130 | 200 | 3420 |
| 4.00 | 5.50 | 9.4 | 41 | 73.8 | 74.3 | 71.2 | 0.88 | 0.86 | 0.8 | 11.2 | 140 | 210 | 3420 |
| 5.50 | 7.50 | 12.8 | 62 | 75.1 | 75.7 | 72.8 | 0.88 | 0.85 | 0.79 | 15.4 | 160 | 230 | 3400 |

## 3 × 440 V, 60 Hz, T60

| Power | | Current | | Motor efficiency η | | | Cos φ | | | Torque | | | Speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FL | | FL | LR | FL | ¾FL | ½FL | FL | ¾FL | ½FL | FL | LR | BD | FL |
| [kW] | [hp] | [A] | [A] | [%] | [%] | [%] | [-] | [-] | [-] | [Nm] | [%] | [%] | [rpm] |
| 1.50 | 2.00 | 3.6 | 18 | 70.7 | 68.1 | 61.8 | 0.84 | 0.80 | 0.73 | 4.1 | 170 | 270 | 3460 |
| 1.90 | 2.60 | 4.4 | 23 | 72.4 | 70.4 | 64.7 | 0.85 | 0.81 | 0.73 | 5.2 | 170 | 270 | 3470 |
| 2.20 | 3.00 | 5.0 | 27 | 73.5 | 71.6 | 66.1 | 0.84 | 0.79 | 0.70 | 6.1 | 180 | 270 | 3470 |
| 2.70 | 3.60 | 6.0 | 33 | 75.2 | 73.7 | 68.7 | 0.86 | 0.82 | 0.74 | 7.4 | 180 | 270 | 3470 |
| 3.00 | 4.00 | 6.6 | 38 | 75.9 | 74.3 | 69.3 | 0.84 | 0.79 | 0.70 | 8.2 | 190 | 290 | 3470 |
| 3.70 | 5.00 | 8.0 | 43 | 75.6 | 74.5 | 70.0 | 0.86 | 0.83 | 0.76 | 10.2 | 170 | 260 | 3460 |
| 4.00 | 5.50 | 8.4 | 47 | 76.2 | 75.1 | 70.7 | 0.86 | 0.83 | 0.76 | 11.0 | 180 | 260 | 3460 |
| 5.50 | 7.50 | 11.6 | 70 | 77.1 | 76.2 | 71.9 | 0.86 | 0.81 | 0.72 | 15.2 | 210 | 290 | 3480 |

# 7. Accessories

## CSCR motor starters

SA-SPM motor starters are required to connect the MS4000 200-240V single-phase 3-wire CSCR motors.

The motor starter boxes incorporate start/run capacitors for motor operation and start/run overload relays together with a surge arrestor for motor protection.

TM077903

| CSCR motor starters | CS [µF] | CR [µF] | Product number | | |
|---|---|---|---|---|---|
| | | | Standard | Delux [8] | Approval |
| 2.20 kW/50 Hz | 260 | 60 | 98582401 | - | CE |
| 2.00 hp/60 Hz | 115 | 20 | 98315256 | 98315257 | $_{c}UL_{us}$ |
| 3.00 hp/60 Hz | 220 | 35 | 98315258 | 99053534 | $_{c}UL_{us}$ |
| 5.00 hp/60 Hz | 300 | 80 | 98315260 | 99053535 | $_{c}UL_{us}$ |

8) Deluxe versions with on/off pressure switch relay for extra protection.

### Related information

*Connection of single-phase motors*

## PSC motor capacitors

The MS4000 220-240V single-phase 3-wire PSC motors must be connected to the mains through a run capacitor permanently connected in series with the auxiliary winding.

### Product numbers

| Motor | Size [µF] / Voltage [V] | Product number |
|---|---|---|
| 0.27 kW/50 Hz | 16/400 | 96279800 |
| 0.55 kW/50 Hz | 20/400 | 96279732 |
| 0.75 kW/50 Hz | 30/400 | 96279808 |
| 1.10 kW/50 Hz | 40/400 | 96279810 |
| 1.50 kW/50 Hz | 60/400 | 96279809 |
| 2.20 kW/50 Hz | 100/400 | 96279810 + 96279809 [9] |

9) 40 µF + 60 µF capacitors in parallel

### Related information

*Connection of single-phase motors*

## CUE frequency converter

TM077504

*The CUE range*

The Grundfos CUE is a series of external frequency converters designed for speed control of a wide range of Grundfos pumps.

The CUE offers quick and easy setup and commissioning compared to a standard frequency converter because of the startup guide. The startup guide takes the installer through all the necessary settings, requiring few inputs from which the CUE automatically sets all necessary parameters.

The CUE enables gentle pumping and thereby protects the water reservoir and the rest of the distribution system, as water hammer can be avoided by adjusting ramp times up and down.

When a CUE is installed, the motor requires no further overload protection. If needed, a Pt100 or Pt1000 together with the MCB 114 provides overheat protection of the motor windings.

If the motor has a built-in Tempcon sensor, this sensor is disconnected when it is exposed to the frequency converter drive. An internal fuse in the motor melts and it cannot be replaced. The motor will work without the sensor, but it is not possible to restore the functionality of the Tempcon sensor.

## Overview of CUE range

| Supply voltage [V] | Power range | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | [kW] | 0.55 | 0.75 | 1.1 | 7.5 | 11 | 45 | 90 | 250 |
| | [hp] | 0.75 | 1.0 | 1.5 | 10 | 15 | 60 | 125 | 350 |
| 1 × 200-240 | | | | • | • | | | | |
| 3 × 200-240 | | | • | • | • | • | • | | |
| 3 × 380-500 | | • | • | • | • | • | • | • | • |
| 3 × 525-600 | | | • | • | • | • | • | • | |
| 3 × 525-690 | | | | | | • | • | • | • |

CUE is available in two enclosure classes:

- IP20/21
- IP54/55.

## RFI filters

To meet the EMC requirements, the CUE comes with the following types of built-in radio frequency interference filter (RFI).

| Voltage [V] | Typical shaft power, P2 [kW (hp)] | RFI filter type | Application |
|---|---|---|---|
| 1 × 200-240 | 1.1 - 7.5 (1.5 - 10) | C1 | Domestic |
| 3 × 200-240 | 0.75 - 45 (1.0 - 60) | C1 | |
| 3 × 380-500 | 0.55 - 90 (0.75 - 125) | C1 | |
| | 110 - 250 (150 - 350) | C3 | Industry |
| 3 × 525-600 | 0.75 - 90 (1.0 - 125) | C3 | |
| 3 × 525-690 | 11 - 250 (15 - 350) | C3 | |

## Inputs and outputs

The CUE is equipped with a number of inputs and outputs:

- 1 RS-485 GENIbus connection
- 1 analog input, 0-10 V, 0/4-20 mA
  - external setpoint
- 1 analog input, 0-10 V, 0/4-20 mA
  - sensor input, feedback sensor
- 1 analog output, 0-20 mA
- 6 digital inputs
  - 2 inputs can be changed to digital outputs
  - all digital inputs and outputs are programmable
- 2 signal relays (C/NO/NC)
  - programmable.

## Features

CUE has the following features:

- startup guide for easy setup of the CUE with Grundfos pumps. It is automatically launched the first time the CUE is connected to the power supply (and relaunched on request)
- checking of direction of rotation
- duty or standby operation
- dry-running protection
- ow-flow stop function
- pump-specific functions.

## Functions

The CUE has a wide range of pump-specific functions, such as the following:

- constant pressure
- constant level
- constant flow rate
- constant temperature
- constant curve.

## Accessories

Grundfos offers various accessories for the CUE.

### MCB 114 sensor input module

The MCB 114 is required to connect a Pt100/Pt1000 sensor, and offers additional analog inputs for the CUE:

- one analog input, 0/4-20 mA
- two inputs for the Pt100/Pt1000 temperature sensors.

### Sensors

The following sensors can be used in connection with the CUE:

- pressure sensors, up to 25 bar
- temperature sensors
- differential-pressure sensors
- differential-temperature sensors
- flowmeters
- potentiometer box for external setpoint setting.

For further information, see the *CUE frequency converter data booklet* at the *Grundfos Product Center*.

### Output filters

Output filters protect the motor from voltage peaks and increased operating temperature due to the frequency converter driving supply. The filters also decrease acoustic noise from the frequency converter driven motor.

Always use sine-wave filters for a submersible motor installation.

See section *Frequency converter operation* for further information.

### Related information

*Frequency converter operation*

# RSI frequency converter

The Grundfos Renewable Solar Inverter (RSI) is an off-grid solar inverter converting the DC power output from the solar panel to AC power supply for the pump operation.

RSI can be used in both new and existing systems as long as the motor specifications are compatible and the motor is suitable for use with a variable frequency drive.

TM066678

## Applications

The RSI is designed for continuous as well as intermittent operation, and it is suitable for various water supply systems, including irrigation.

The RSI can be used in existing systems with submersible pumps or dry-installed pumps, thus providing a very wide range of applications, allowing you to leverage renewable energy sources with the ability to back up the system with grid or generator power.

## Features and benefits

### Enclosure classes up to IP66

RSI is available in three enclosure classes:

- IP21: The RSI must be installed in a cabinet with sufficient ingress protection.
- IP54: The RSI can be installed without a cabinet, but must not be installed outdoors without additional protection against water and sun.
- IP66: The RSI can be installed without a cabinet. It can be installed outdoors, but should be shaded from direct sunlight.

### Setup wizard with Grundfos product library

The RSI has a built-in Grundfos product library that allows a plug-and-pump experience. The motor library contains all related parameters, which simplifies the setup process to just a few clicks and a setup time of a few minutes.

### Detachable operating panel

The operating panel is mounted on the RSI via a magnet and a communication plug. The inverter can operate without the operating panel, and operates with the last saved setup data.

### AC/DC compatibility

The RSI is AC and DC compatible. The RSI can be connected to the grid or a generator as back-up power during solar panel disruptions.

The AC compatibility also allows the end user to connect the inverter to three-phase power in their workshop for off-site setup, which enables a fast and simple on-site installation.

### Maximum power point tracking (MPPT)

The inverter has built-in electronics with four MPPT algorithms. The inverter continuously optimises the power output according to available solar irradiation as well as various environmental conditions.

### Overvoltage and undervoltage protection

Overvoltage and undervoltage may occur in case of faulty installation. The inverter disconnects the power connection to the motor if the voltage falls outside the allowed voltage limits. The inverter stays in fault mode with the error code displayed until reset.

### Overload protection

Overload may occur if the maximum allowed frequency is set too high or a wrong pump is used. The inverter disconnects the power connection to the motor if overload occurs. The inverter stays in fault mode with the error code displayed until reset.

### Overcurrent protection

Overcurrent may occur if, for example, a wrong cable size is used. The inverter disconnects the power connection to the motor if the current falls outside the allowed current limits. The inverter stays in fault mode with the error code displayed until reset.

Note that for the overcurrent protection to function properly, the actual current value must be adjusted according to the maximum current attained at the maximum frequency shortly after the system started operation.

### Overtemperature protection

The inverter may be overheated if appropriate ventilation is not available or the ambient temperature is too high. The inverter disconnects the power connection to the motor if overtemperature occurs in the inverter. The inverter stays in fault mode with the error code displayed until reset.

By installing the Pt100/Pt1000 option board, the motor temperature can be monitored via the Pt100/Pt1000, and the motor can be protected against overtemperature.

## No-load protection

No-load situation of the inverter may occur if, for example, the cable to the motor is broken. The inverter disconnects the power connection to the motor if no-load occurs. The inverter stays in fault mode with the error code displayed until reset.

For the no-load protection to function properly, check the RSI installation and operating instructions carefully and follow them during operation.

## Operating history

The inverter stores historical operating data. The data can be retrieved through the menu on the inverter.

## Communication

Built-in:

- Serial communication (RS485/Modbus)
- Ethernet IP
- Profinet IO
- Modbus TCP
- Bacnet IP.

Via optional boards:

- CanOpen
- Profibus DP
- DeviceNet
- Lonworks
- AS-interface.

## Product numbers

### Low voltage range (3 × 208-240 V)

| Power [kW (hp)] | Product number | Electrical data | | Frame size | IP class |
|---|---|---|---|---|---|
| | | Max. P2 [kW (hp)] | Rated output current [A] | | |
| 1.5 (2) | 99090622 | 1.5 (2) | 8 | A | IP66 |
| 2.2 (3) | 99090633 | 2.2 (3) | 11 | A | |
| 3.0 (4) | 99090634 | 3.0 (4) | 12.5 | A | |
| 4.0 (5) | 99090635 | 4.0 (5) | 18 | B | |
| 5.5 (7.5) | 99090636 | 5.5 (7.5) | 24.2 | B | |
| 7.5 (10) | 99090637 | 7.5 (10) | 31 | B | |
| 11 (15) | 99090638 | 11 (15) | 48 | C | |
| 15 (20) | 99090639 | 15 (20) | 62 | C | |

### High voltage range (3 × 380-440 V)

| Power [kW (hp)] | Product number | Electrical data | | Frame size | IP class |
|---|---|---|---|---|---|
| | | Max. P2 [kW (hp)] | Rated output current [A] | | |
| 2.2 (3) | 99044348 | 2.2 (3) | 5.6 | A | IP66 |
| 3.0 (4) | 99044349 | 3.0 (4) | 8 | A | |
| 4.0 (5) | 99044350 | 4.0 (5) | 9.6 | A | |
| 5.5 (7.5) | 99044351 | 5.5 (7.5) | 12 | A | |
| 7.5 (10) | 99044352 | 7.5 (10) | 16 | B | |
| 11 (15) | 99044363 | 11 (15) | 23 | B | |
| 15 (20) | 99044364 | 15 (20) | 31 | B | |
| 18.5 (25) | 99044365 | 18.5 (25) | 38 | C | |
| 22 (30) | 99044366 | 22 (30) | 46 | C | |
| 30 (40) | 99044367 | 30 (40) | 61 | C | |
| 37 (50) | 99044368 | 37 (50) | 72 | C | |
| 45 (60) | 99648886 | 45 (60) | 87 | MR7 | IP54 |
| 55 (75) | 99648887 | 55 (75) | 105 | MR7 | |
| 110 (150) | 99648888 | 110 (150) | 205 | MR8 | |
| 132 (200) | 99648889 | 132 (200) | 261 | MR9 | |
| 160 (250) | 99648890 | 160 (250) | 310 | MR9 | |
| 200 (300) | 99648891 | 200 (300) | 385 | ED | |
| 250 (350) | 99648892 | 250 (350) | 460 | ED | |

### Option boards

| Description | Product number |
|---|---|
| Digital | 99295565 |
| Analog | 99295551 |
| Pt100/Pt1000 | 99295525 |

For further information, see the *RSI documentation* available at the *Grundfos Product Center*.

# MP 204

TM073846

*MP 204 motor protector*

The MP 204 is an electronic motor protector designed for the protection of an asynchronous motor.

The MP 204 cannot be used in installations where a frequency converter is installed.

The MP 204 operates with the following set of limits:

- warning limits
- trip limits.

If one or more of the warning limits are exceeded, the motor continues to run, but the warnings appear on the MP 204 display.

If one of the trip limits is exceeded, the trip relay stops the motor and an alarm appears on the MP204 display. At the same time, the signal relay is operating to indicate that the limit is exceeded.

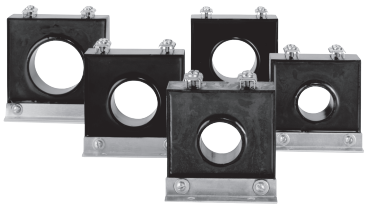
The details of warnings and alarms can be reviewed on the Grundfos GO remote app.

The MP 204 primarily protects the motor by measuring the motor current utilizing a true RMS measurement. The trip relay stops the motor if, for example, the current exceeds the preset value. Secondarily, the motor can be protected via temperature measuring by a Tempcon sensor, a Pt100/Pt1000 sensor, or a PTC sensor/thermal switch.

The MP 204 is designed for single- and three-phase motors. In single-phase motors, the starting and run capacitors are also measured. Cos ϕ is measured in both single- and three-phase systems.

The MP 204 directly measures motor currents up to 120A on the motor cables passing through the device. Five sizes of external single-turn transformers are available for motor currents higher than 120A and up to 999A.

**Note:** the monitoring of motor temperature via Tempcon is not possible when single-turn transformers are used.



TM032033

*Single-turn transformers*

## Applications

Applications of the MP 204 are the following:

- stand-alone motor protector
- remote monitoring via a Grundfos GENIbus.

## Benefits

The MP 204 offers the following benefits:

- suitability for both single-phase and three-phase motors
- dry-running protection
- overload protection
- very high accuracy
- made for submersible pumps
- insulation resistance measurement
- monitoring motor temperature throughout the motor cable in MS motors with a built-in Tempcon sensor (without the need for a control cable)
- direct measurement of motor currents up to 120A
- settings and status reading via the Grundfos GO remote app and GENIbus.

## Monitoring options

The MP 204 monitors the following parameters:

- insulation resistance before startup
- temperature (Tempcon, Pt100/Pt1000 sensor and PTC/thermal switch)
- overload/underload
- overvoltage/undervoltage
- current unbalance
- phase sequence
- phase failure
- power factor
- power consumption
- harmonic distortion
- operating hours and number of starts.

## Grundfos GO remote app

The Grundfos GO Remote app offers easy access to setting options, status information, control and fault finding of the MP 204 on your Android or iOS-based device.

The Grundfos MI 301 mobile interface is required to connect the MP 204 with your device via Bluetooth. It has a rechargeable Li-ion battery that must be charged separately. Currently, Grundfos supports Android devices using Android 5.0 and later, and Apple devices using iOS 6.1.2 and later, but cannot guarantee operation on all devices in the market.

TM053890

*MI 301*

## Communication

The MP 204 is prepared for the following communication modules:

- Grundfos Connect cellular (CIU 900 + CIM 280)
- Grundfos Connect IP (CIU 900 + CIM 550)
- PROFIBUS DP (CIU 900 + CIM 150)
- Modbus RTU (CUI 900 + CIM 200)
- Modbus Cellular/SMS (CUI 900 + CIM 260)
- Modbus TCP/PROFINET IO (CIU 900 + CIM 500).

### Related information

*Communication*

## Product numbers, MP 204

### Product numbers

| Product | Product number |
|---|---|
| MP 204 | 96079927 |
| Grundfos Go Remote interface | |
| Grundfos MI 301 (with charger) | 98046408 |
| Grundfos MI 301 (without charger) | 98761178 |
| Single-turn transformers | |
| Current transformer ratio: 200:5, $I_{max.}$ = 200 A | 96095274 |
| Current transformer ratio: 300:5, $I_{max.}$ = 300 A | 96095275 |
| Current transformer ratio: 500:5, $I_{max.}$ = 500 A | 96095276 |
| Current transformer ratio: 750:5, $I_{max.}$ = 750 A | 96095277 |
| Current transformer ratio: 1000:5, $I_{max.}$ = 1000 A | 96095278 |

For further information about motor protection via the MP 204, see the *MP 204 documentation* available at the *Grundfos Product Center*.

## Technical data, MP 204

| | |
|---|---|
| Enclosure class | IP20 |
| Ambient temperature | -20 °C to +60 °C |
| Relative humidity | 5-95 % |
| Voltage range | 100-480 VAC |
| Current range | 3-999 A |
| Frequency | 50 to 60 Hz |
| IEC trip class | 1-45 |
| Special Grundfos "P" trip class delay | 0.1 to 30 s |
| Marking | CE, cUL, UKCA |
| Consumption | Max. 5 W |
| Plastic type | Black PC/ABS |

## Electrical data, MP 204

| | Measuring range | Accuracy | Resolution |
|---|---|---|---|
| Current without external current transformers | 3-120 A | ± 1 % | 0.1 A |
| Current with external current transformers | 120-999 A | ± 1 % | 1 A |
| Phase-to-phase voltage | 80-610 VAC | ± 1 % | 1 V |
| Frequency | 47-63 Hz | ± 1 % | 0.5 Hz |
| Insulation resistance | 10-1 MΩ | ± 10 % | 10 kΩ |
| Temperature via Pt100/ Pt1000 | 0-180 °C | ± 1 °C | 1 °C |
| Temperature via Tempcon | 0-125 °C | ± 3 °C | 1 °C |
| Power consumption | 0-16 MW | ± 2 % | 1 W |
| Power factor (cos φ) | 0 - 0.99 | ± 2 % | 0.01 |
| Run capacitor (single-phase) | 10-1000 µF | ± 10 % | 1 µF |
| Start capacitor (single-phase) | 10-1000 µF | ± 10 % | 1 µF |
| Number of starts | 0-65535 | - | 1 |
| Energy consumption | 0-4 × $10^9$ kWh | ± 5 % | 1 kWh |

# Control MP 204

TM049512

The Control MP 204 cabinets are supplied with all necessary components to regulate the water supply by switching on and off a Grundfos SP pump when the installation reaches a certain water pressure or level. For this purpose, an external pressure switch or a flow switch must be connected to the control.

The integrated MP 204 device monitors and protects the motor of the pump. Additionally, the control cabinets have a built-in main switch and a thermal-magnetic circuit breaker. The MP 204 device is factory-set to an average current value, and must be adjusted during commissioning. Different models are available, depending on the motor size and the starting method.

## Functions

- Demand-depend on/off control
- Protection against dry running
- Fault indication
- Motor monitoring and protection

## Communication

See the section on MP 204 communication options.

**Related information**

*Communication*

## Technical data

| | |
|---|---|
| Number of pumps | 1 |
| Enclosure class | IPS4 |
| Ambient temperature | 0° C to 40 °C |
| Relative humidity | 95 % |
| Supply frequency | 50 Hz |
| Supply voltage | 3 × 400 |
| Rated power | 3.0 - 132 kW |
| Rated current | DOL 3 - 290 A |
| | SD 3 - 290 A |
| Motor connection | DOL or SD |
| Motor protection device | MP 204 |
| Marking | CE, UKCA |

## Product numbers

| Motor rated characteristics | | | | |
|---|---|---|---|---|
| Voltage / Hz | Power | Current | Connection | |
| [V/Hz] | [kW] | [A] | Direct on line (DOL) | Star-Delta (YD) |
| 3 × 400 V / 50 Hz | 3.0 | 3-8 | 99175449 | 99175463 |
| | 5.5 | 8-14.5 | 99175450 | 99175464 |
| | 9.2 | 14.5-23.5 | 99175451 | 99175465 |
| | 13.0 | 23.5-30.0 | 99175452 | 99175466 |
| | 15.0 | 30.0-36.5 | 99175453 | 99175467 |
| | 18.5 | 36.5-45 | 99175454 | 99175468 |
| | 26.0 | 45-61 | 99175455 | 99175469 |
| | 30.0 | 61-75 | 99175456 | 99175470 |
| | 37.0 | 75-85 | 99175457 | 99175471 |
| | 55.0 | 85-120 | 99175458 | 99175472 |
| | 75.0 | 120-156 | 99175459 | 99175473 |
| | 90.0 | 156-180 | 99175460 | 99175474 |
| | 110.0 | 180-235 | 99175461 | 99175475 |
| | 132.0 | 235-290 | 99175462 | 99175476 |

For further information, see the *Control MP 204 documentation* available at the *Grundfos Product Center*.

# Communication Interface Units (CIU)

TM078946

*Grundfos Communication Interface Unit (CIU)*

A CIU is required for complete control of the pump system and the data communication between the main network and an SP pump with a CUE, RSI or MP 204. The CIU enables data communication via open and interoperable networks, such as the following:

- LON
- PROFIBUS DP
- PROFINET IO
- Modbus RTU
- Modbus TCP
- Ethernet/IP
- BACnet MS/TP
- BACnet IP
- Grundfos Connect.

**Applications**

The range of Grundfos CIUs offers ease of installation and commissioning as well as user-friendliness. All units are based on standard functional profiles for an easy integration into the network.

The CIU devices enable communication of operating data, such as measured values and setpoints, between the pump and the following:

- PLC
- SCADA system
- building management system
- Grundfos Connect.

**Benefits**

- Open communication standards
- complete process control
- one concept for the Grundfos products
- 24-240 VAC/DC power supply in CIU devices
- simple configuration and easy installation
- readiness for DIN rail or wall mounting.

**Product numbers**

To create the necessary control unit, a CIU must be ordered with the Communication Interface Module (CIM) required for the Fieldbus protocol. Use the table below for the selection.

**Product numbers**

| Product | Product number |
|---|---|
| CIU 900 | 99448387 |
| CIM 100 | 96824797 |
| CIM 150 | 96824793 |
| CIM 200 | 96824796 |
| CIM 300 | 96893770 |
| CIM 500 | 98301408 |
| CIM 260-EU [10)] | 99439302 |
| CIM 260-US [10)] | 99439306 |
| CIM 280-EU [11)] | 99895383 |
| CIM 280-US [11)] | 99895386 |
| Antenna EU | 99518079 |
| Antenna US | 98851149 |
| Mounting bracket for Antenna-US | 99606614 |
| Grundfos Connect subscription - one year | 92711156 |

10) 3G/4G SIM card required

11) eUICC SIM card included

## Fieldbus support protocols

| Products | | MP 204 | CUE | RSI |
|---|---|---|---|---|
| GENIbus | | built-in | built-in | N/A |
| LON | | N/A | CIU 900 + CIM 100 | Optional |
| PROFIBUS DP | | CIU 900 + CIM 150 | | Optional |
| PROFINET IO | | CIU 900 + CIM 500 | | built-in |
| Modbus RTU | | CIU 900 + CIM 200 | | built-in |
| Modbus TCP | | CIU 900 + CIM 500 | | built-in |
| Modbus cellular [12) 13)] | EU | CIU 900 +CIM 260-EU + Antenna-EU + SIM card [14)] | | N/A |
| | US | CIU 900 +CIM 260-US + Antenna-US + SIM card [14)] + Mounting bracket for Antenna-US | | N/A |
| EtherNet/IP | | N/A | CIU 900 + CIM 500 | built-in |
| BACnet MS/TP | | N/A | CIU 900 + CIM 300 | N/A |
| BACnet IP | | N/A | CIU 900 + CIM 500 | built-in |
| Grundfos Connect IP [15) 16)] | | CIU 900 + CIM 500 | | |
| Grundfos Connect cellular [12) 16)] | EU | CIU 900 + CIM 280-EU [17)] + Antenna-EU | | |
| | US | CIU 900 + CIM 280-US [17)] + Antenna-US + Mounting bracket for Antenna-US | | |

12) Available in regional versions for EU and USA (consider needed 3G/4G frequency bands)

13) Cellular data connection for SCADA and SMS

14) 3G/4G SIM card required for the CIM 260

15) Wired internet connection required

16) One-year subscription required for the Grundfos Utility Connect service

17) eUICC SIM card included in the CIM 280

For further information about data communication via CIU units and fieldbus protocols, see the *CIM/CIU documentation* available at the *Grundfos Product Center*.

# PR 5714

The PR 5714 is a programmable LED indicator with relay output. In combination with a Pt100/Pt1000 sensor, it offers the following features:

- continuous monitoring of the motor temperature
- protection against too high temperature.

The Pt100/Pt1000 sensor measures the temperature, and the PR 5714 monitors it, ensuring that the preset values are not exceeded. If the preset values are exceeded, the PR 5714 relay disconnects the contactor.

Monitoring and protection employing a Pt100/ Pt1000 require the following parts:

- Pt100 or Pt1000 sensor
- PR 5714 relay
- sensor cable.

The following temperature limits are preset on delivery:

- 60 °C warning limit
- 75 °C stop limit.

The temperature limits must be adjusted during the start-up based on the temperature at maximum load in normal operation. The following temperature limits are preset in the PR 5714 on delivery:

- 60 °C warning limit
- 75 °C stop limit.

### Technical data

| | |
|---|---|
| Enclosure class | IP65 (fitted in a control panel) |
| Ambient temperature | -20 °C to +60 °C |
| Relative air humidity | 95 % (condensating) |
| Voltage variation | 1 × 24-230 VAC ± 10 %, 50-60 Hz<br>24-250 VDC ± 20 % |
| Approvals | UL, DNV |
| Marking | CE |

### Product number

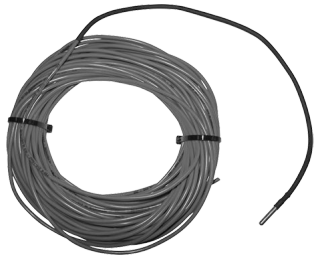
| PR 5714 relay | Voltage | Product number |
|---|---|---|
| | 24-230 VAC, 50/60 Hz / 24-250 VDC | 96913234 |

## Pt1000

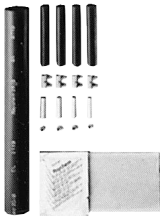
A Pt1000 temperature sensor can be installed to monitor the temperature of the MS4000 motors without Tempcon or fed by a frequency converter. To mount the Pt1000, one of the motor studs must be replaced by a staybolt for the Pt1000.

Temperature monitoring is done via the PR 5714, MP 204, CUE, RSI or another monitoring device prepared for reading the Pt1000 signal. The temperature limits for motor protection must be set on the monitoring device during start-up according to the normal operating conditions, see section Temperature monitoring.

If a sensor cable longer than 100 m is required, a 20 m sensor with cable can be spliced to the required drop sensor cable length. The recommended maximum total length is 350 m.

| Pt1000 sensor, including cable | Cable length [m] | Product number |
|---|---|---|
|  | 20 | 96804042 |
| | 40 | 96804044 |
| | 60 | 96804064 |
| | 80 | 96804065 |
| | 100 | 96804067 |

| Staybolt kits for Pt1000 in MS 402 and MS 4000 | Description | Product number |
|---|---|---|
| | Staybolt kit for Pt1000<br>Material: EN 1.4401 / AISI 316 | 98090278 |
| | Staybolt kit for Pt1000.<br>Material: EN 1.4539 / AISI 904L | 98090341 |

| Extension kit for sensor cable for Pt100/Pt1000 | Description | Product number |
|---|---|---|
|  | Extension kit for Pt100/Pt1000 sensor cable [18]<br>For watertight shrink-joining of the sensor cable | 99039717 |

18) Extra sensor cable must be ordered separately

| Drop sensor cable | Description | Product number |
|---|---|---|
| | Drop cable for extension: 4 × 1 $mm^2$ [19]<br>Maximum recommended length: 350 m. | 00RM5271 |

19) Mention length when ordering

### Related information

*Temperature monitoring*

# Model C motor cables

| Motor type | Motor plug material | Length | | Jacketed flat cable (for CE-approved motors) | | Single leads (for cCSAus-approved motors) | Single leads (for booster applications) | PTFE jacketed cable |
|---|---|---|---|---|---|---|---|---|
| | | | | EPR rubber | | XLPE rubber | | PTFE |
| | | [m] | [ft] | 4 G 1.5 mm$^2$ | 4 G 1.5 mm$^2$ 2-plugs [20] | 4 × 1 × 14 AWG | 3 × 1 × 14 AWG BM | 4G 2.5mm$^2$ |
| MS4000 Model C [21] | EN 1.4401 | 1.7 | 5.6 | 99412628 | 99412615 | - | - | - |
| | | 1.8 | 5.9 | - | - | 99412552 | - | - |
| | | 2.5 | 8.2 | 99410241 | 99410222 | 99412580 | - | - |
| | | 5 | 16.4 | 99410115 | 99410429 | 99412836 | - | - |
| | | 10 | 33 | 99410245 | 99410147 | - | - | - |
| | | 20 | 66 | 99410252 | 99410168 | - | - | - |
| | | 25 | 82 | - | - | 99413964 | - | - |
| | | 30 | 99 | 99410264 | 99410189 | - | - | - |
| | | 40 | 131 | 99410266 | 99410194 | - | - | - |
| | | 50 | 164 | 99410301 | 99410198 | - | - | - |
| | | 60 | 197 | 99410303 | 99410214 | - | - | - |
| | | 70 | 230 | 99410425 | 99410219 | - | - | - |
| MS4000R Model C [21] | EN 1.4539 | 1.7 | 5.6 | 99412529 | 99411278 | - | - | - |
| | | 1.8 | 5.9 | - | - | 99413021 | - | - |
| | | 2.0 | 6.6 | - | - | | 99412545 | 99414026 |
| | | 2.5 | 8.2 | 99411201 | 99411260 | 99412872 | - | - |
| | | 3.8 | 12.5 | - | - | - | 99414019 | - |
| | | 4.0 | 13.1 | - | - | - | - | 99413950 |
| | | 5.0 | 16.4 | 99413823 | 99411413 | 99412863 | - | - |
| | | 8.0 | 26.3 | - | - | 99414037 | - | - |
| | | 10 | 33 | 99412525 | 99411421 | - | 99414022 | 99410476 |
| | | 20 | 66 | 99412497 | 99411434 | - | - | 99411153 |
| | | 30 | 99 | 99412464 | 99411472 | - | - | 99411155 |
| | | 40 | 131 | 99412383 | 99411488 | 99413967 | - | 99411156 |
| | | 50 | 164 | 99411611 | 99411505 | - | - | 99411158 |
| | | 60 | 197 | 99412782 | 99412700 | - | - | 99411160 |
| | | 70 | 230 | 99412702 | 99412650 | - | - | 99411183 |
| | | 80 | 263 | - | - | - | - | 99411188 |
| | | 90 | 295 | - | - | - | - | 99411193 |
| | | 100 | 328 | - | - | - | - | 99411196 |
| | | 110 | 361 | - | - | - | - | 99413762 |
| | | 120 | 394 | - | - | - | - | 99413722 |
| | | 150 | 492 | - | - | - | - | 99414036 |
| | | 200 | 656 | - | - | - | - | 99413755 |
| | | 220 | 722 | - | - | - | - | 99413882 |
| | | 400 | 1312 | - | - | - | - | 99413760 |

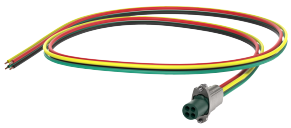
20) It is a motor cable with additional plug for drop cable. A submersible drop cable with plug is required for easy connection.

21) Model B motor cables are available as service kits.

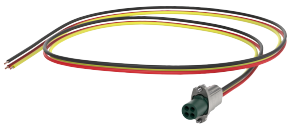
**Note:** Long motor cables that will not work completely submerged in water must be sized to work in the air. Otherwise, motor cables must work submerged in water.

Jacketed flat cable
(for CE-approved motors)

Single leads
(for cCSAus-approved motors)



Single leads
(for booster applications)



PTFE jacketed cable

Jacketed flat cable 2-plugs
(for CE approved motors)

**Related information**

*Cables*

*Moment of inertia*

## Submersible drop cable

| Product | Description | Number of leads and nominal cross-section [mm²] | Outer cable diameter min./max. [mm] | Weight [kg/m] | Product number |
|---|---|---|---|---|---|
| | Suitable for the following applications:<br>• continuous application in groundwater and potable water (approved for potable-water applications)<br>• connection of electrical equipment (e.g., submersible motors)<br>• installation depths up to 2000 m and average loads.<br>Ozone-, water- and weather-resistant insulation and sheath of special EPR-based elastomer materials.<br>Maximum permissible water temperature:<br>60 °C.<br>Maximum permissible lead service temperature:<br>90 °C.<br>Further cable sizes are available. For more information, see the *Grundfos Product Center*. | 1 × 16 | 11.0 / 14.5 | 0.280 | 00ID4071 |
| | | 1 × 25 | 12.5 / 16.5 | 0.365 | 00ID4072 |
| | | 1 × 35 | 14.0 / 18.5 | 0.490 | 00ID4073 |
| | | 1 × 50 | 16.5 / 21.0 | 0.690 | 00ID4074 |
| | | 1 × 70 | 18.5 / 23.5 | 0.920 | 00ID4075 |
| | | 1 × 95 | 21.0 / 26.5 | 1.210 | 00ID4076 |
| | | 1 × 120 | 23.5 / 28.5 | 1.455 | 00ID4077 |
| | | 1 × 150 | 26.0 / 31.5 | 1.825 | 00ID4078 |
| | | 1 × 185 | 27.5 / 34.5 | 2.160 | 00ID4079 |
| | | 4G1.5 | 10.5 / 13.5 | 0.165 | 00ID4063 |
| | | 4G2.5 | 12.5 / 15.5 | 0.235 | 00ID4064 |
| | | 4G4.0 | 14.5 / 18.0 | 0.335 | 00ID4065 |
| | | 4G6.0 | 16.5 / 22.0 | 0.460 | 00ID4066 |
| | | 4G10 | 22.5 / 24.5 | 0.800 | 00ID4067 |
| | | 4G16 | 26.5 / 28.5 | 1.165 | 00ID4068 |
| | | 4G25 | 32.0 / 34.0 | 1.650 | 00ID4069 |
| | | 4G35 | 33.0 / 42.5 | 2.200 | 96432949 |
| | | 4G50 | 38.0 / 48.5 | 3.260 | 96432950 |
| | | 4G70 | 43.0 / 54.5 | 4.149 | 96432951 |

**Related information**

*Cables*

*8. Cable sizing*

## Submersible drop cables with plug

It is used for an easy connection to the 2-plug motor cable of a MS402 or MS4000 motor.

| Product | Cable length [m] | Product number |
|---|---|---|
| | 4 × 1.5 $mm^2$ | |
| | 15 | 96737342 |
| | 20 | 96737343 |
| | 25 | 96737344 |
| | 30 | 96737345 |
| | 40 | 96737347 |
| | 50 | 96737349 |
| | 70 | 96737351 |
| | 100 | 96737354 |
| | 4 × 2.5 $mm^2$ | |
| | 15 | 0079H021 |
| | 20 | 0079H022 |
| | 25 | 0079H023 |
| | 30 | 0079H024 |
| | 40 | 0079H025 |
| | 50 | 0079H026 |
| | 70 | 0079H028 |
| | 100 | 0079H029 |
| | 4 × 4 $mm^2$ | |
| | 15 | 0079H041 |
| | 20 | 0079H042 |
| | 25 | 0079H043 |
| | 30 | 0079H044 |
| | 40 | 0079H045 |
| | 50 | 0079H046 |
| | 70 | 0079H048 |
| | 100 | 0079H049 |

### Related information

*Cables*

*8. Cable sizing*

## Female plug kit for drop cable

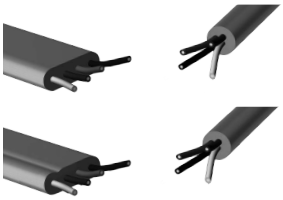
| Product | Description | Version | Product number | |
|---|---|---|---|---|
| | | | N-version | R-version |
| TM007883 | It is a female plug for watertight joining of the MS402/ MS4000 with 2-plug motor cable and submersible drop cable in an acrylic tube filled with resin. It is used for both single- and multi-core cables during installation of submersible pumps. 24 hours of hardening is required. | For cables up to 4 × 2.5 $mm^2$ | 00799901 | 00799918 |
| | | For cables up to 4 × 6 $mm^2$ | 00799902 | 97937466 |

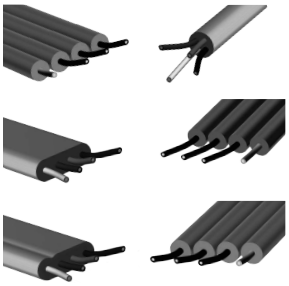
### Related information

*Cables*

# Cable termination kit, type KM

For instruction on how to make the cable termination between motor cable and drop cable, see the *KM quick guide* available at the *Grundfos Product Center*.

| Possible cable termination | | Content of kit | Motor cable [mm²] | Drop cable [mm²] | Number of leads | Product number |
|---|---|---|---|---|---|---|
| Motor cable | Drop cable | | | | | |
|  | | | KM kits with pressed connections: | | | |
| | | | 1.5 - 6 | 1.5 - 6 | 4 | 00116251 |
| | | | 6 - 16 | 6 - 16 | 4 | 00116252 |
| | | | 10 - 25 | 10 - 25 | 4 | 00116255 |
| | | | KM kits with screw connectors: | | | |
| | | | 6 - 35 | 6 - 35 | 4 | 96636867 |
| | | | 25 - 70 | 25 - 70 | 4 | 96636868 |

| Possible cable termination | | Content of kit | Motor cable [mm²] | Drop cable [mm²] | Number of leads | Product number |
|---|---|---|---|---|---|---|
| Motor cable | Drop cable | | | | | |
|  | | | KM kits with pressed connections: | | | |
| | | | 1.5 - 6 | 1.5 - 6 | 4 | 00116257 |
| | | | 6 - 16 | 6 - 16 | 4 | 00116258 |
| | | | 10 - 50 | 10 - 50 | 4 | 96637330 |
| | | | 16 - 70 | 16 - 70 | 4 | 96637332 |
| | | | 1.5 - 6 | 1.5 - 6 | 3 | 00116253 |
| | | | 10 - 25 | 10 - 25 | 3 | 00116254 |
| | | | 10 - 50 | 10 - 50 | 3 | 96637318 |
| | | | 16 - 70 | 16 - 70 | 3 | 96637331 |

| Possible cable termination | | Content of kit | Motor cable [mm²] | Drop cable [mm²] | Number of leads | Product number |
|---|---|---|---|---|---|---|
| Motor cable | Drop cable | | | | | |
| | | | KM kits with pressed connections: | | | |
| | | | 10-70 | 10-70 | 1 | 96828296 |
| | | | 32-120 | 32-120 | 1 | 00116256 |
| | | | KM kits with screw connectors: | | | |
| | | | 70-240 | 70-240 | 1 | 96637279 |

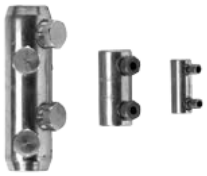
**Note:** A KM termination kit for single leads only consists of material for one connection. When ordering, keep in mind how many kits are needed for a complete cable termination.

## Related information

*Cables*

## Cable termination kit, types M0 to M4

| Product | Description | Version | | | |
|---|---|---|---|---|---|
| | | Type | Diameter of cable joint [mm] | Outer cable diameter [mm] | Product number |
| | This kit is for watertight joining of motor cable and submersible drop cable.<br>The joint is encapsulated by the glue which is part of the kit.<br>**Note:** When ordering this cable termination kit, the screw connectors are not included. For ordering suitable screw connectors, see the table below. | M0 | Ø40 | Ø6 - Ø15 | 00ID8903 |
| | | M1 | Ø46 | Ø9 - Ø23 | 00ID8904 |
| | | M2 | Ø52 | Ø17 - Ø31 | 00ID8905 |
| | | M3 | Ø77 | Ø26 - Ø44 | 00ID8906 |
| | | M4 | Ø97 | Ø29 - Ø55 | 91070700 |

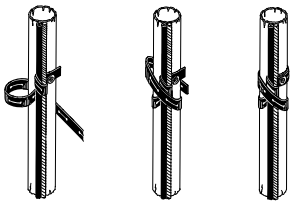
| | | Cross-section of leads [$mm^2$] | Number of connectors | Product number |
|---|---|---|---|---|
|  | These are accessories for cable kits M0 to M4.<br>These are the screw connectors only. | 6-25 | 4 | 96626021 |
| | | 16-95 | | 96626022 |
| | | 35-185 | | 96626023 |
| | | 70-240 | | 96626028 |

In high pressure, high temperature or sea water applications, use the type KM cable termination kits.

**Related information**

*Cables*

## Cable clips

| Product | Description | Product number |
|---|---|---|
|  | The clips are for fastening of the cable and straining wire to the riser pipe.<br>They must be fitted every 3 meters.<br>One set for approximately 45 m riser pipe:<br>• 16 cable buttons.<br>• 7.5 m rubber band. | 00115016 |

**Related information**

*Cables*

## Zinc anodes

Cathodic protection by zinc anodes protects submersible motors against corrosion in chloride-containing liquids, such as brackish water and seawater.

The sacrificial anodes are placed on the outside of the motor, increasing the motor diameter by a minimum of 40 mm. To ensure proper electric and metallic contact between the clamp and the motor surface, the motor surface must be cleaned thoroughly before a new anode string is fitted, and the anode strings must be fastened tightly. The distance between the anode strings must be identical.

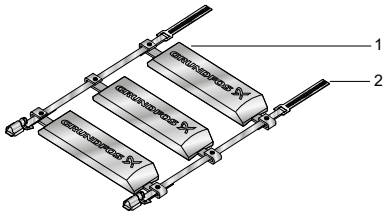
During operation, the material of the zinc anodes is reduced, and the anodes become gradually covered by corrosion products, which obstructs the direct metallic contact between the anode and the motor surface. To counteract this, the metallic contact must be ensured through the clamp, keeping tight contact with the motor by adjusting the clamp.

Inspection must be made at regular intervals to ensure the functioning of the galvanic cathodic protection system. The first inspection is recommended after six months, while the following subsequently approximately once a year.

The life of a zinc anode is 1 to 4 years, depending on the operation conditions.

The number of anodes required depends on the operation conditions, the pump, and the motor in question.

The following tables show the available kits and the number of anode strings recommended for each motor model. Contact Grundfos for further details.




TM078809

*Anode string components*

| Pos. | Description | Part |
|---|---|---|
| 1 | 1 pc. zinc anode cast around the clamp | PN 99722879 |
| 2 | 2 pcs. stainless steel clamps | PN 99812400 |

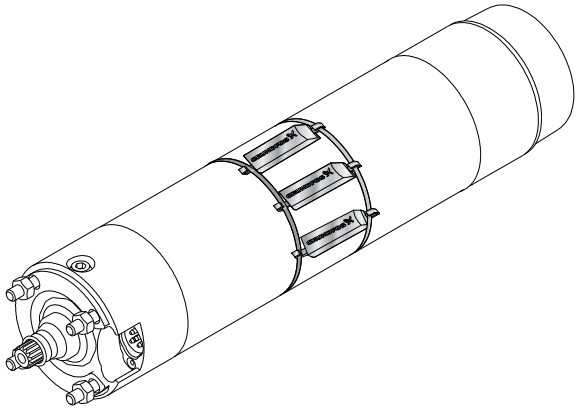
### Anode string kits

| Kit no. | Kit contents |
|---|---|
| 92836041 | 1 × 99812400, 2 × 99722879 |
| 92836043 | 1 × 99812400, 4 × 99722879 |
| 92621545 | 1 × 99812400, 6 × 99722879 |

TM083095

*Anode string kit with 2 anodes*

| Zinc anodes for MS - MMS motors | | | |
|---|---|---|---|
| Motor | Length | Recommended number of anode strings | Ordering |
| | | | Product number |
| MS402 | Up to 350 mm | 1 | 92836041 |
| MS4000 | Up to 1000 mm | 1 | 92836041 |
| MS6000 | Up to 1000 mm | 1 | 92836043 |
| MMS6 | Up to 1000 mm | 1 | 92836043 |
| MMS6 | 1000 to 1400 mm | 2 | 92836043 |
| MMS8000 | Up to 1000 mm | 1 | 92836043 |
| MMS8000 | 1000 to 2060 mm | 2 | 92836043 |
| MMS10000 | Up to 2000 mm | 2 | 92621545 |
| MMS10000 | 2000 to 2500 mm | 3 | 92621545 |
| MMS12000 | Up to 2000 mm | 2 | 92621545 |
| MMS12000 | 2000 to 2300 mm | 3 | 92621545 |



TM086185

*One anode string on the MS submersible motor*

## Flow sleeves

Grundfos offers a complete range of stainless steel flow sleeves for vertical and horizontal operation. The selection depends on the combined SP pump and motor assembly. For more information, review the *SP Accessories Data Booklet* available at the *Grundfos Product Center*.

TM078415

*Flow sleeves*

# 8. Cable sizing

Grundfos offers round single-core and multicore submersible drop cables. Drop cables for Grundfos 4" submersible motors are available without or with a plug for easy connection to the motor cable.

The submersible drop cable must be sized to work in air based on:

- the allowed voltage drop in its entire length (from the motor starter to the motor terminals)
- the current capacity of its cross-section at the maximum air temperature at which the cable will work.

The voltage supplied at the motor terminals must always be within the motor voltage tolerances. For this reason, the voltage drop in the cable and other systems between the voltage source and the motor terminals must be considered or limited.

Grundfos recommends allowing a maximum voltage drop of 1 % to 3 % in the drop cable.

Additionally, the energy losses in the cable are proportional to the voltage drop, thus limiting the voltage drop may save energy.

Local and national requirements regarding cable sizing and selection must always be followed and can demand the use of a bigger cross-section than the one computed by the following formulas and tables.

The cable properties used in the following formulas and tables correspond to Grundfos submersible drop cables. If other brands are used, contact your cable supplier for guidance.

**Related information**

*Submersible drop cable*

*Submersible drop cables with plug*

## Calculation of the cross section

Use the following formula to calculate the minimum required cross-section of a cable as a function of the installation length for a three-phase submersible pump.

**Single-phase motor**

$$q = \frac{I \times 2 \times 100 \times L \times \rho \times \cos\varphi}{U \times \Delta U - (I \times 2 \times 100 \times L \times X_L \times \sin\varphi)}$$

**Three-phase motor (DOL)**

$$q = \frac{I \times 1.73 \times 100 \times L \times \rho \times \cos\varphi}{U \times \Delta U - (I \times 1.73 \times 100 \times L \times X_L \times \sin\varphi)}$$

**Three-phase motor (Star-delta)**

$$q = \frac{I \times 100 \times L \times \rho \times \cos\varphi}{U \times \Delta U - (I \times 100 \times L \times X_L \times \sin\varphi)}$$

## Calculation of the maximum cable length

The maximum cable length for a given cable cross-section can be calculated by the following formulas.

**Single-phase motor**

$$L = \frac{U \times \Delta U}{I \times 2 \times 100 \times \left(\cos\varphi \times \frac{\rho}{q} + \sin\varphi \times X_L\right)} \quad [m]$$

**Three-phase motor (DOL)**

$$L = \frac{U \times \Delta U}{I \times 1.73 \times 100 \times \left(\cos\varphi \times \frac{\rho}{q} + \sin\varphi \times X_L\right)} \quad [m]$$

**Three-phase motor (Star-delta)**

$$L = \frac{U \times \Delta U}{I \times 100 \times \left(\cos\varphi \times \frac{\rho}{q} + \sin\varphi \times X_L\right)} \quad [m]$$

The maximum cable length may be based on the maximum operating current when it is lower than the rated current of the motor. For example, if the maximum operating current is 10 % lower than the rated current of the motor, the cable may be 10 % longer than the value computed with the formula or indicated in the tables.

## Calculation of the energy loss

The cross-section of the cable determines the energy losses throughout its length. To minimize operating losses, the cable cross-section may be increased compared to what is calculated based on the maximum allowed voltage drop. This is only economical if the borehole provides the necessary space and if the operational time of the pump is long. The energy loss is computed by multiplying the power loss with the operating hours.

The following formula can be used to calculate the power loss in the cable of a three-phase system.

$$\Delta p = \frac{3 \times L \times \rho \times I^2}{q} \quad [W]$$

| Formula designations | | |
|---|---|---|
| ΔU | = | Allowed voltage drop in the cable: 1 to 3 [%] |
| U | = | Rated voltage [V] |
| I | = | Rated current [A] |
| cos φ | = | Rated power factor |
| q | = | Cable cross-section [$mm^2$] |
| ρ | = | Cable specific resistance: 0.025 [$\Omega.mm^2/m$] |
| $X_L$ | = | Cable inductive resistance: $0.078 \times 10^{-3}$[Ω/m] |
| L | = | Cable length [m] |
| Δp | = | Power loss [W] |
| sin φ | = | $\sqrt{1 - \cos^2\varphi}$ |

## Selection tables

For an easy selection, the tables below indicate the maximum length of Grundfos submersible drop cables for different cross-sections. The cable length is calculated based on a maximum voltage drop of 3 % and air temperature of 30 °C.

**T4 motors at 3 × 400V, 50 Hz, DOL**

**Voltage drop 3 %**

**Maximum cable lengths [m]**

| P2 | P2 | Imax | Cos φ | Cable cross-section [mm²] | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| kW | hp | [A] | - | 1.5 | 2.5 | 4.0 | 6.0 | 10 | 16 | 25 | 35 | 50 |
| 0.75 | 1.00 | 2.0 | 0.84 | 247 | 411 | 655 | 979 | | | | | |
| 1.10 | 1.50 | 2.8 | 0.81 | 183 | 304 | 485 | 724 | | | | | |
| 1.50 | 2.00 | 4.0 | 0.78 | 133 | 221 | 352 | 526 | 868 | | | | |
| 2.20 | 3.00 | 6.0 | 0.74 | 93 | 155 | 247 | 369 | 608 | 956 | | | |
| 3.00 | 4.00 | 7.9 | 0.81 | 65 | 108 | 172 | 257 | 424 | 670 | | | |
| 4.00 | 5.50 | 9.6 | 0.80 | 54 | 90 | 143 | 214 | 353 | 557 | 853 | | |
| 5.50 | 7.50 | 13.0 | 0.81 | 39 | 66 | 104 | 156 | 258 | 407 | 624 | 855 | |
| 7.50 | 10.00 | 19.3 | 0.77 | 28 | 46 | 74 | 110 | 182 | 287 | 438 | 599 | 827 |
| Cable current capacity in air at 30 °C [A] | | | | 23 | 30 | 41 | 53 | 74 | 99 | 131 | 162 | 202 |

For motors with star-delta starting, the cable length can be calculated by multiplying the relevant cable length from the above tables by $\sqrt{3}$.

**T4 motors at 3 × 460V, 60 Hz, DOL**

**Voltage drop 3 %**

**Maximum cable lengths [m]**

| P2 | P2 | Imax | Cos φ | Cable cross-section [mm²] | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| kW | hp | [A] | - | 1.5 | 2.5 | 4.0 | 6.0 | 10 | 16 | 25 | 35 | 50 |
| 0.75 | 1.00 | 2.2 | 0.86 | 252 | 420 | 670 | | | | | | |
| 1.10 | 1.50 | 3.0 | 0.80 | 199 | 330 | 527 | 787 | | | | | |
| 1.50 | 2.00 | 4.1 | 0.82 | 142 | 236 | 376 | 562 | 929 | | | | |
| 2.20 | 3.00 | 5.8 | 0.78 | 105 | 175 | 279 | 417 | 688 | | | | |
| 3.00 | 4.00 | 7.5 | 0.83 | 77 | 127 | 203 | 304 | 502 | 793 | | | |
| 3.70 | 5.00 | 8.6 | 0.80 | 69 | 115 | 184 | 274 | 453 | 715 | | | |
| 4.00 | 5.50 | 9.4 | 0.82 | 62 | 103 | 164 | 245 | 405 | 640 | 981 | | |
| 5.50 | 7.50 | 12.6 | 0.82 | 46 | 77 | 122 | 183 | 302 | 477 | 732 | | |
| 7.50 | 10.00 | 18.6 | 0.78 | 33 | 55 | 87 | 130 | 215 | 338 | 517 | 708 | 977 |
| Cable current capacity in air at 30 °C [A] | | | | 23 | 30 | 41 | 53 | 74 | 99 | 131 | 162 | 202 |

For motors with star-delta starting, the cable length can be calculated by multiplying the relevant cable length from the above tables by $\sqrt{3}$.

## Examples

### Maximum cable length

| **Example** | | |
|---|---|---|
| Allowed voltage drop (ΔU) | = | 3 % |
| Motor | = | 5.5 kW, MS4000, T6 |
| Connection method | = | Direct on line |
| Rated voltage (U) | = | 3 × 400 V, 50 Hz |
| Rated current (I) | = | 14.4 A |
| Rated power factor (cos φ) | = | 0.78 |
| Cable cross-section (q) | = | 4 mm² |
| Cable specific resistance (ρ) | = | 0.025 [Ω.mm²/m] |
| Cable inductive resistance ($X_L$) | = | 0.078 × $10^{-3}$[Ω/m] |
| sin φ | = | 0.63 |

$$L = \frac{400 \times 3}{14.4 \times 1.73 \times 100 \times \left(0.78 \times \frac{0.025}{4} + 0.63 \times 10^{-3}\right)} \quad [m]$$

L = 98 m

### Energy loss comparison

| **Example** | | |
|---|---|---|
| Motor | = | 5.5 kW, MS4000, T6 |
| Connection method | = | Direct on line |
| Rated voltage (U) | = | 3 × 400 V, 50 Hz |
| Rated current (I) | = | 14.4 A |
| Cable length (L) | = | 80 m |
| Cable specific resistance (ρ) | = | 0.025 [Ω.mm²/m] |
| Operating hours/year | = | 4000 h |
| Cable cross-section (q) options | | |
| Choice A: | = | 4 mm² |
| Choice B: | = | 6 mm² |

**Choice A**

$$\Delta p_A = \frac{3 \times 80 \times 0.25 \times 14.4^2}{4}$$

$\Delta p_A = 311\,W$

$Energy\ loss_A$ = 311 W × 4000 h = 1444 kWh

**Choice B**

$$\Delta p_B = \frac{3 \times 80 \times 0.25 \times 14.4^2}{6}$$

$\Delta p_B = 207\,W$

$Energy\ loss_B$ = 207 W × 4000 h = 829 kWh

**Savings**

| | | |
|---|---|---|
| Annual savings | = | $Energy\ loss_A$ - $Energy\ loss_B$ |
| | = | 1244 kWh - 829 kWh |
| | = | 415 kWh |

By choosing the 6 $mm^2$ instead of the 4 $mm^2$ cable, an annual saving of 415 kWh can be achieved. Considering an evaluation time of 10 years, the total energy saving can be:

| | | |
|---|---|---|
| Saving energy after 10 years | = | 415 × 10 |
| | = | 4150 kWh |

The amount saved can be calculated in local currency by multiplying the saved energy by the local energy cost.

# 9. Further product documentation

*MS - MMS Installation and operating instructions*
*MS - MMS Safety instructions and other important information*
*MS Service Kit Catalogue*
*MS4000 model C Service Instruction*

# 10. Grundfos Product Center

Online search and sizing tool to help you make the right choice.
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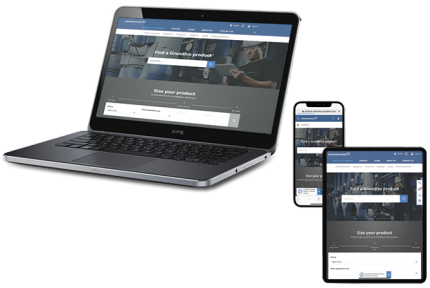

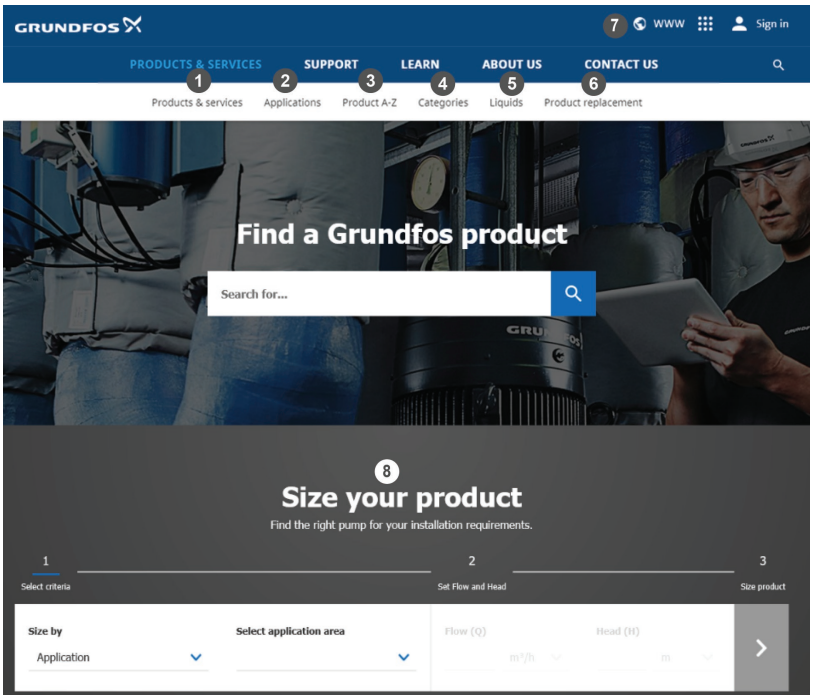


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